



School of Public Policy

PP413 – Final Report

Systematic Country Diagnostics of
Vietnam



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***Note:** The graphs for Economic Growth History, The Binding Constraint and The Growth Syndrome are a part of the report. All the other graphs are added in the appendix.*

No

Executive Summary

In 1975, as the last American helicopters lifted off from the U.S. embassy in Saigon, Vietnam was a nation in ruins. Its infrastructure had been devastated by war, and its economy was predominantly reliant on subsistence farming under the shadow of French colonial exploitation. Few could have predicted that, within a few decades, this war-torn country would transform into one of the world's fastest-growing economies.

Vietnam, a middle-income country with a real GDP of \$377 billion and a GDP per capita of \$3,760 in 2023, has undergone a transformative economic journey since the 1986 *Đổi Mới* reforms, which shifted the economy from a centrally planned model to a mixed economy incorporating socialist values. The country's growth since the economic reforms has not only been high, but also stable. The country is export-oriented with exports standing at 87.1% of its GDP in 2023 - a number higher than all its counterparts except Singapore - and dominated by the industry sector. These statistics and reforms, combined with the increase in FDI influx, present Vietnam to be a recipe for success. Yet, its future growth is under threat.

Demographically, Vietnam's population is aging rapidly, and the country is out of its demographic window of opportunity. Its labour force is also starting to shrink. The threat of shrinking labor force is compounded by a fall in the relative labor productivity in the secondary and tertiary sectors. Additionally, Vietnam's gross capital formation has fallen post-COVID and now shows stagnation, indicating slower capital deepening. According to Solow's (1957) model, this indicates growth led by total factor productivity (TFP). The concern of sustaining Vietnam's growth in circumstances of the TFP growth plateauing then arises.

This report summarizes the causes that may prove to be primary barriers to Vietnam's long term economic growth. It starts by laying out detailed growth trajectory of the country, delves deep into identifying binding constraints in consideration with the growth tree, diagnosis a syndrome for the country and concludes by providing comprehensive, specific recommendations.

Upon analysis, inadequate **transport infrastructure** is identified as the primary hindrance to Vietnam's long-term growth. The report adopts a two-pronged approach to identify roads and ports as separate but equally critical binding constraints. Inadequate road infrastructure in Vietnam causes congestion, delaying work hours and reducing labour productivity due to exhaustion. Statistics also show low road quality of Vietnam and its poor performance on the logistics performance index. On the other hand, inadequate port infrastructure is resulting in export delays. Given Vietnam's dependence on exports, these delays are alarming. Transport dependent sectors also experience lower growth rates across Vietnam. The report also touches upon aspects that may be concerning in the near future but are not currently binding to the nation's progress.

To foster growth, the report recommends a policy package to support the current infrastructure strategy of Vietnam, which includes **integrating a multimodal transport network** and **digitisation and integration of technology** in the existing logistic chains. These are proposed

with the aim of **reducing congestion, costs** and importantly, **streamlining the transport network of Vietnam**

1. Economic Growth History

1.1 Growth Trajectory

Vietnam, a middle-income country, has undergone a remarkable transformation. Its modern history began with a break from French colonial rule, followed by a Cold War-era division between the communist North and capitalist South. This ideological split culminated in a devastating war, marked by immense civilian loss and widespread destruction. After the withdrawal of American forces, Vietnam was reunified under the communist leadership of Ho Chi Minh. Today, the country is governed by a single-party communist state that oversees key public services and state-owned enterprises. Yet, Vietnam's impressive rise is not just a political story—it is, above all, an economic one. Over the past few decades, the country has transitioned from an agrarian base to a dynamic, export-driven economy, positioning itself as one of Asia's most promising emerging markets.

Vietnam with a real GDP of \$35 billion in 1985 experienced a stable average annual growth of 6.2% to reach its current real GDP of \$377 billion (Figure 1.1a). The country lifted 30 million people out of poverty – a remarkable feat – with GDP per capita reaching \$3,760 in 2023 and achieving an average growth rate of 4.7% annually (Figure 1.1b). But not only is this growth amongst the highest in the region, but it is also highly stable (Figure 1.1c). Vietnam's GDP growth rates are the least volatile among peers and one of the least volatile between 1994-2014 in the world (World Bank, 2020). Moreover, in COVID, the country maintained positive growth unlike all its peers except Bangladesh, despite facing one of the strictest lockdowns in the world (PwC, 2022).

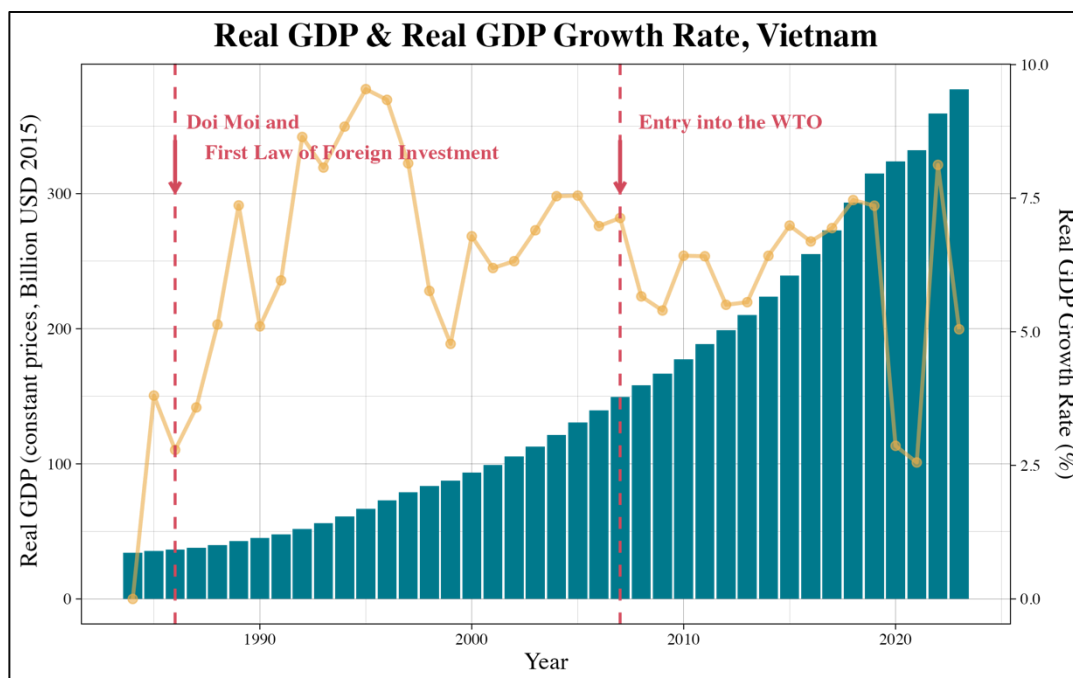


Figure 1.1a: Real GDP and Real GDP Growth Rate, Vietnam

Source: World Development Indicators

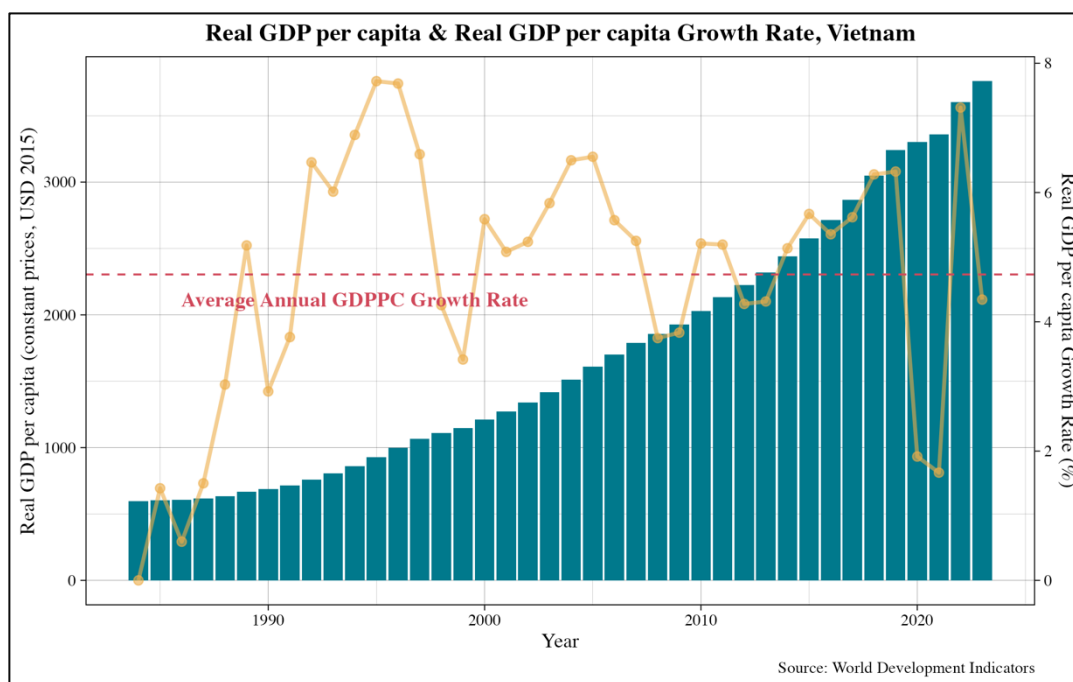


Figure 1.1b: Real GDPPC and Real GDPPC Growth Rate, Vietnam

Source: World Development Indicators

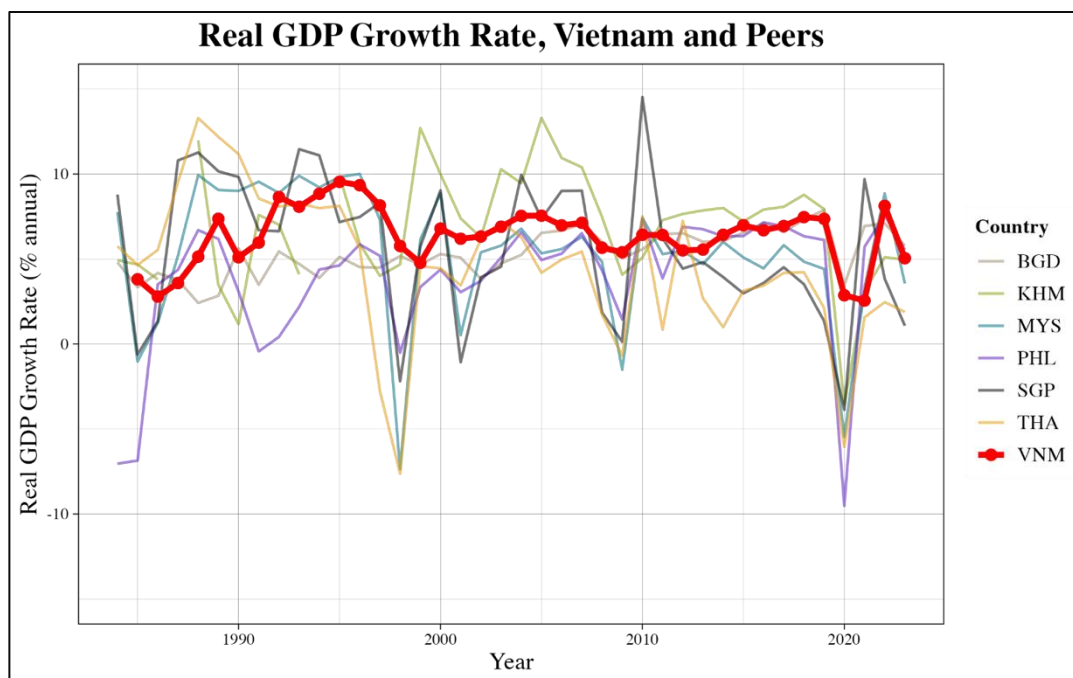


Figure 1.1c: Real GDP Growth Rate, Vietnam and Peers

Source: World Development Indicators

In 1986, the Vietnamese Communist Party passed a bundle of market-oriented reforms – Doi Moi reforms. Bureaucratic centralized management based on state-subsidies was abolished, and a multi-sector market-first approach was adopted. For instance, land ownership was incentivized – moving away from collective farming. Farmers could now produce more and invest in their piece of land. Furthermore, the Law on Investment (1988) allowed for full foreign ownership of companies and created export processing zones. This business-friendly environment allowed for growing investment in the country, especially foreign direct investment (FDI). Increase in FDI inflows from less than 0.01% of GDP in 1986 to 4.3% in 2023, presented Vietnam with key capital and technological spillovers to kickstart its industrial base.

Vietnam's demographic transition is particularly striking, as the country faces the dual challenge of a rapidly aging population and a narrowing window for demographic advantage. In 2023, Vietnam's population reached 100 million, with 12% aged 65 and above, and around 24% aged 0–14 - leaving approximately 64 million working-age adults (Figure 1.1d). This structure, coupled with a low fertility rate of just 2 births per woman and a steadily rising life expectancy of 75 years (2023), signals a swift shift toward an aging society (Figure 1.1e). As a result, Vietnam's labor force is beginning to contract, and its age dependency ratio is rising sharply – resulting in a relatively brief window of opportunity (Figure 1.1f). According to UNFA (2021), Vietnam will

enter a “super-aged population period” in 2055 with individuals above 65 years of age comprising 21% of the population. These demographic trends make it increasingly important for Vietnam to increase its unit-labor productivity and sustain long-run growth rates.

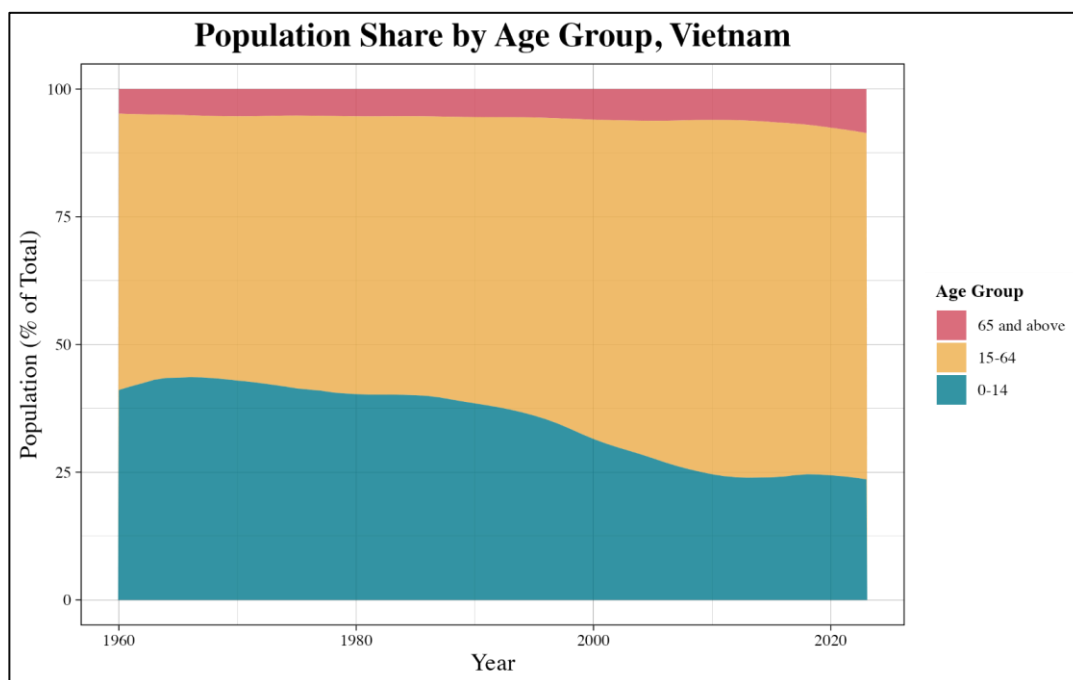


Figure 1.1d: Population Structure, Vietnam

Source: World Development Indicators

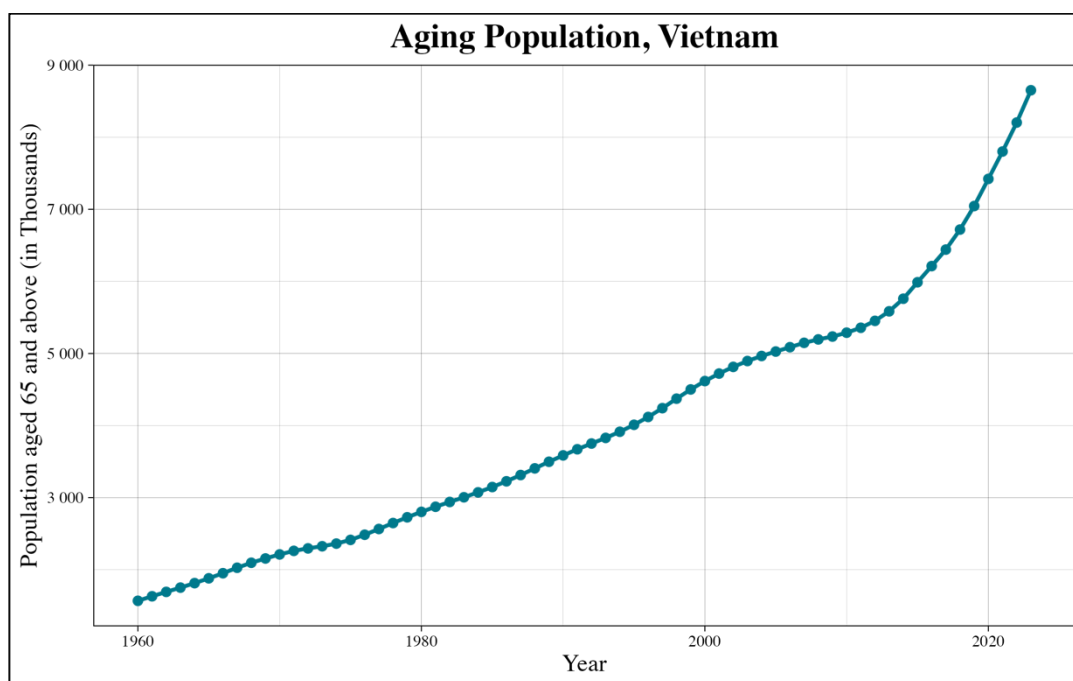


Figure 1.1e: Aging Population, Vietnam

Source: World Development Indicators

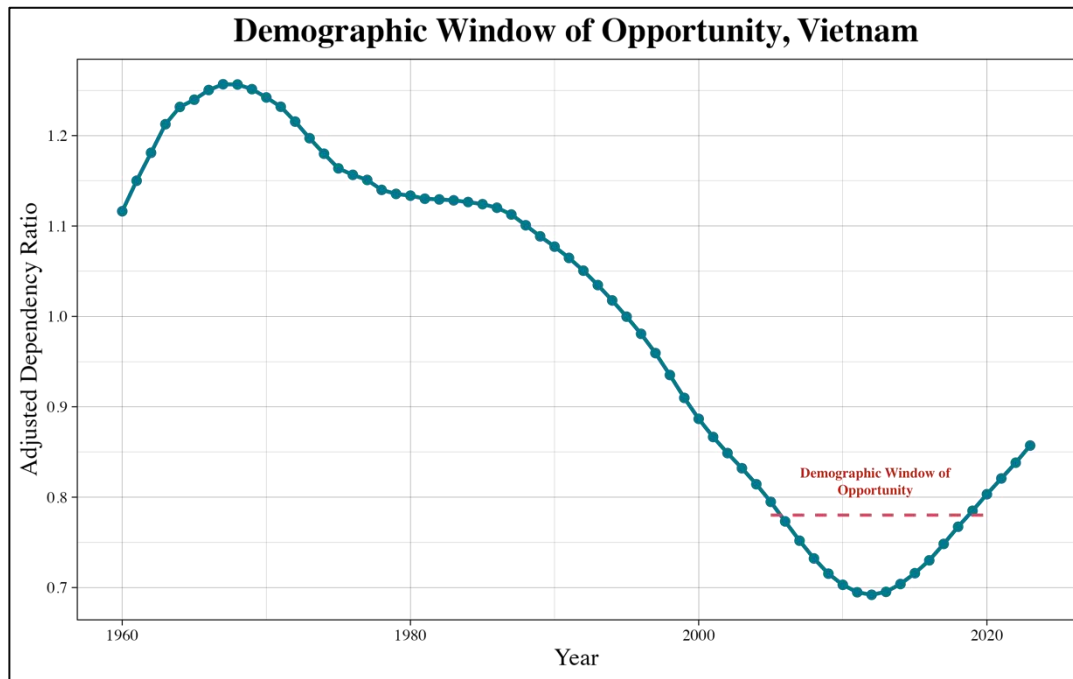


Figure 1.1f: Demographic Window of Opportunity, Vietnam

Source: World Development Indicators

Currently, Vietnam's working population is predominantly employed across agriculture, manufacturing, and services, with the labor force roughly evenly distributed among the three sectors (Figure 1.1g). However, services and manufacturing collectively contribute around 90% of GDP. Worryingly, these are the very sectors now experiencing a decline in output per worker—an alarming trend given the country's growing need to offset demographic constraints. (Figure 1.1h). As productivity weakens, each working-age individual is producing less than in previous years, reversing the trend Vietnam needs to achieve to maintain economic momentum in the face of an aging population. Alongside falling labour productivity is a plateauing capital stock. Capital accumulation rose significantly since the Doi Moi Reforms but fell dramatically during the global financial crisis. Gross capital formation fell from 32% of GDP to just 23% between 2008-2013. The gross capital formation rate has since plateaued at 23% (Figure 1.1i). The stagnancy in capital formation is concerning, for it indicates slower capital deepening. In other words, if investment remains lower relative to GDP, Vietnam may experience slower capital accumulation, reducing productivity gains from new infrastructure, machinery, and technology. This could limit industrial upgrading and the country's ability to transition into more advanced, high-value industries. Given

Vietnam's aging population and decelerating growth in labor force, labor productivity growth may not be fast enough to offset the capital formation stagnancy, leading to slower overall GDP growth.

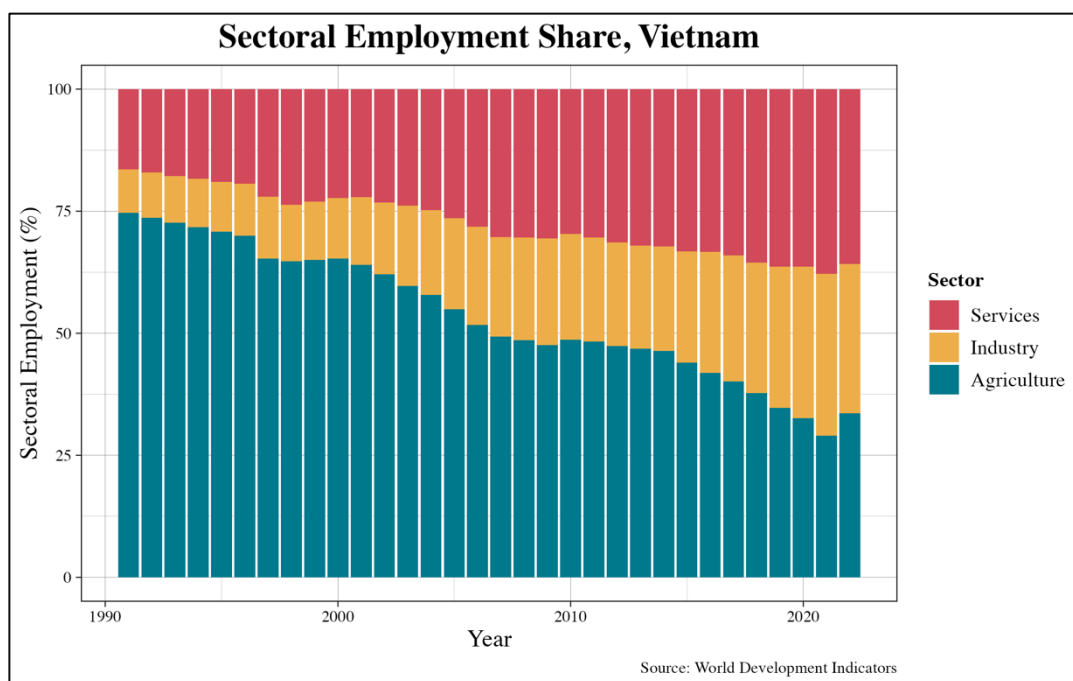


Figure 1.1g: Sectoral Employment, Vietnam

Source: World Development Indicators

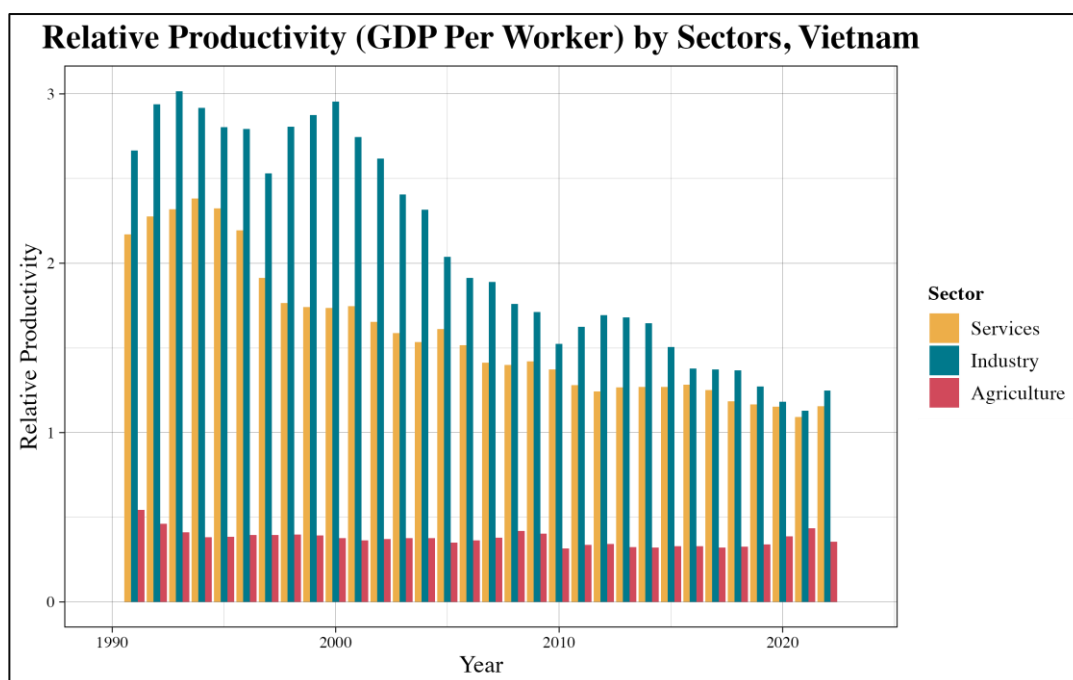


Figure 1.1h: Relative Productivity (GDP Per Worker) by Sectors, Vietnam

Source: World Development Indicators

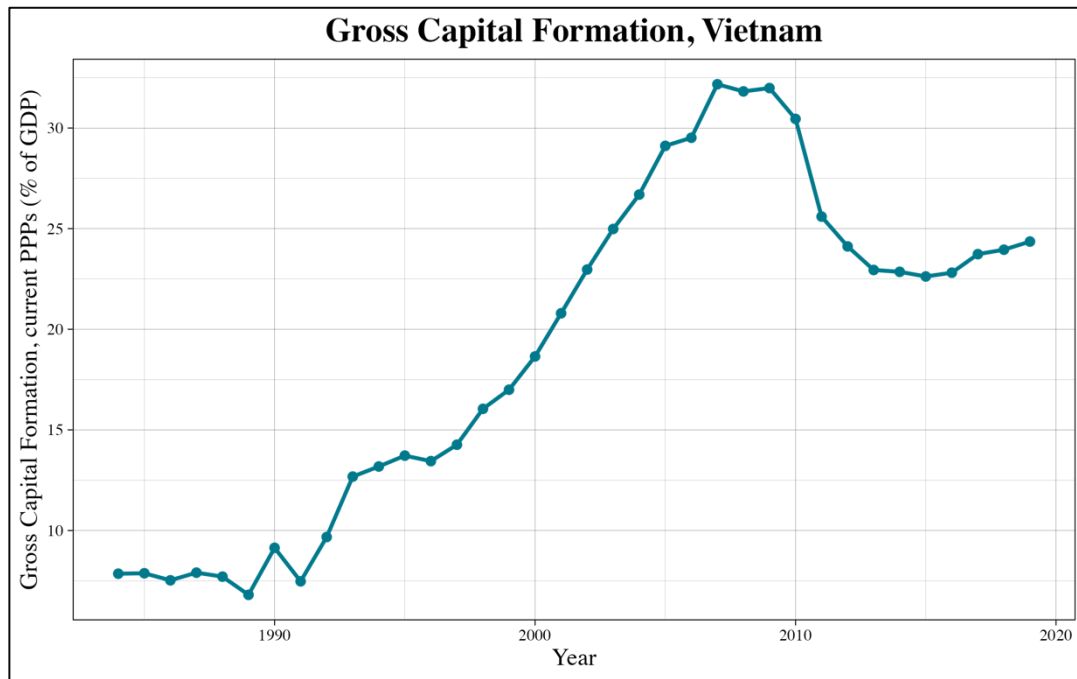


Figure 1.1i: Gross Capital Formation, Vietnam

Source: Penn World Tables

Vietnam's impressive economic growth has been driven by total factor productivity (TFP). The country's TFP has faced multiple fluctuations over the last two decades (Figure 1.1j). The TFP peaked in 2008 with a value of 1.8. This is in line with a time of strong export growth, increased FDI, and Vietnam's integration into the global value chain with its WTO accession in 2007. The sharp rise in TFP was followed by a sharper fall, explained by the disrupted export demand due to the global financial crisis (2007-2009) and internal economic challenges such as banking weaknesses and a property market bubble (Van, 2017). Vietnam's TFP has since recovered, stabilised and is now exhibiting a gradual climb. Confirming Solow's (1957) seminal claims, Vietnam's high TFP values display technological progress' role in driving its 6% average annual GDP growth, given the backdrop of a stagnation in capital stock formation and declining relative productivity across sectors. As TFP growth is keeping Vietnam's current growth rates elevated, it is increasingly important to question how to sustain this growth in the long run so that Vietnam can continue growing and emerge as a developed economy.

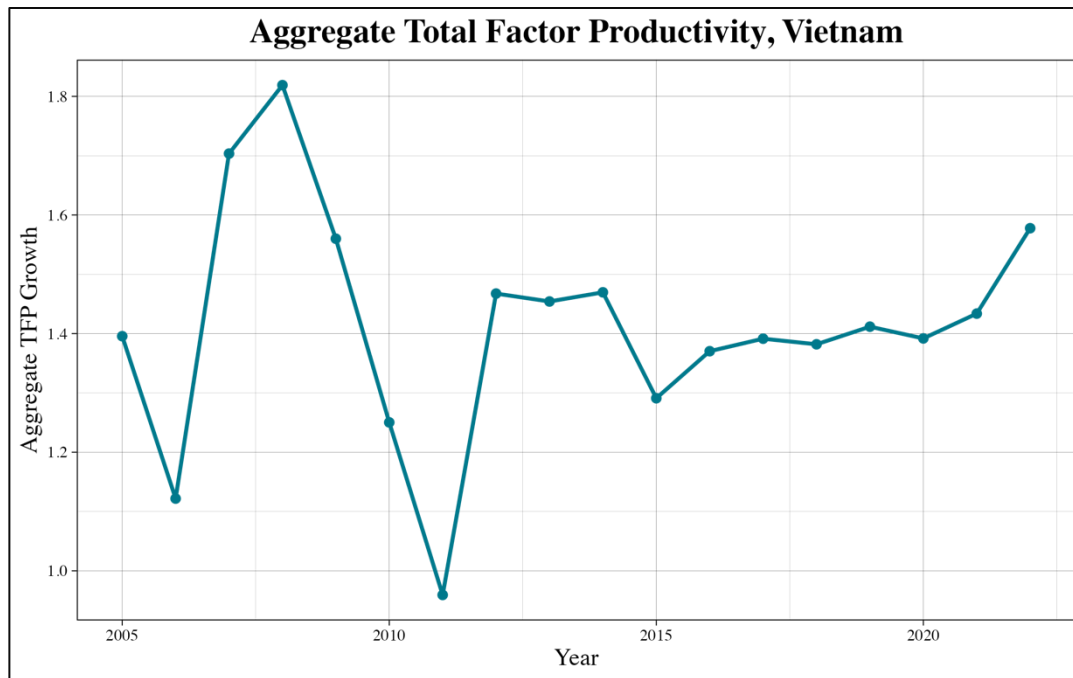


Figure 1.1j: Aggregate Total Factor Productivity, Vietnam

Source: Own Computation Using GSO Data

1.2 Economic Complexity and Product Space

Vietnam’s economic transformation over the past two decades is closely tied to its growing position in the global product space and steady improvements in economic complexity. The country has successfully diversified its exports beyond traditional agriculture into key sectors such as electronics, machinery, and chemicals. Vietnam’s Economic Complexity Index (ECI) stands at 0.19, reflecting moderate capability in producing and exporting complex products. This diversification has allowed the country to engage in global value chains, particularly as firms shift production under the “China Plus One” strategy (Reed, 2020). Yet, despite this progress, Vietnam’s export basket still shows a heavy reliance on low-Product Complexity Index (PCI) goods like textiles, cashew nuts, and coconuts, which may constrain its ascent into high productivity sectors. While peer countries such as Thailand and the Philippines have gradually reduced their textile dependence, Vietnam’s textile exports have remained sticky at 20–28% (Figure 1.2a)

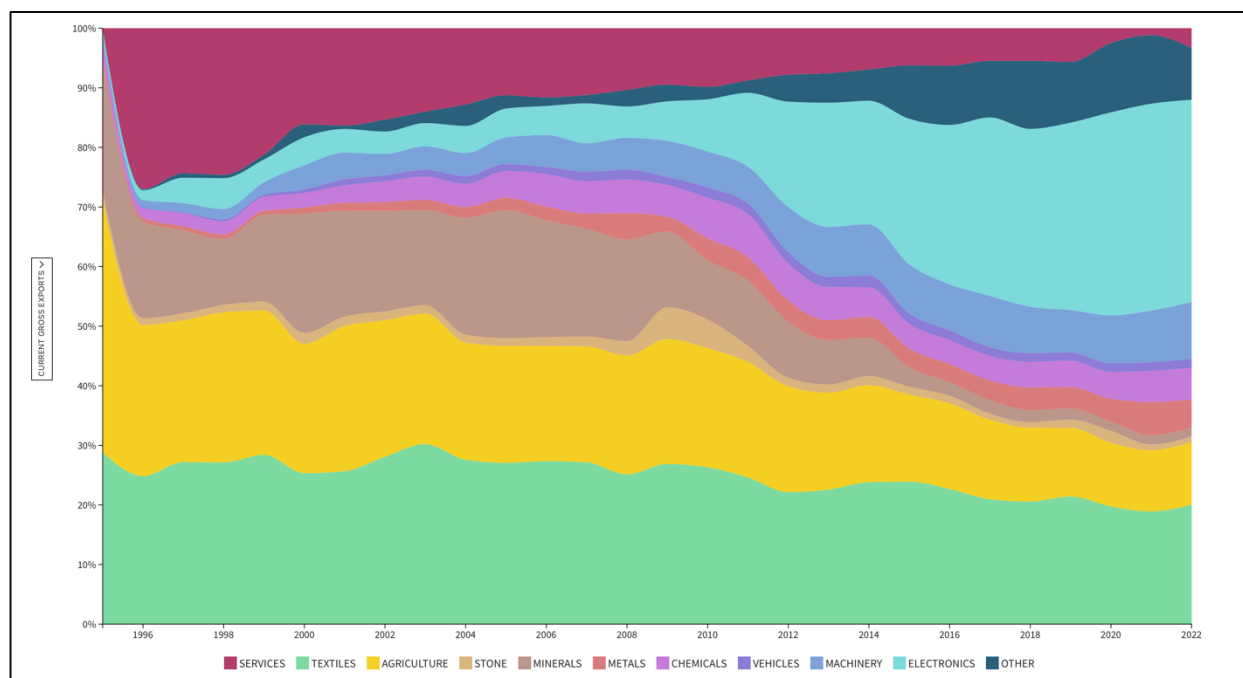


Figure 1.2a: Export Share by Sector, Vietnam

Source: Atlas of Economic Complexity

Electronics have become the standout sector in Vietnam's export portfolio. Transmission apparatus, electronic integrated circuits, and photographic cameras are among the country's top exports, with strong demand from global markets. New product introductions have also predominantly emerged in the electronics category, with far fewer developments in machinery or chemicals—sectors that carry higher PCI values and greater potential for long-term productivity gains. This imbalance underscores the challenge of translating export growth into sustained complexity upgrades.

Nevertheless, Vietnam is well-positioned within the product space to transition into more complex industries, thanks to its strong base of capabilities and wide product diversification. The Complexity Outlook Index of 0.75 indicates favourable conditions as Vietnam's current exports are close to high-PCI products, allowing for jumps into new product lines that the country can build expertise around and potentially export (Figure 1.2b). Products such as electrical capacitors and pumps, compressors, and fans are especially promising. For instance, electrical capacitors represent a \$32 billion market closely related to Vietnam's existing exports, while the latter category—already generating \$700 million in exports for Vietnam—is linked to an \$82 billion

global market. These products not only offer high complexity but are also connected to fast-growing, knowledge-intensive value chains.

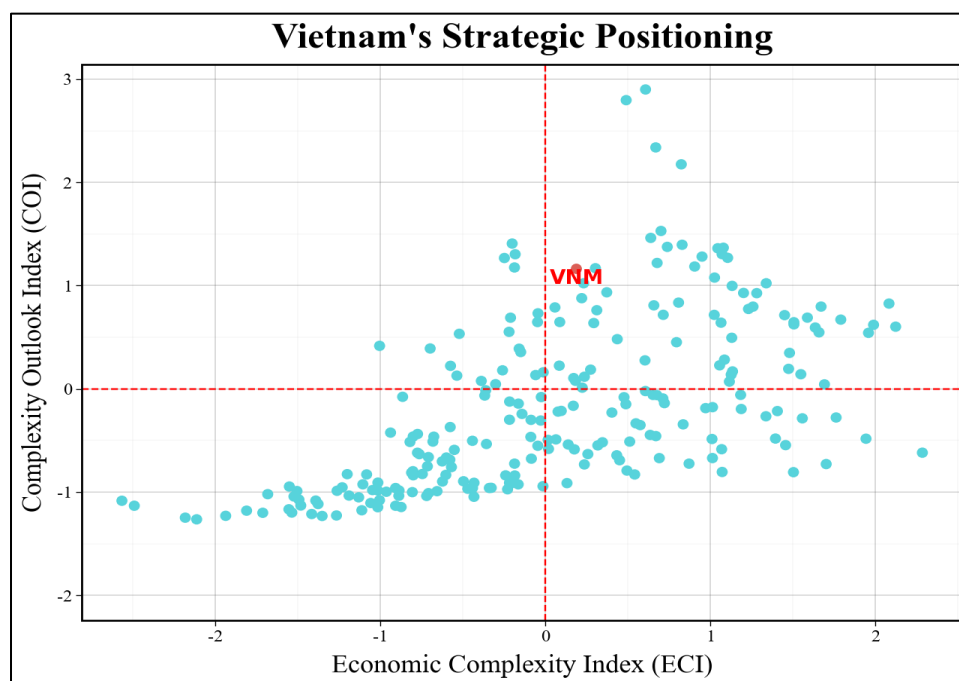


Figure 1.2b: Vietnam's Strategic Positioning

Source: Atlas of Economic Complexity

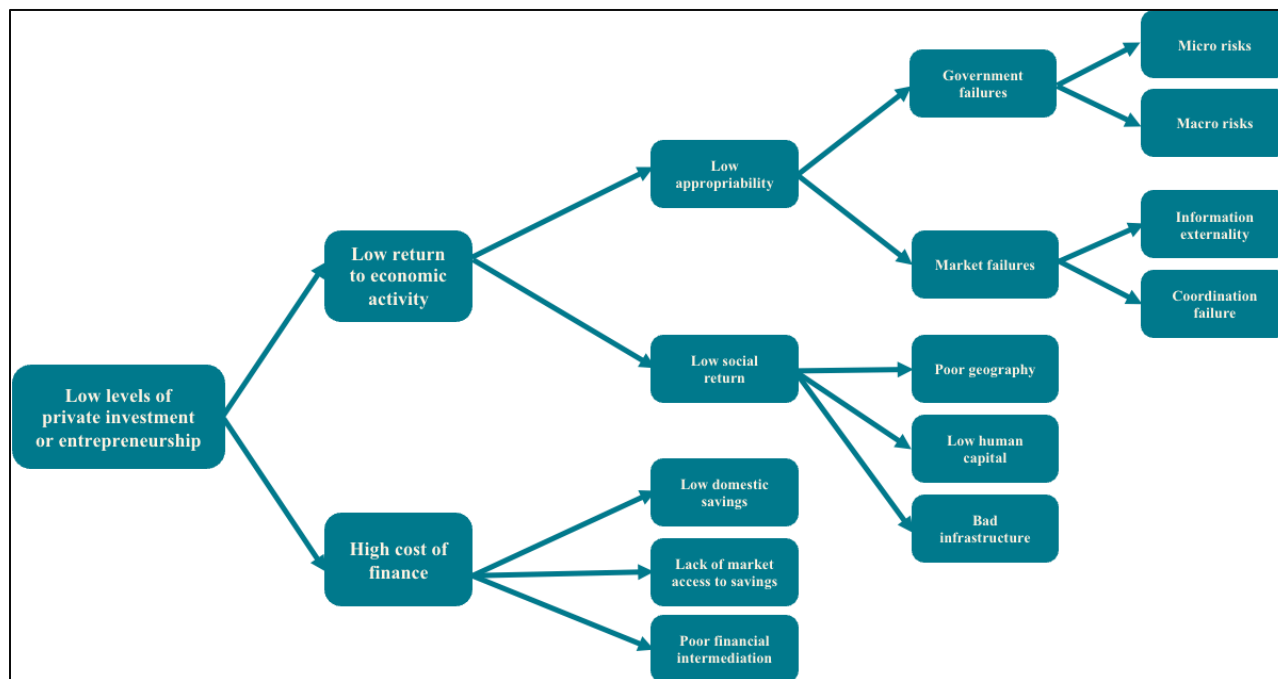
In summary, Vietnam's economic complexity landscape reveals a country with substantial diversification and promising pathways for growth.

1.3 Growth Question

Vietnam's growth has often been described as miraculous—an economy that has rapidly ascended to become the latest Asian tiger. Yet, beneath this impressive trajectory lie emerging challenges that could constrain future growth. An aging population, a shrinking labor force, and declining productivity in both manufacturing and services raise concerns about the sustainability of Vietnam's development model. To maintain or improve current levels of GDP per capita, the economy will need to become significantly more productive. So far, Total Factor Productivity (TFP) has played a central role in supporting high growth, but sustaining this momentum will require moving into more complex and high-value products. The core issue now is not just how fast Vietnam grows, but how sustainably it grows. This brings us to the most pressing question for Vietnam's economic future: **“How can Vietnam sustain its current growth rate in the long run?”**

2. Growth Diagnostics

In this section, we follow the growth tree to identify the binding constraints to Vietnam's long-term growth.



To identify pressing areas where analysis must be conducted, we referred to the World Bank Enterprise Survey (WBES) and focused upon aspects that a higher proportion of firms consider to be their main obstacle.

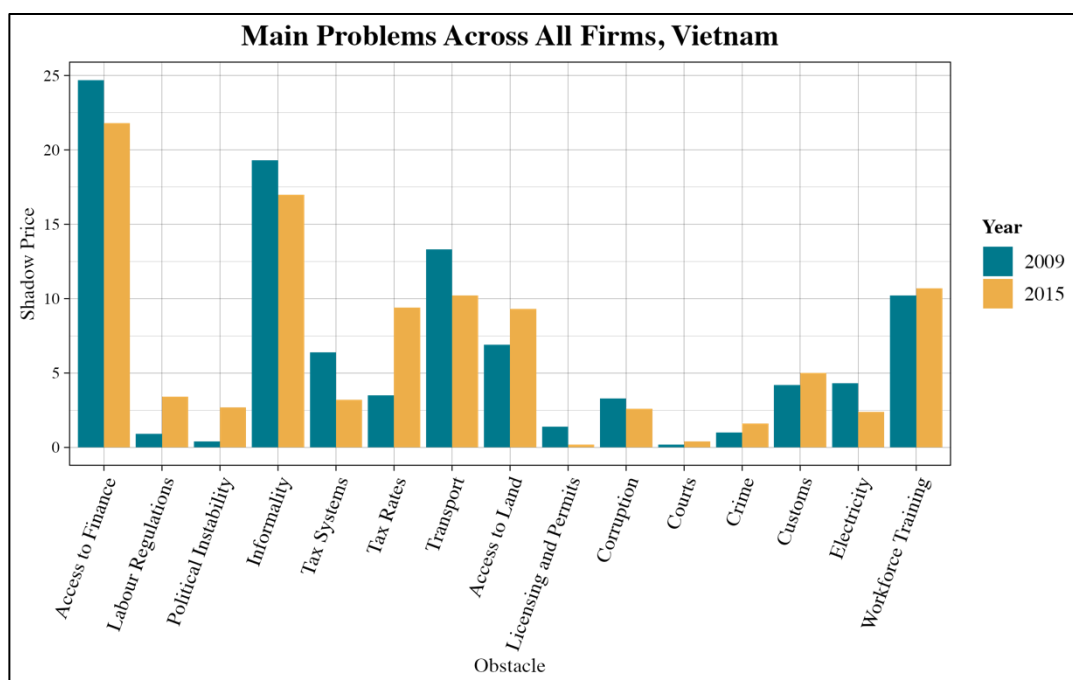


Figure 2a: Main Problems Across All Firms, Vietnam

Source: World Bank Enterprise Survey

2.1 Transport Infrastructure

General Aspects of Transport Infrastructure

Vietnam is an export intensive nation. As of 2021, the country exports 87% of its GDP, with its export basket primarily comprising electronics and textiles. Given its export status, it is imperative for Vietnam to possess an efficient transport infrastructure to transport goods from manufacturing plants to export destinations. To facilitate exports, Vietnam is heavily reliant its roads and ports. According to time-series data from the Asian Transport Outlook, Vietnam transported only 216 thousand tonnes of freight via flights and 560 thousand tonnes via railways, juxtaposed to 1,303,328 thousand tonnes of freight via roadways and 242,366 thousand tonnes via shipping or waterways (Figure 2.1a).

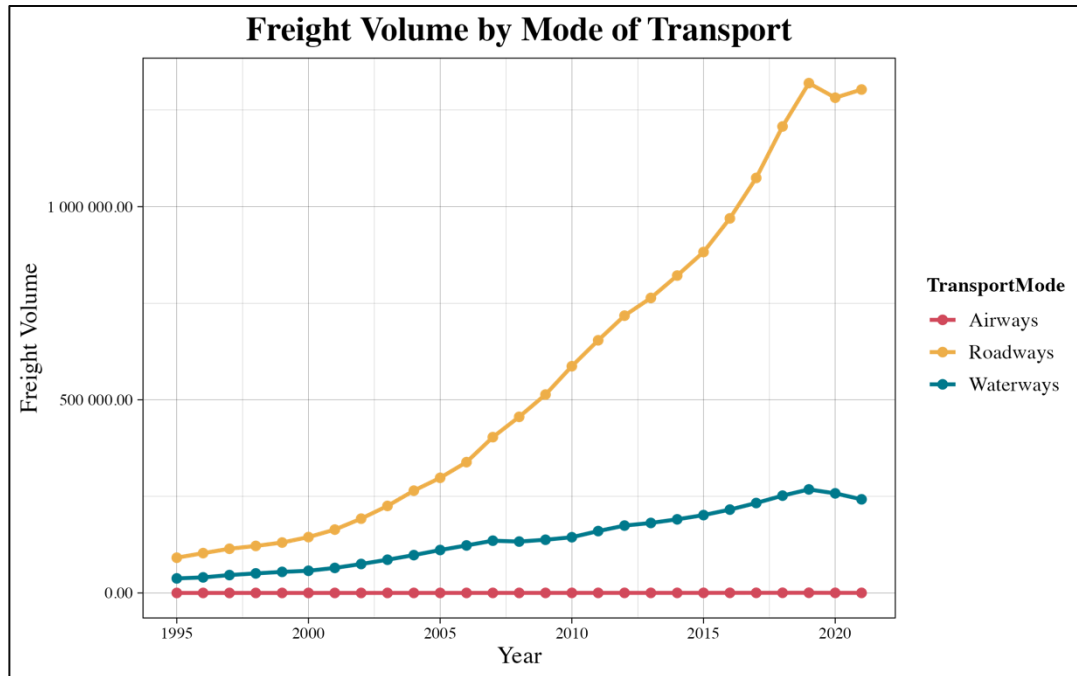


Figure 2.1a: Freight Volume by Mode of Transport, Vietnam

Source: Asian Transport Outlook

Despite its dependence on exports and dire need for efficient roads and ports, the country performs worse than all its peers, including Cambodia, on World Bank's logistics performance index. With an overall score of 2.8, Vietnam clearly struggles to match the logistical capabilities expected of an export-intensive economy. We took Vietnam's poor performance on the LPI as the first indication of problems in its transport infrastructure. This inclination was backed by Vietnam's high logistics costs, when taken as a percentage of GDP. The authoritarian nation's logistics costs have remained consistently higher than all its peers (UNCTAD, 2021). Infact, the costs, which were declining between 2016-2020, showcase a sharp upward trend post-COVID with costs being a staggering 19.1% of GDP in 2021 (Figure 2.1b).

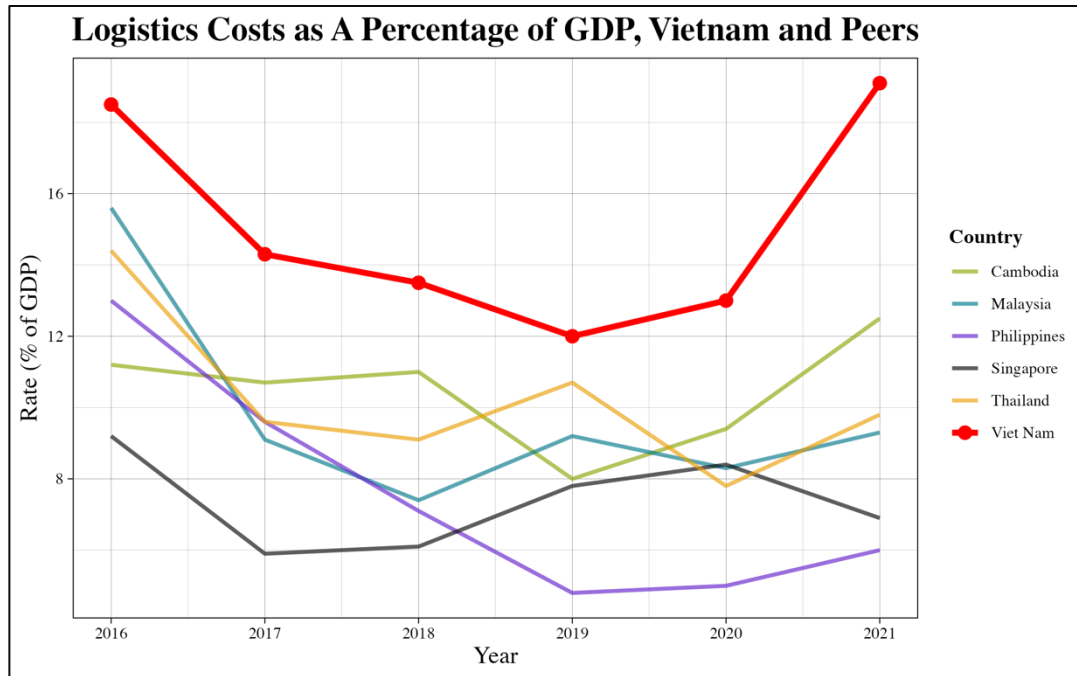


Figure 2.1b: Logistics Costs as A Percentage of GDP, Vietnam and Peers

Source: UNCTAD

The World Bank Enterprise Survey shows that over 8% of manufacturing firms consider transport infrastructure to be the primary problem (Figure 2.1c). These values shoot up to approximately 15% when all firms are taken into consideration (Figure 2a) and approximately 18% when only small firms are considered (Figure 2.1d). Only large foreign firms that make enough profit in international markets to use expensive and fast modes to transport their goods do not complain about the country's transport situation (Figure 2.1e). These are also the firms located either in Hanoi or in Ho Chi Minh – the major cities closest to major ports. These statistics indicate that transport of goods in Vietnam is a problem, especially for domestic, manufacturing and small firms. All these initial indications provide a strong base for delving deeper into transport infrastructure as a binding constraint.

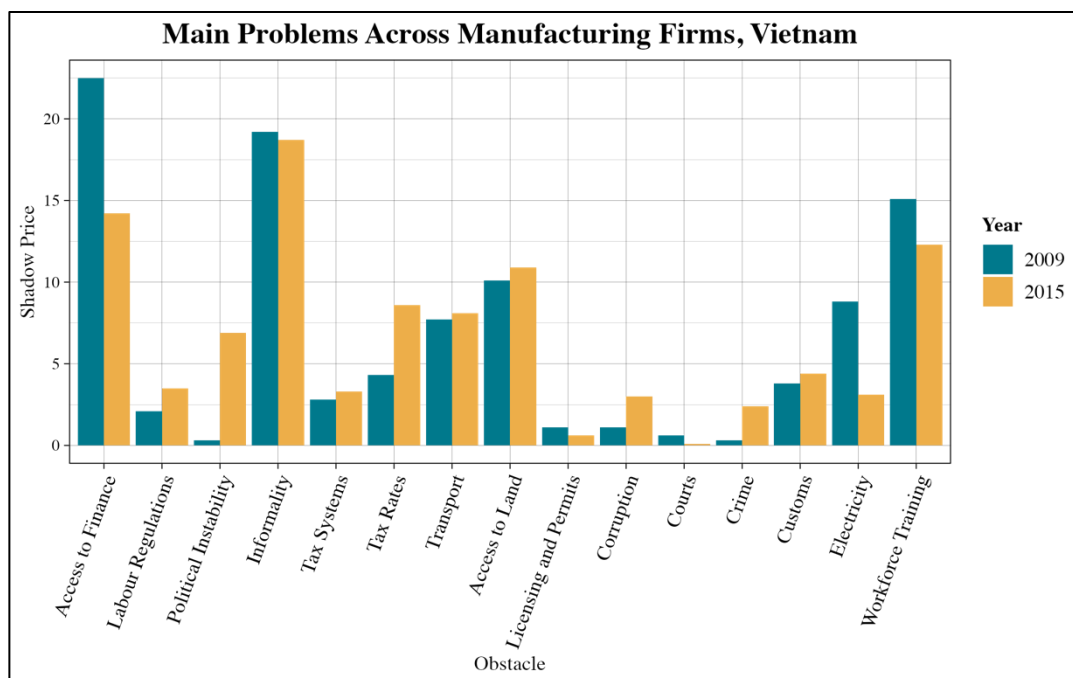


Figure 2.1c: Main Problems Across Manufacturing Firms, Vietnam

Source: World Bank Enterprise Survey

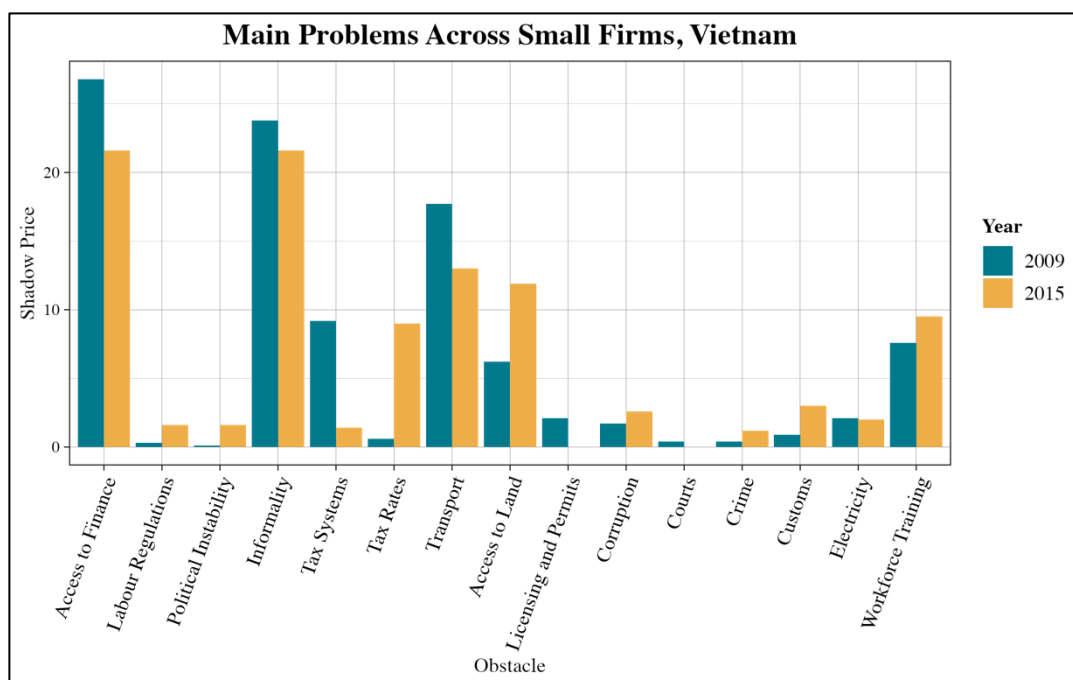


Figure 2.1d: Main Problems Across Small Firms, Vietnam

Source: World Bank Enterprise Survey

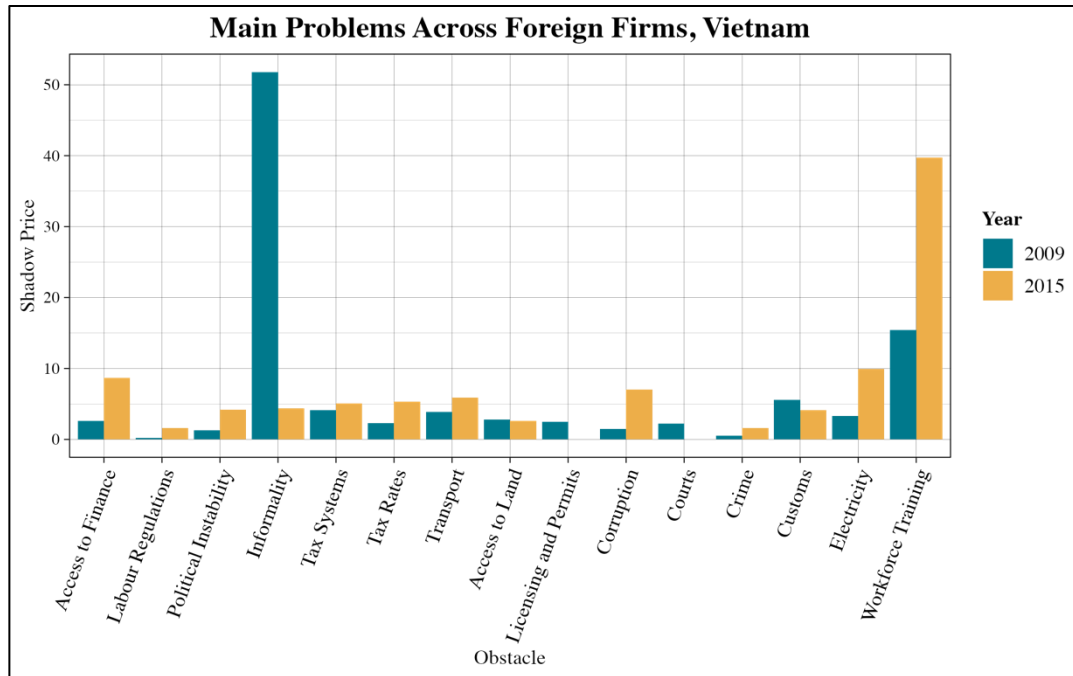


Figure 2.1e: Main Problems Across Foreign Firms, Vietnam

Source: World Bank Enterprise Survey

We follow a two-pronged approach, diagnosing roadways and waterways as two separate constraints.

2.1.1 Roadways

Vietnam currently has an estimated 570,448 kilometers of roads, of which approximately 24,000 kilometers are national highways, and 816 kilometers are expressways (Open Development Vietnam, 2022). Given this extensive spread, roadways remain the backbone of Vietnam's freight network, used to transport about 84.27% of the total freight. Yet, its overall road quality rating at 3.2 out of 7 sits well below the trend implied by both volume of goods exported by Vietnam (log exports) (Figure 2.1.1a) and income levels of Vietnam (log GDP) (Figure 2.1.1b). This underperformance is striking, given that motorcycles, a mode of transport that imposes low loads and relatively less pavement stress, dominate the urban and rural travel in the country. Specifically, 96% of all passenger vehicles in Vietnam are motorcycles – a proportion much higher than Thailand, Singapore, Malaysia and Philippines (Figure 2.1.1c). The data for the remaining peers (Cambodia and Bangladesh) was not available in this case.

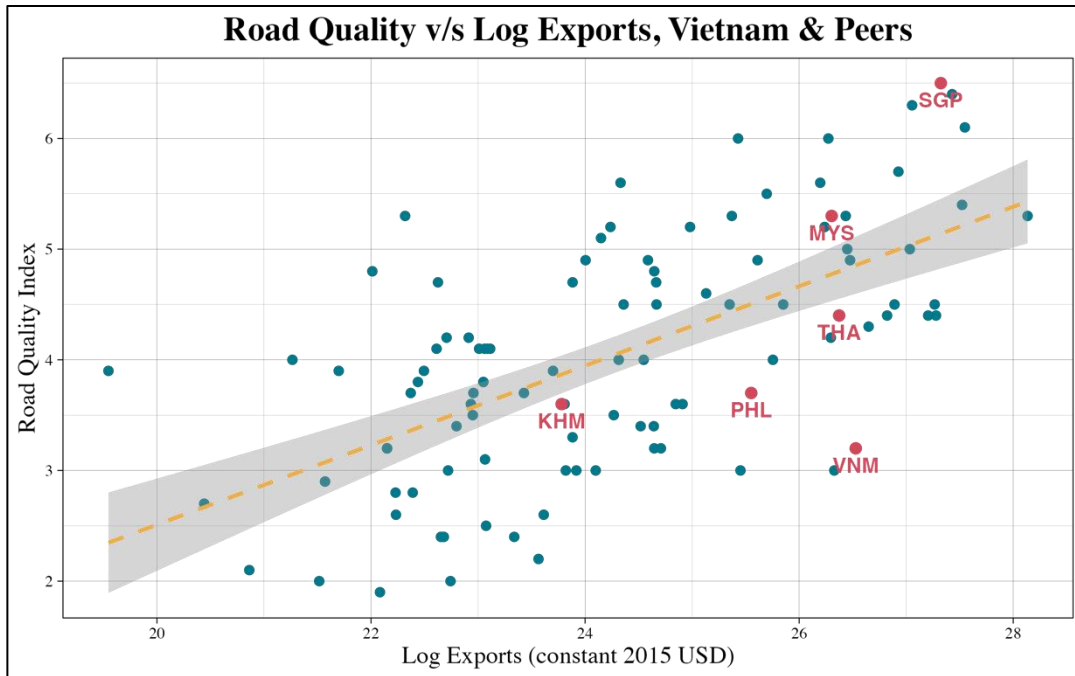


Figure 2.1.1a: Road Quality v/s Log Exports, Vietnam and Peers

Source: World Development Indicators and World Bank

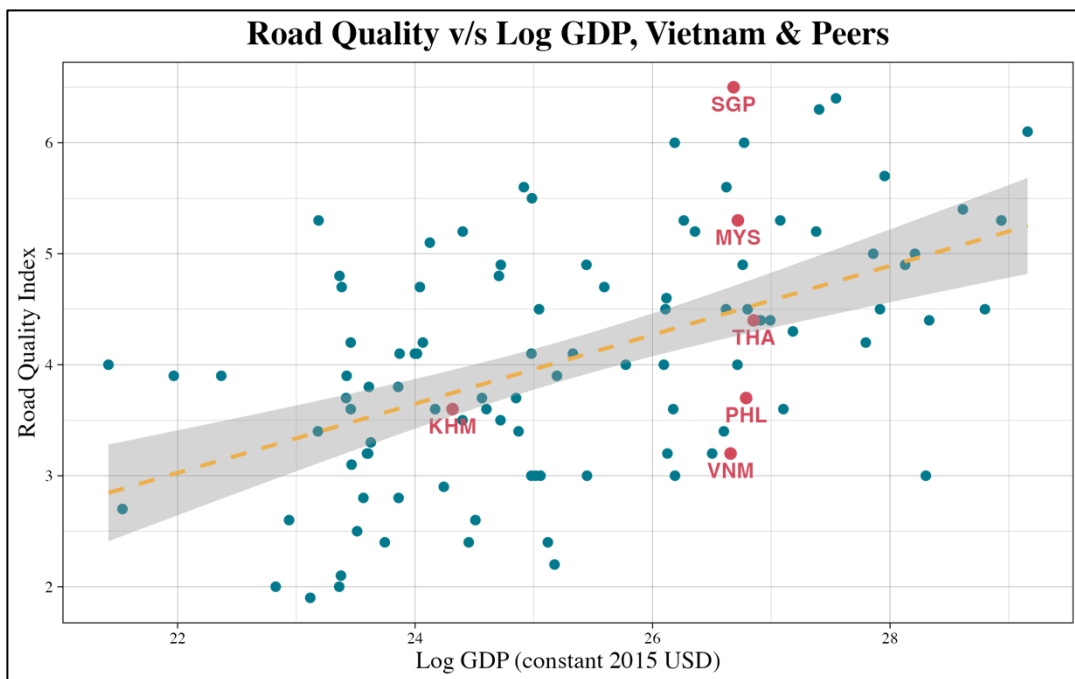


Figure 2.1.1b: Road Quality v's Log GDP, Vietnam and Peers

Source: World Development Indicators and World Bank

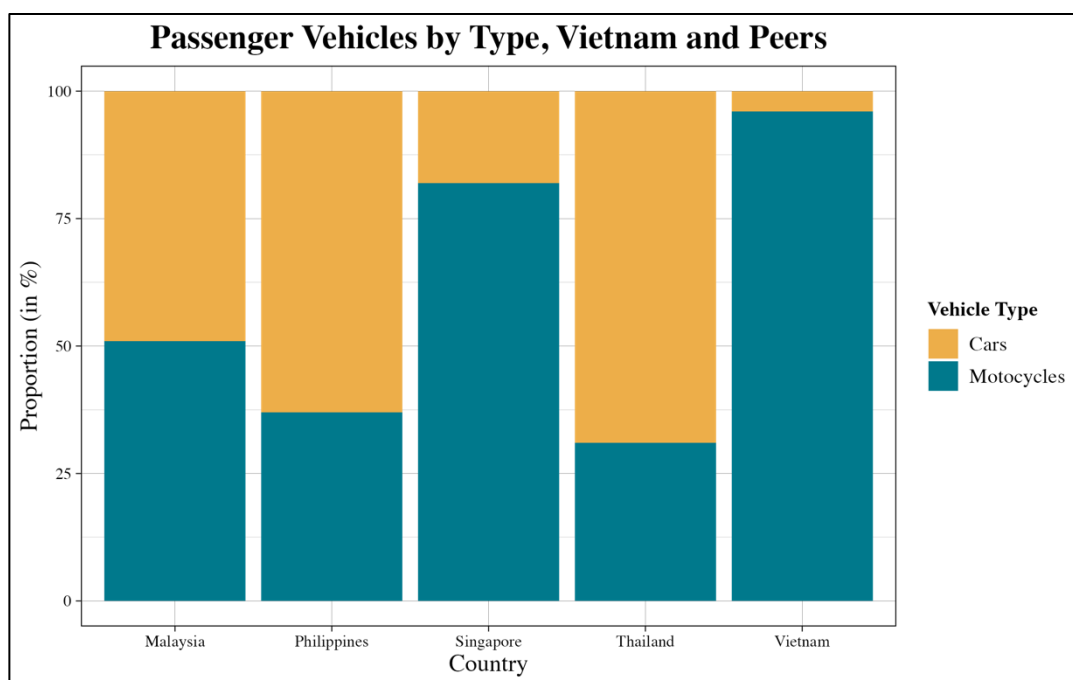


Figure 2.1.1c: Passenger Vehicles by Type, Vietnam and Peers

Source: Bain and Company, Standard Chartered Bank

This, however, is changing now. Vietnam's sustained growth of over 6% per annum is fueling a rapid surge in car ownership, with car sales showing a year-on-year growth rate of 25% over the past decade (Figure 2.1.1d). As the car-fleet share rises, heavier axle loads and worsening congestion will accelerate pavement deterioration, amplify maintenance backlogs, and heighten logistic bottlenecks. Thereby, Vietnam's road network risks becoming the very constraint that chokes its next stage of export-led expansion.

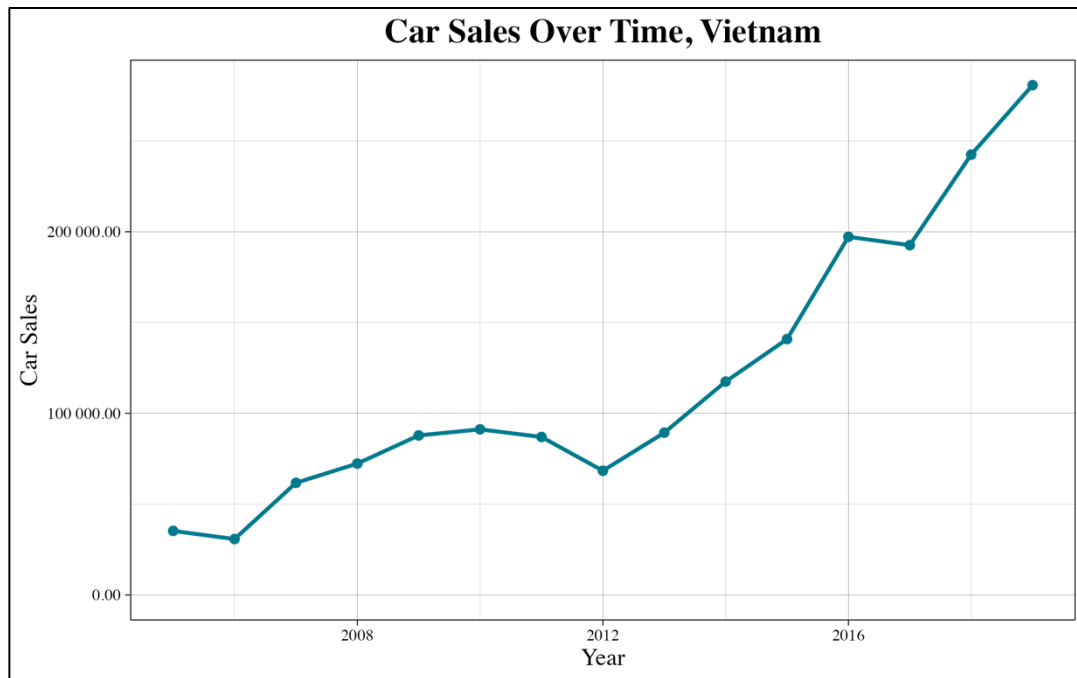


Figure 2.1.1d: Car Sales Over Time, Vietnam

Source: Asian Transport Outlook

Shadow Prices

To assess the shadow price of road transport in Vietnam, we measured the average travel time for a 10-kilometer commute in its two largest cities. The journey takes an average 30.47 minutes in Ho Chi Minh and 30.23 minutes in Hanoi. These numbers are only next to Manila, the capital city of Philippines infamous for its traffic, where it takes just over 32 minutes to travel the distance. Evening rush-hour congestion further inflates these times to roughly 40 minutes. On the flip side, Vietnam's peers enjoy a much faster commute. In both Singapore (Vietnam's aspirational peer) and Thailand (one of Vietnam's most developed peers), drivers can cover 10 kilometers in a little over 12 minutes (Figure 2.1.1e)

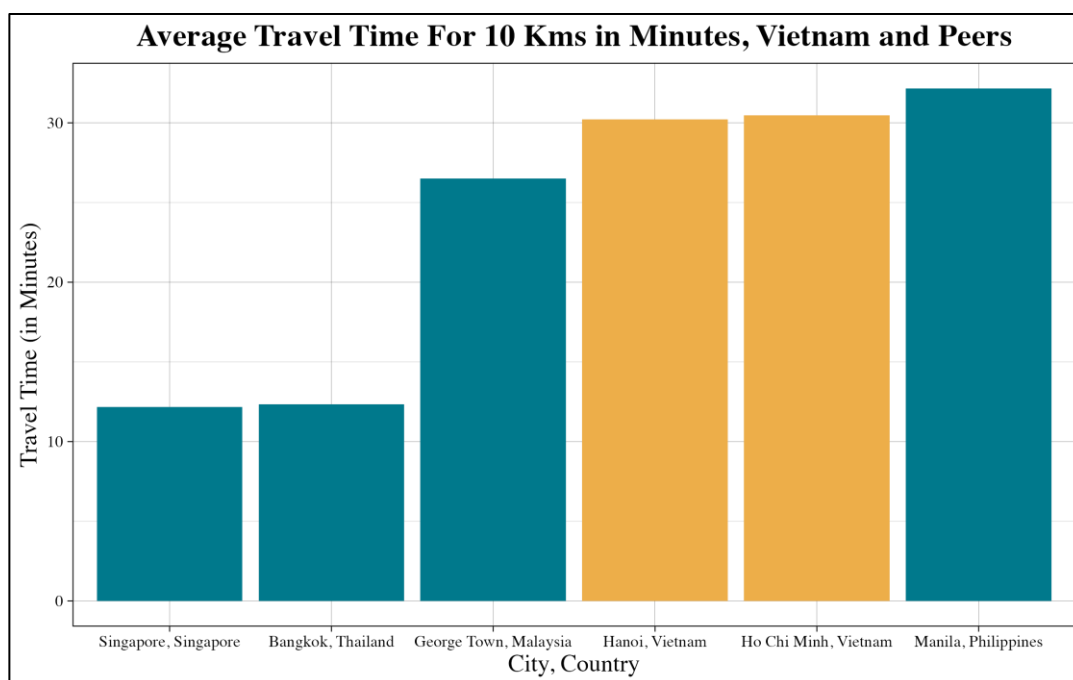


Figure 2.1.1e: Average Travel Time for 10 Kms, Vietnam and Peers

Source: Tomtom Index, 2024

Data from Tomtom’s 2024 traffic index further underscores these delays. For instance, the congestion levels in Ho Chi Minh touched an average of 116% on weekends, with an average traffic jam count of 105 per day. The condition of Hanoi is also only slightly better, with average congestion levels of 81% in 2024 and a mean of 43 traffic jams per day. These elevated travel-time and reliability costs attest that Vietnam’s road network carries a high implicit price—imposing significant time penalties on households and freight operators alike.

Camels and Hippos

To identify whether road-transport dependent industries struggle to thrive in Vietnam, we compared their growth rates to those of road-transport independent industries. To pick the dependent industries, we looked at Vietnam’s Input-Output tables for 2022 and computed the percentage of total road transport inputs consumed by each industry. Post the computation, we classified every sector which consumes more than 5% of road transport inputs as ‘road-transport dependent’ (Figure 2.1.1f). By this methodology, 4 industries – Food, beverages and tobacco; Agriculture, hunting, forestry and fishing; Wholesale trade and commission trade (except of motor vehicles and motorcycles) and Construction – were classified as road-transport dependent. These industries have showcased an average annual growth rate of 2.9% per annum, juxtaposed to road-

transport independent industries, that have grown at an annual rate of 6.98% per annum (Figure 2.1.1g). This confirms the existence of camels and hippos for road infrastructure in Vietnam.

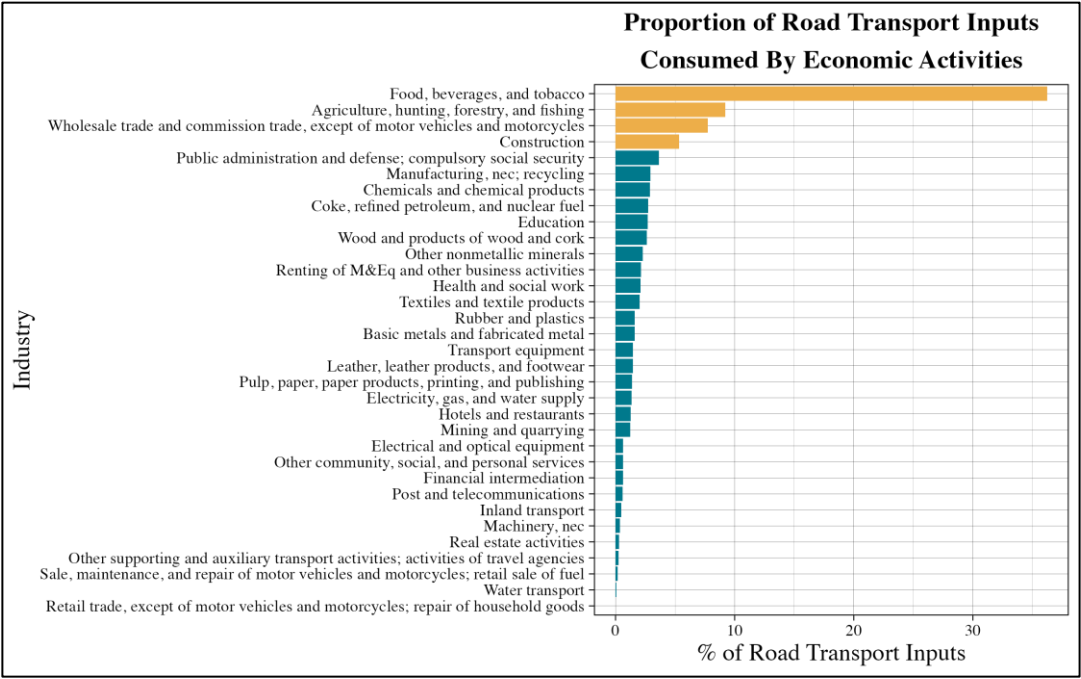


Figure 2.1.1f: Road-Transport Dependent Sectors

Source: Input-Output Tables, 2022

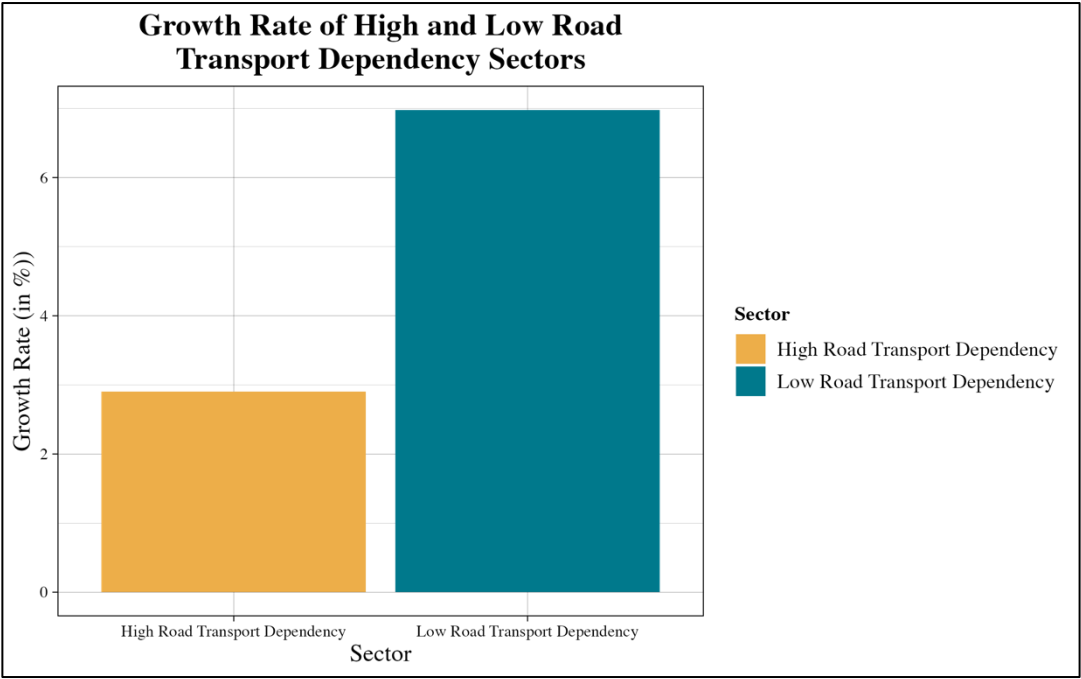


Figure 2.1.1g: Growth Rate of Road Transport Dependent and Road Transport Independent Sectors

Source: General Statistics Office, Vietnam

Movements-in-Movements

Figure 2.1.1h showcases the correlation between Vietnam's road quality and its GDP growth over the years. Between 2006 and 2012, gradual improvements in road conditions aligned with steady economic growth. A significant boost came between 2013 and 2015, coinciding with major infrastructure projects like the improvement of NH1A and the Northern Delta Transport Development Project (World Bank, 2024), leading to accelerated GDP growth. Again, between 2018-2019, road quality improved, especially due to Phase 1 construction of the North-South Eastern Expressway, leading to a jump in Vietnam's growth rate.

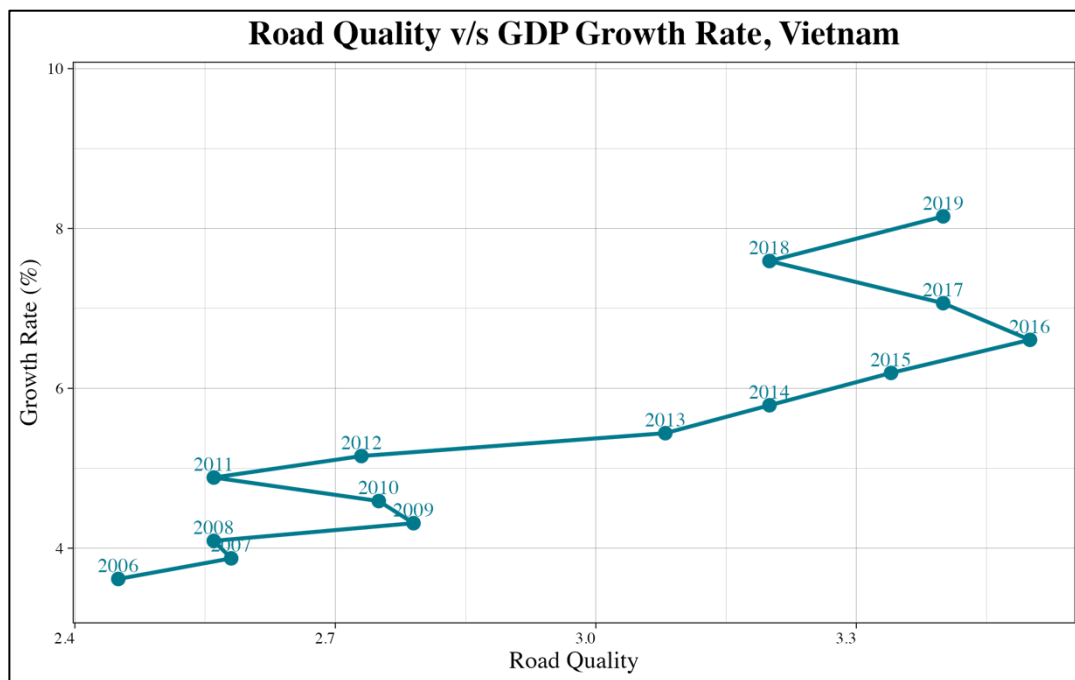


Figure 2.1.1h: Movements in Movements (GDP), Vietnam

Source: Asian Transport Outlook and World Bank

A companion plot of road quality against freight volumes shows the same pattern: better roads unleash heavier cargo flows (Figure 2.1.1i). This pattern underscores that road infrastructure is not just correlated with growth and adds gravity to the claim of roadways being a binding constraint.

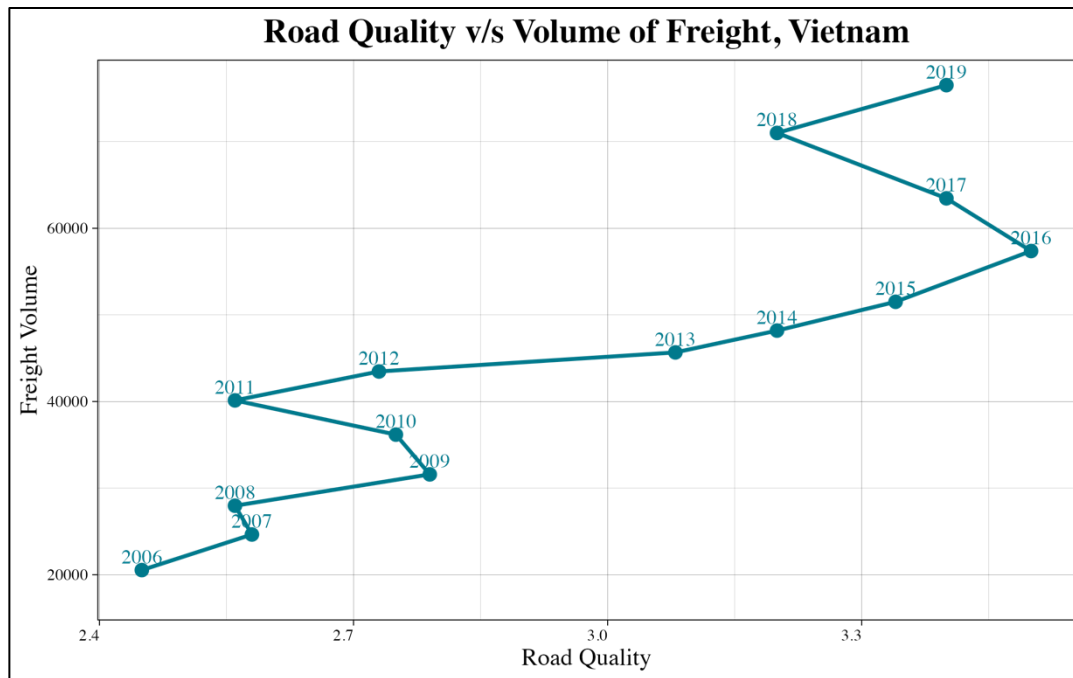


Figure 2.1.Ii: Movements in Movements (Freight Volume), Vietnam

Source: Asian Transport Outlook and World Bank

Agents Attempting to Overcome the Constraint

Vietnamese authorities and private firms have begun to tackle road-infrastructure bottlenecks through both massive investment and modal diversification. In 2024, the prime minister of Vietnam announced \$27 billion in public investments, 65% of which was exclusively for road infrastructure (Figure 2.1.1j). Additionally, in 2023 alone, \$4 billion were invested in the construction of the Hanoi Ring Road 4 and the Ho Chi Minh City Ring Road 3, signifying a targeted focus on enhancing road infrastructure and connectivity within Vietnam's major metropolitan areas (BW Industrial, 2024).

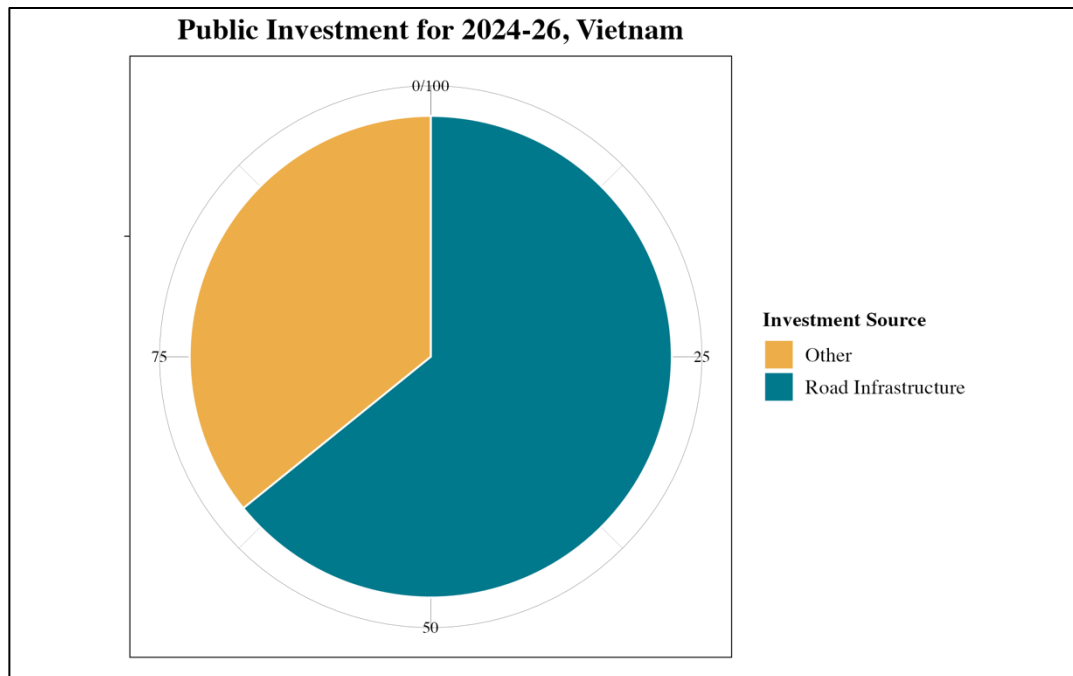


Figure 2.1.1j: Investment in Road Infrastructure

Source: Hanoi Times, 2024

Firms are also moving to alternative means to transport freight within the country. For instance, inland waterways showcased a 47% year-on-year increase in cargo throughout between 2011 and 2017 (Vietnam Briefing, 2022). This growth, however, proved to be unsustainable, with road transport continuing to dominate, accounting for about 84% of goods transported (Figure 2.1a). Railways for freight transportation remain an untapped aspect due to the inadequate interior of trains. Regardless, the road investments and exploration of freight transport using inland waterways provide sufficient evidence that agents are actively seeking ways to navigate through Vietnam's poor road infrastructure.

Conclusion

Based on the evidence, Vietnam's road transport infrastructure clearly constitutes a binding constraint on its economic development. Although roads account for over 84% of freight movement and span more than 570,000 km, their poor quality, chronic congestion, and accelerating pavement deterioration impose steep time and reliability costs on households and businesses. This is reflected in the underperformance of road-dependent industries, which grow at just 2.9% annually versus 6.98% for other sectors. In the absence of substantial improvements in pavement quality, congestion control, and multimodal integration, Vietnam's roadways will hinder

the next phase of its export-led growth. Addressing these infrastructure deficits is therefore vital to ensure sustainable long-term economic development.

2.1.2 Waterways

As of 2024, Vietnam has a network 320 ports, 50% of which are international shipping points. The nation's port system has experienced significant growth, with total cargo volume reaching approximately 864.4 million tons in 2024, marking a 14% increase from the previous year (Xinhua, 2024). Despite this extensive infrastructure, most of these ports are limited to handling coastal vessels, with only a select few equipped to manage large-scale international trade and trans-shipment activities (Parker, 2024). Additionally, his rapid expansion has led to capacity challenges. Major ports like Cat Lai in Ho Chi Minh City and Hai Phong are operating at over 90% capacity, resulting in congestion and increased logistics costs (MN Shipping, 2023).

Vietnam ranks 85 out of 139 countries on The World Bank's port quality index with a score of 3.45 out of 7. While this is better than its world standing on the road quality index, it lags on the trend implied by both volume of goods exported by Vietnam (log exports) (Figure 2.1.2a) and income levels of Vietnam (log GDP) (Figure 2.1.2b). Infact, the country performs even worse than Cambodia – a country whose GDP is 10% of Vietnam's and exports goods that worth less than 20% to that of Vietnam (WDI, 2024). Moreover, our country of focus trails too far from Singapore, its aspirational peer, which ranks 1st in both road and port quality indices.

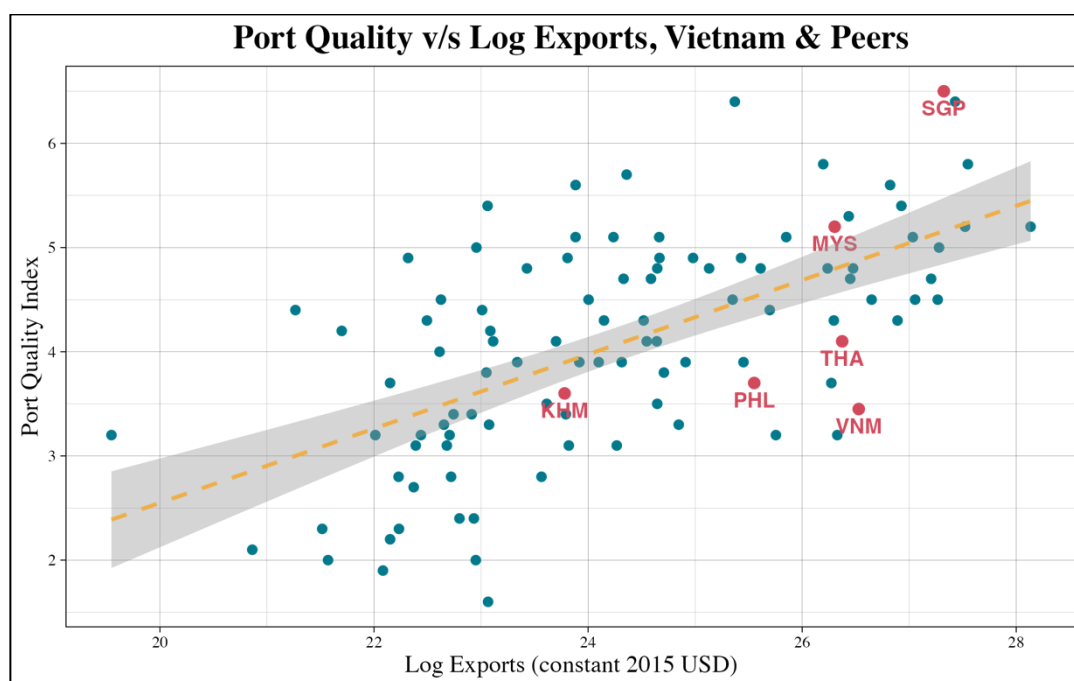


Figure 2.1.2a: Port Quality v/s Log Exports, Vietnam and Peers

Source: World Development Indicators and World Bank

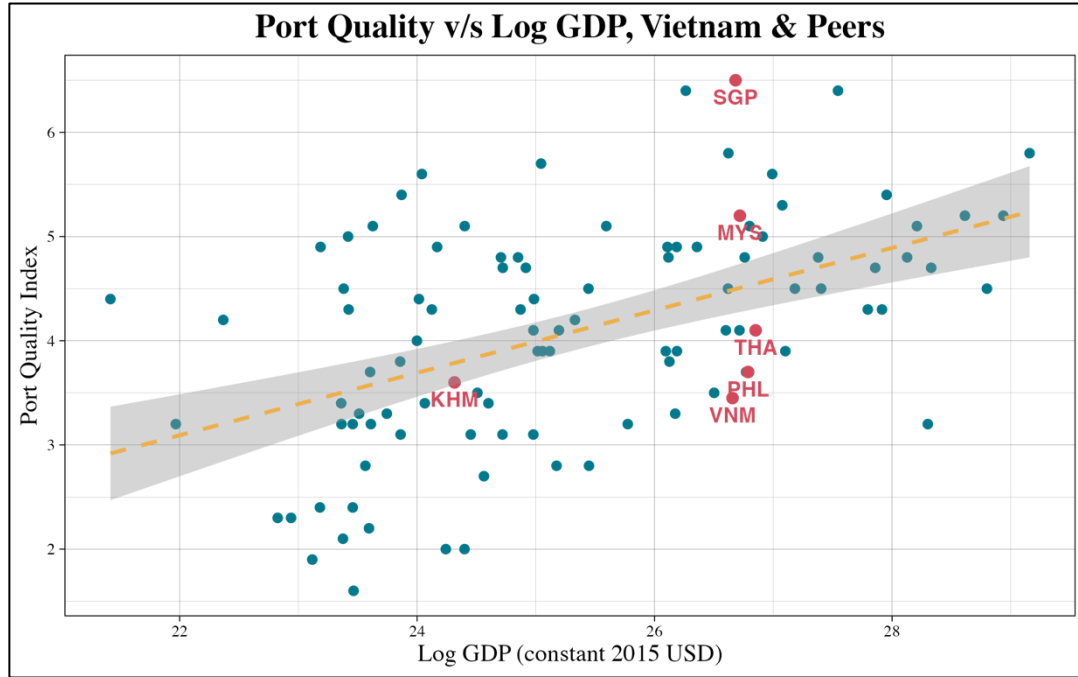


Figure 2.1.2b: Port Quality v/s Log GDP, Vietnam and Peers

Source: World Development Indicators and World Bank

Shadow Prices

To check whether ports demand a high shadow price in Vietnam, we looked at the country-wise export delays by waterways. Export shipments experience average delays of 13 days in Vietnam. This number is much higher than all its peers. Thailand, the peer country with longest delays also faces only a 5.8-day delay on an average. (Figure 2.1.2c)

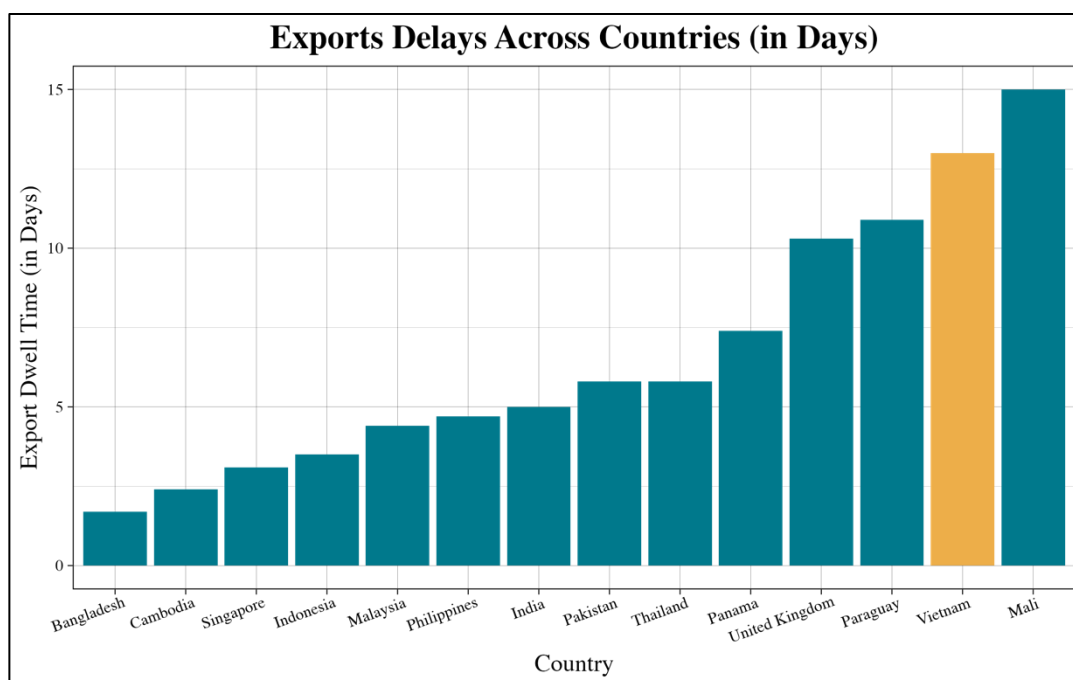


Figure 2.1.2c: Export Delays Across Countries

Source: Supply Chain Indicators, World Bank

According to Saigon Phong News (2024), these delays have significantly hampered Vietnam’s \$1 trillion dollar trade goal, as outdated port warehouses and container shortages create bottlenecks. Tran Anh Duc of the Investment and Trade Task Force (Vietnam Business Forum Alliance) notes that infrastructural inadequacies and steep logistics costs, around 25% of GDP, far exceed global norms. Moreover, procedural inefficiencies - including manual paperwork and outdated cargo-handling equipment - prolong vessel turnaround times. Prolonged shipment durations disrupt supply chains, ramp up inventory carrying expenses, and undermine export reliability. In this light, the shadow price of maritime transport in Vietnam is high.

Camels and Hippos

To identify whether port dependent industries struggle to thrive in Vietnam, we compared their growth rates to those of port independent industries. To pick the dependent industries, we looked at Vietnam’s Input-Output tables for 2022 and computed the percentage of total water transport inputs consumed by each industry. Post the computation, we classified every sector which consumes more than 5% of road transport inputs as ‘port dependent’ (Figure 2.1.2d). By this methodology, 4 industries – Food, beverages and tobacco; Agriculture, hunting, forestry and fishing; Chemicals and chemical products and Construction – were classified as port dependent.

These industries have showcased an average annual growth rate of 2.64% per annum, juxtaposed to road-transport independent industries, that have grown at an annual rate of 6.81% per annum (Figure 2.1.2e). This confirms the existence of camels and hippos for port infrastructure in Vietnam.

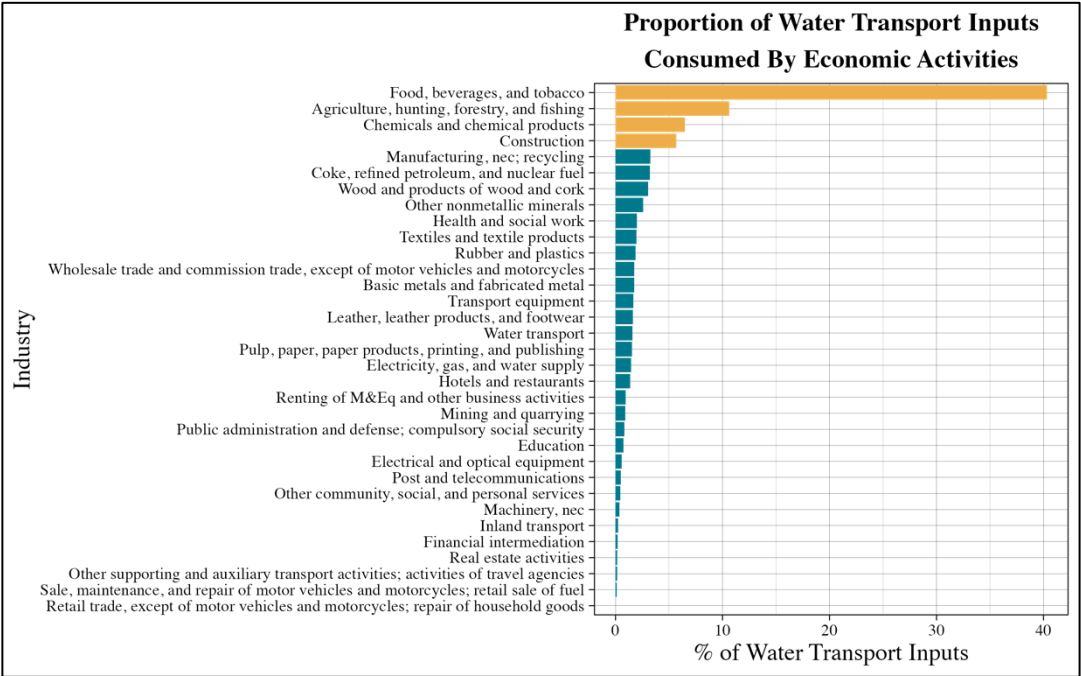


Figure 2.1.2d: Port Dependent Sectors, Vietnam

Source: Input-Output Tables, 2022

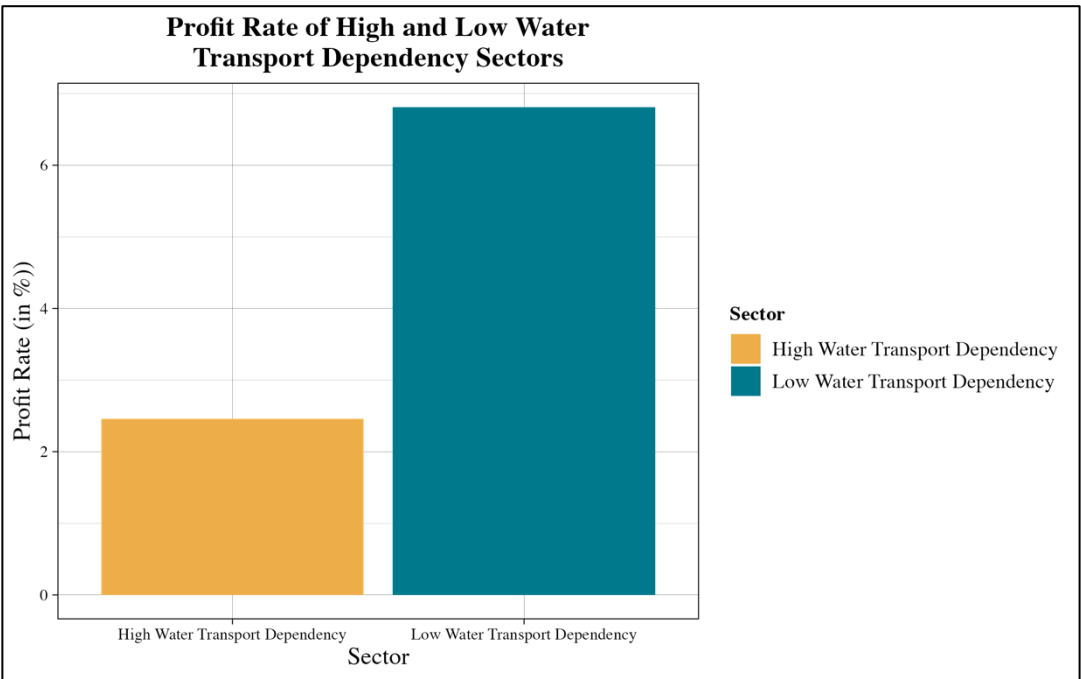


Figure 2.1.2e: Growth Rate of Port Dependent and Port Independent Sectors

Source: General Statistics Office, Vietnam

Movements-in-Movements

Figure 2.1.2f showcases the correlation between Vietnam's port quality and its GDP growth over the years. Between 2006 and 2008, slow improvements in port conditions aligned with slow but steady economic growth. In the next two years, the establishment of the Cai Mep–Thi Vai Deep-Water Port, Van Phong Port and the Hon La Port led to a rapid increase in port quality with an annual 4.5% economic growth. The 2011 push back in port quality came due to suspension of the construction of the Van Phong Port. Since then, port quality in Vietnam has improved by 0.6 points on the index and remains highly correlated with its GDP growth rate.

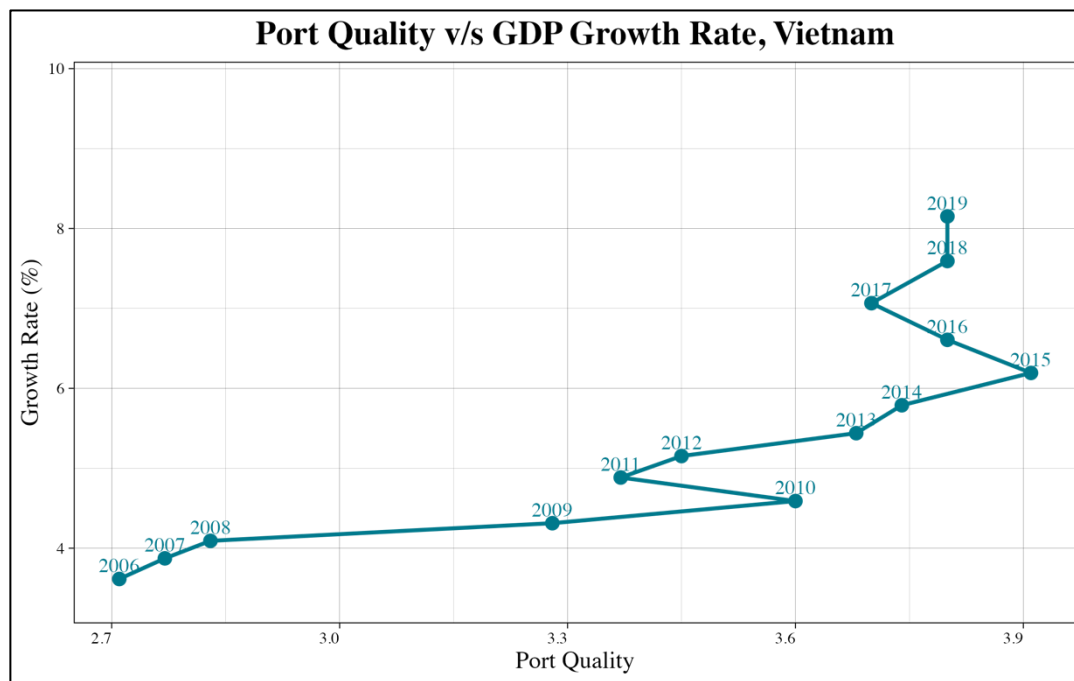


Figure 2.1.2f: Movements in Movements (GDP), Vietnam

Source: World Development Indicators and World Bank

A corresponding plot of port quality against freight volumes shows the same pattern: better ports unleash heavier cargo flows (Figure 2.1.2g). This pattern strengthens the claim that port infrastructure is a binding constraint.

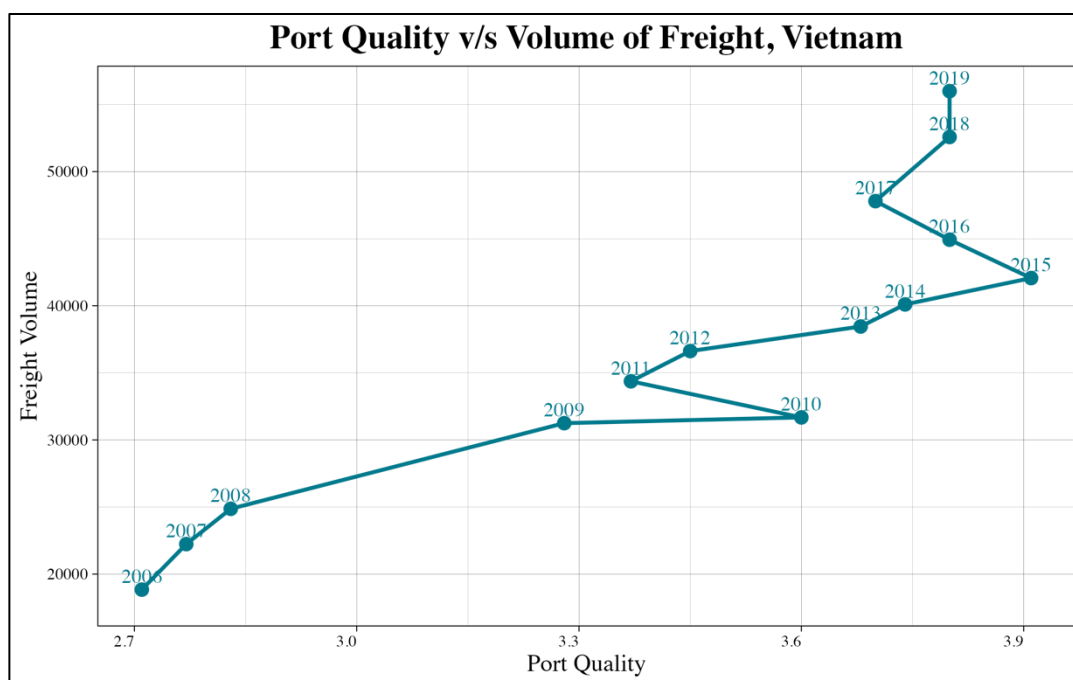


Figure 2.1.2g: Movements in Movements (Freight Volume), Vietnam

Source: Asian Transport Outlook and World Bank

Agents Attempting to Overcome the Constraint

Just like roads, the Vietnamese private firms are also trying to combat problems with ports through modal shifts as well as heavy investments. Given the delays in Vietnam, the Ministry of Transport (2022) estimated the international air cargo transport to and from Vietnam to increase at a staggering 50% per annual 2025 onwards. This is a worrying trend, given that in some cases, air freight costs (\$2.5 - \$5.0/kg) are over sixty times higher than road transport (\$0.043 - \$0.067/kg) (Freightos, 2021).

At the same time, private players are investing heavily in ports to reduce the delays. Between 2011 and 2020, 86% of the total \$8.5 billion in seaports came from private firms (Figure 2.1.2h). Additionally, the Vietnam Port Association is expecting non-governmental firms to invest \$11.7 billion in country's seaport infrastructure by 2030 (Portcoast, 2023). Prime Minister Pham Minh Chinh is also making active efforts to attract international giants to invest in ports. For instance, in 2024, he visited the gulf to invite UAE firms to invest (Tuoi Tre News, 2024) and personally acknowledged Adani Group's \$2 billion investment in Lien Chieu port (Hanoi times, 2024).

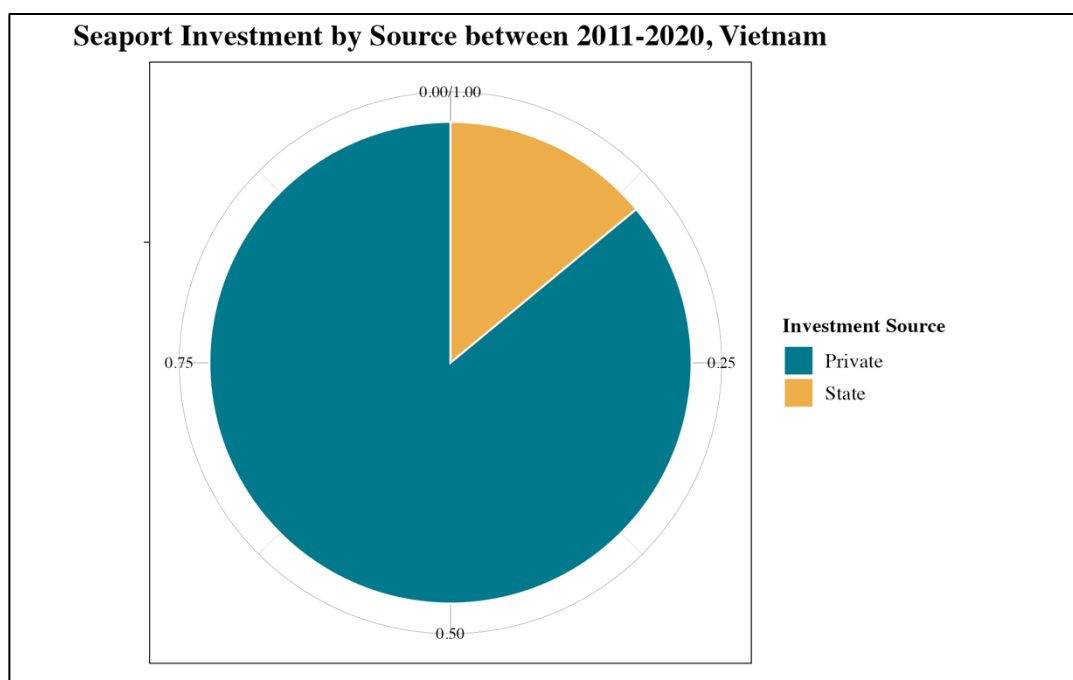


Figure 2.1.2h: Investment in Seaports by Source

Source: Vietnam Chamber of Commerce and Industry

Conclusion

Based on the evidence, Vietnam's port infrastructure constitutes a binding constraint on its economic development. Although the nation operates 320 ports handling 864.4 million tons of cargo, most facilities lack depth and scale for large vessel trans-shipment. Moreover, export delays more than double regional peers translate into steep logistics costs and slow growth of port transport dependent sectors. Without prompt expansion of deep-water terminals, modernization of cargo-handling equipment, and streamlined procedures, Vietnam's ports will stifle its next phase of export-led expansion. Mobilising both public and private investment to upgrade capacity and reduce turnaround times is therefore essential to sustain long-term economic growth.

2.2 Human Capital

General Aspects of Human Capital

Vietnam has made significant strides in education, with near-universal primary and secondary enrolment. However, tertiary enrolment remains low at 40%, and only 39% of the workforce over 25 have completed secondary education, with 11.2% achieving tertiary education (Figures 2.2a - 2.2c). Average years of schooling stand at 8 years, which aligns with GDP per capita but lags peer

countries (Figure 2.2d). Despite relatively high learning-adjusted years of schooling (11 years), education quality has not translated into technical skills, as 61.92% of the labor force is unskilled (Figures 2.2e - 2.2f; ILO & ILSSA, 2018). Low unemployment rates, which rise with higher education levels, alongside the large informal sector, indicate a labor market dominated by low-skilled workers (Figures 2.2g - 2.2h).

There is also growing evidence of a misalignment between the education system and the needs of the labor market, particularly in terms of technical and vocational skills. Firms frequently report dissatisfaction with the quality of workforce training, and many struggle to find workers with the right skill sets (Mori & Stroud, 2020). Given this disconnect, our analysis considers both education and workforce training data to assess whether human capital is a binding constraint under the Growth Diagnostics framework. On testing all four signals of the framework, we find that human capital, while a constraint, is not binding on Vietnam's growth.

Shadow Prices

Studies of returns to education in Vietnam indicate a rise until 2008, peaking at 8%, before declining to 5.2% by 2014 (Table 2.2a), below global (10%) and East Asia-Pacific (9.4%) averages (Montenegro & Patrinos, 2014). While these declining returns could suggest an oversupply of educated labor, they also point to deeper labor market distortions, skill mismatches, and gaps in technical qualifications (Montenegro & Patrinos, 2014).

The public sector absorbs most of Vietnam's high-skilled labor, particularly university graduates. Approximately 70% of wage workers with a university degree are employed in the public sector, where jobs offer better security and social status, reinforcing the challenges faced by private firms in attracting skilled workers. This segmentation limits the availability of high-skilled labor to the private sector, which has little incentive to pay higher wages to attract workers from the public sector. Furthermore, the dominance of low-skilled and informal jobs in the private sector reduces firms' willingness to invest in training, as there is less demand for advanced skills. This could partly explain the low returns to education: While the public sector offers relatively higher returns for higher education, the larger private sector (employing 82% of the workforce) remains focused on unskilled labor, especially in low-complexity sectors, thus offering lower returns to education (Figure 2.2i; Demombynes & Testaverde, 2018).

A study on wage-education mismatch found that 14.8% of university graduates are employed in mismatched jobs, with 64% of these in the private sector, earning 30% less than those in matched jobs (Table 2.2b; Tran & Pham, 2019). This reinforces the idea that the labor market undervalues higher educational attainment, especially in sectors where the demand for skilled labor is limited. Furthermore, 74% of the labor force lacks technical training (Demombynes & Testaverde, 2018), highlighting a disconnect between the education system and the labor market. The supply of skills is misaligned with the needs of industries, contributing to lower demand for higher education and, consequently, contributing to lower returns to education across both sectors.

However, if all firms were truly substituting education with more directly relevant vocational training, we would expect to see a strong premium for such training—especially within the private sector. Yet, this is not the case. Returns for both basic and professional vocational training have been declining in the private sector. In contrast, we see an increasing trend in returns to secondary and professional vocational training in the public sector (Figure 2.2i; Demombynes & Testaverde, 2018). This divergence further supports the idea that private firms operate predominantly in low-skill, low-complexity environments, where advanced training is not necessary. Public sector institutions, which are more skill-intensive but constitute only 8% of the workforce, seem to be the ones driving demand for vocationally trained workers. This pattern reinforces the argument that the private sector’s weak demand for skilled labor—rather than a shortage of human capital per se—is what drives the low returns to education and training.

Camels and Hippos

Sectors that contribute most to GDP growth, such as manufacturing, agriculture, and wholesale/retail trade, have the lowest proportions of trained labor. In contrast, skilled labor-intensive sectors like professional services and finance contribute minimally. However, this does not necessarily suggest that human capital constraints are binding. Industries that are more prevalent in the economy by nature do not require as much training, which explains the low percentages of trained labor in these sectors (Figures 2.2j - 2.2l). The apparent lack of human capital constraints might reflect the type of industries dominating the economy, which rely more on unskilled labor.

Movements-in-Movements

Vietnam's years of schooling have steadily increased since 1985, in line with global trends. Although the relationship between education and GDP is somewhat flatter in Vietnam compared to its peers, there is still evidence of positive growth elasticity to education as well as workforce training (Figures 2.2m - 2.2o). This suggests that human capital accumulation could have contributed to growth to some extent. However, it is important to note that correlation does not imply causation. Vietnam's growth has been driven by multiple factors, and human capital may have simply grown alongside other factors rather than being a key driver of economic expansion.

Agents Attempting to Overcome Constraints

In this section, we examine whether agents - particularly firms and the government—are making efforts to overcome potential human capital constraints. While education outcomes and investments have been improving, we shift our focus here to workforce training. This is because the key issue appears to lie not in the quantity of education but in the misalignment between formal education and the specific technical skills required by the labor market. Therefore, we assess the extent to which training programs - especially firm-led and government-supported vocational education - are bridging this gap between supply and demand in skills.

Vietnam's Enterprise Surveys show that a low percentage of firms offer worker training compared to its regional peers. While the proportion of employees receiving training is slightly better, it remains lower than in countries such as Bangladesh and Cambodia (Figures 2.2q). This low level of training investment by firms suggests that government programs, such as Technical and Vocational Education Training (TVET), might be supplementing private sector efforts.

However, while the Vietnamese government has made strides in improving TVET, systemic issues remain. Mori and Stroud (2020) highlight that Vietnam's TVET system has been ineffective in providing the intermediate-level skills necessary for the evolving labor market. Despite policy emphasis on aligning TVET curricula with employer demands, there remains a significant gap between the skills provided by these institutions and those required by the private sector. There is a mismatch between the skills taught by TVET institutions and the skills needed by industries undergoing technological innovation. This misalignment is particularly noticeable in intermediate-skilled workers, such as mechanical technicians, needed for more advanced industrial processes.

Despite these shortcomings, the lack of significant employer investment in training suggests that the constraint posed by workforce skills is not pressing enough to drive substantial changes in labor market dynamics. Employers, particularly in foreign-invested sectors and most private firms, continue to prefer unskilled labor (Mori & Stroud, 2020), indicating that the demand for more advanced skills is not significant. As a result, firms are not investing heavily in workforce training, and the market remains relatively stagnant in terms of labor force development.

Vietnam's migrant stock is 0.08%, suggesting low high-skilled labor imports (Figure 2.2r). This could again be due to the dominance of low-skilled industries in the private sector, which creates little demand for skilled foreign workers. Consequently, there is no active effort to increase the proportion of high-skilled labor in Vietnam.

Conclusion

Based on the evidence presented, human capital—both education and workforce training—does not represent a binding constraint in Vietnam's economic development. While the country has made significant improvements in education, the mismatch between the skills provided by the education system and the needs of the labor market, particularly in high-skill industries, limits the effectiveness of education in driving economic growth. Despite this, the dominance of low-skilled sectors, both public and private, and the preference for unskilled labor in the private sector further diminish the incentive for firms to invest in workforce training or hire highly educated workers.

The relatively low returns to education, particularly in the private sector, thus reflect a broader issue of labor market inefficiencies rather than a lack of skilled labor. The lack of demand for highly skilled workers, coupled with low employer investment in training programs, suggests that the primary constraint lies in the nature of the labor market itself, not the availability of human capital.

In summary, Vietnam's human capital challenges are more about misalignment and inefficiencies within the labor market than about the absolute scarcity of educated or trained workers. Moreover, the low returns to education and training coupled with the low firm-level efforts to improve human capital, reflects that this constraint is not binding on Vietnam's growth. Human capital might not be a binding constraint at present but addressing these inefficiencies through better alignment between education and labor market needs, and improving employer investment in workforce

training, could help diversify into more complex industries and unlock further potential for economic growth.

2.3 Access to Finance

Overview of Financial System

Vietnam's financial system has grown significantly over the past two decades, supported by strong economic performance and progressive regulatory reforms. The sector comprises 50 foreign bank subsidiaries, 28 commercial banks, and 1,181 People's Credit Funds, reflecting a diverse and competitive structure (Statista, 2024). Domestic credit to the private sector stood at 126% of GDP in 2022, suggesting moderate financial depth. Despite this, Vietnam's firms are complaining the most about access to finance within the World Bank Enterprise Survey (Figure 2a). Building on this domestic foundation, Vietnam looked outward to integrate within global financial markets.

Since its 2007 WTO accession, the country has steadily attracted FDI, reaching USD 18.5 billion in 2023—exhibiting greater stability than many peers. Investor confidence is further evidenced by Vietnam's BB+ sovereign bond rating with a stable outlook and increasing participation in government and corporate bond markets since 2022 (AsianBondsOnline, 2025). A net financial account surplus of USD 8.47 billion in 2023 highlights the growing role of foreign investors in Vietnamese assets.

At the domestic level, Vietnam benefits from a high savings rate, supporting financial system liquidity and macroeconomic stability. With gross domestic savings at 37% of GDP—second only to Singapore—and a 2.5% fiscal deficit in 2023, the country maintains a healthy balance between private thrift and public spending. Customer deposits grew by 13.2% year-on-year in 2023, despite low interest rates, reinforcing depositor confidence and creating a solid foundation for credit expansion (FiinGroup, 2024).

Nevertheless, intermediation inefficiencies persist. Non-performing loans rose sharply to 7.91% by end of 2023, driven by economic stress and weak collateral enforcement (FiinGroup, 2024). Though regional comparisons show similar NPL levels (Figure 2.3k), the trend exposes increased banking risk. These concerns are compounded by a relatively low Capital Adequacy Ratio of 11.7% in 2023—well below peers such as Thailand (20.4%) and Cambodia (22.5%)—which signals limited loss-absorbing buffers (CEIC, 2024). Alongside modest bank capitalization and a

Net Interest Margin (4.5%) below the global average, these vulnerabilities pose risks to the financial stability of large Vietnamese banks and may hamper effective intermediation.

Shadow Prices

Testing access to finance as a binding constraint involves establishing that there is an insufficient supply of credit. If this were the case, we would expect to see high real interest rates coupled with low quantities of credit. In 2023, Vietnam's real interest rates reached 7.3%, relatively high when compared to Thailand (3%) and Bangladesh (0.63%).¹ Only Malaysia reported a similar real interest rate as Vietnam. Between 2015-2023, Vietnam's average real interest rate of 5.2% proved to be higher than the 3% average of peers. However, there is great volatility in Southeast Asia. For instance, from 2022 to 2023, real interest rates jumped 2.3% to 7.3% in Malaysia, -1.6% to 3% in Thailand, and 3.8% to 7.3% in Vietnam. Given a standard deviation of 3.6% among peer countries, Vietnam's average of 5.2% does not represent an extreme outlier. It falls within a typical range of variation, suggesting that while elevated, it is not alarmingly high relative to regional variation.

The State Bank of Vietnam (SBV) – the country's central bank – engaged in monetary policy expansion to support growth in 2024. The average lending rate decreased by 0.6-1 percentage point compared to 2023, while credit growth reached 15.1% (ADB, 2025). Coupled with this, average inflation increased to 3.6% in 2024 compared to the 3.3% figure in 2023. Using the fisher equation, as nominal rates decrease and inflation increases, we can expect real interest rates numbers to be lower in 2024, potentially more aligned with peers.

Vietnam's credit to the private sector stands at 150% of GDP — significantly above the global trend line (Figure 2.3a). However, many of its Southeast Asian peers also exhibit similarly high levels. This suggests that while Vietnam's credit intensity is elevated, it aligns with a regional pattern and follows a comparable credit market structure. The SBV have ambitious plans of increasing credit growth by 16% in 2025 (ADB, 2025). Moreover, the World Bank Enterprise Survey reveals that the working capital financed via bank loans is one of the highest in Vietnamese firms, relative to peers (Figure 2.3c).

In sum, while Vietnam's real interest rates are somewhat higher than those of its peers, there is insufficient evidence to suggest that access to finance is a binding constraint based on price signals alone.

Camels and Hippos

To test for Camels and Hippos, it is important to first identify those economic sectors that are most reliant on accessing finance. There is a lack of official statistics and aggregated data for all economic sectors, making it difficult to identify the sectors of interest. Therefore, a proxy is used – sectors most dependent on trade finance – as this data was more readily available in the paper by AMRO (2024). The paper identifies Agriculture & Fishing (27%), Construction (25%), Wholesale & Retail Trade (15.4%), and Metals (16.4%) as those sectors with the highest share of trade finance usage. However, this data overlooks some important industries. After a thorough literature review, real estate and manufacturing are two further industries that rely on finance. The report from ADB (2024) states that real estate contributes to 21.3% of total outstanding bank credit, while a separate report precludes that manufacturing is driving credit growth (FiinRatings, 2024).

Given that these industries are finance-dependent, if finance was a binding constraint we would expect these industries to exhibit the lowest GDP growth rates. Figure 2.3d depicts that these finance-dependent industries exhibit varying growth rates. Some sectors such as mining, real estate, and agriculture & fishing exhibit less than 4% growth, other sectors such as construction, manufacturing, and wholesale and retail trade experience far greater growth rates in the range of 5-7% annually. As a result, we conclude that the evidence from Camels and Hippos provides little indication that finance-dependent sectors are significantly disadvantaged. Therefore, access to finance does not appear to be a binding constraint, based on this analysis.

Movements-in-Movements

Assessing whether alleviating credit constraints (e.g. reducing interest rates) improves economic growth, we would on a minimal level expect to see a negative correlation between real-interest rates and GDP per capita growth rates. Moreover, on the investment side, we would expect a negative correlation between real interest rates and gross capital formation. Figure 2.3f - 2.3g present the movement-in-movement graphs to test these correlations. In both cases there is no clear correlation, let alone a negative one. This evidence in movements-in-movements therefore substantiates the conclusions made earlier – access to finance is not a binding constraint.

Poor Financial Intermediation

The evidence from price signals, camels and hippos, and movements-in-movements all indicate that finance is not a binding constraint. However, there is a growing concern that banks are becoming riskier as non-performing loan rates increase, leading to more conservative lending and

fewer loans being provided. This section investigates whether poor financial intermediation and underlying financial risk is constraining access to finance in Vietnam.

As noted in the overview, Vietnam has the lowest capital adequacy ratio relative to peers, while experiencing an increase in non-performing loans (NPLs). These trends form a concerning outlook that could render the banking sector too risky. Upon closer inspection, not only are Vietnam's NPL rates lower than global trends but also very much aligned to regional numbers (Figure 2.3k). Therefore, while Vietnam's NPLs have experienced an uptick in 2022 and 2023, it is still at relatively low levels.

The concern then becomes, if this growth in NPL jumps rapidly, is the Vietnamese banking system resilient enough to keep providing credit? We look at equity over total assets for all major Vietnamese banks using the data collated by Thanh Ngo and Tu Le (2017). The 29 banks in the dataset had an equity-to-asset ratio of 9% in 2022 – above the 8% Basel III standards (BIS, 2019). Moreover, of the 10 banks with the highest non-performing loans, 7 of them had maintained sufficient loan loss provisions. Finally, the average ROE for the 29 banks is 17.4 in 2022 and grew by 12% year-on-year. This goes to show that banks profits are not depleting significantly due to NPLs and have enough headroom to circumvent financial issues.

Additionally, the State Bank of Vietnam (SBV) is likely to bailout banks in times of banking failure. A prominent example is the case of Saigon Joint Stock Commercial Bank (SCB), which faced a severe crisis following the arrest of real estate tycoon Truong My Lan in October 2022 for orchestrating a massive financial fraud. In response, the SBV provided SCB with approximately \$26 billion in special loans, equivalent to about 6% of Vietnam's GDP, to support the bank's liquidity and protect depositors (Reuters, 2025).

In essence, poor financial intermediation due to growing banking risk is not a binding constraint for Vietnam. While NPLs are on the rise, the banking systems balance sheets indicate resilience against bank failures. On top of this, the Vietnamese government is unlikely to let a banking crisis unfold, especially given its intervention with Saigon Commercial Bank (SCB).

Conclusion

After evaluating price signals, camels and hippos, movements-in-movements, and financial health of the banking sector, it is evident that access to finance is not a binding constraint for Vietnam. Credit is widely available, real interest rates—while elevated—are within regional norms, and

finance-dependent sectors are not underperforming. Although non-performing loans have risen, they remain manageable, and banks continue to meet capital adequacy standards. Strong deposit growth, high savings rates, and government readiness to intervene, as seen in the SCB case, further reinforce sector resilience.

2.4 Macroeconomic Risks

Vietnam's rapid economic growth has positioned it as a rising star among emerging markets, attracting significant interest from investors, policymakers, and international institutions. However, the pace of this expansion raises important questions about its underlying sustainability. This paper critically examines whether Vietnam's growth is anchored in sound macroeconomic policies or if it is being propelled by unsustainable borrowing and emerging structural vulnerabilities. If the latter, this may hamper long-run growth in the economy. The subsequent analysis finds that macro risks are not a binding constraint for Vietnam.

In 2024, Vietnam's debt-to-GDP ratio lies at 33.8%, much lower than peers: Thailand (65%), Malaysia (68.4%), and Singapore (175%). Demonstrating that Vietnam has maintained a highly sustainable debt path. The Vietnamese Communist Party has been running government deficits for several years. However, these public deficits (\$14bn) are small in comparison to the private savings which amounted to \$171bn in 2023. The government is able to comfortably raise funds to support its low levels of deficits, without worrying the investors about debt sustainability. Vietnam's strong current account position, wherein the country has been running a surplus of 6% of GDP in 2023, entails that the country's public and private accounts combined generate savings net of investment activities. These savings have generally been used to accumulate foreign reserves to mitigate exchange rate pressures, especially vis-à-vis the US dollar. In general, Vietnam's fiscal stance is strong. Its government is mindful with public deficit and has managed the debt-to-GDP ratio well.

Headline inflation rates came to 3.6% in 2024, which is well below the SBV target of 4-4.5%. Inflation has remained at these low levels since 2015 – providing a highly stable and predictable market. On the other hand, the Dong has depreciated vis-à-vis the US Dollar, this has improved competitiveness of exports. However, too much volatility in the FX market would see the SBV intervene with FX reserves to stabilize the market. The SBV holds \$80 billion in FX reserves, demonstrating to markets its credibility to intervene to stabilize the Dong.

So far, Vietnamese fiscal balance, savings rate, and low inflation rates have indicated that the macro environment is business friendly and robust. However, with the new tariffs from the US – along with heightened geopolitical risks, Vietnamese trade can face major setbacks. About 87% of Vietnam’s GDP comes from exports of goods and services, of which 24% goes to the US. With the Trump administration setting 10% baseline tariffs as well as a 46% tariff that is on a 90-day pause, these could significantly hurt Vietnamese manufacturers whose business model and products are catered to serve the US market. While terms of trade with the US are still under negotiations, the heightened risk of tariffs makes trade diversification ever more important.

Vietnam’s strong fiscal discipline, low debt-to-GDP ratio, healthy current account surplus, controlled inflation, and ample FX reserves collectively indicate macroeconomic stability. These fundamentals reassure investors and policymakers that growth is not reliant on unsustainable imbalances. Therefore, macroeconomic risks are not a binding constraint to Vietnam’s long-term growth trajectory.

2.5 Appropriation

Appropriation refers to the risk that firms will not be able to retain the returns to their investment, due to factors like corruption, weak property rights, political instability, or excessive state interference. To evaluate whether this is a binding constraint in Vietnam, we examine indicators related to corruption, political and regulatory uncertainty, legal protections, and state distortions in markets.

While Vietnam exhibits signs of corruption, regulatory hurdles, and occasional state influence in the market, the data suggests that appropriation is not a binding constraint to economic growth. A relatively low percentage of firms identify political instability or licensing requirements as major obstacles, with under 10% citing business licensing and permit issues — a figure even lower among the most productive sectors like large, foreign, and manufacturing firms (Figures 2.5a-2.5b). Regulatory delays are also modest: for instance, operating licenses take just around 11 days to obtain, outperforming many regional peers and global benchmarks (Figure 2.5c). Similarly, complaints about tax administration, court access, and access to land are minimal, with less than 5% of firms seeing them as critical issues in 2023 and virtually none citing courts as a top constraint (Figures 2.5d - 2.5f).

Although corruption remains prevalent (Figures 2.5g - 2.5h), it does not appear to discourage firm growth or investment, nor is it a top concern among businesses (Figure 2.5i - 2.5j). Fewer than 5% of firms report it as their biggest obstacle, including less than 10% of foreign firms, and below 5% of large and manufacturing firms - all groups central to Vietnam's growth story (Figures 2.5k - 2.5m). Moreover, there is no strong correlation between control of corruption and economic growth, indicating that improvements in corruption levels have not driven Vietnam's economic success (Figure 2.5n). In line with this, productive firms continue to expand, and foreign direct investment remains strong, suggesting that while appropriation risks may exist, they are not currently impeding investment, innovation, or expansion in a significant way.

3. Growth Syndrome

Our analysis leads to the conclusion that the low levels of private investment or entrepreneurship are a result of low returns to social and economic activity. These symptoms are rooted in the lack of complimentary factors of infrastructure, specifically in roadways and waterways that prevent the investments from being fully productive. The logical connections are presented below in Figure 3a.



Figure 3a: Growth Diagnostics Tree

The constraint situated in the Vietnamese context produces the following syndrome: **Vietnam has outgrown its transport infrastructure.** Vietnam has sustained miraculous levels of growth over the last 30 years, that has not been complimented with the necessary transport infrastructure to support the economy because of the imperative of social distributive policies. As a socialist country their priority in spreading investment is to eliminate the gap between richer and poorer regions, but this has come at the cost of their own growth especially since its growth is export driven. This has manifested in over-investment into regional ports that provide low returns, cabotage restrictions to protect the domestic shipping industry, and highway investments in regional cites that are not used (Minh, 2016). This causes congestion and bottlenecks on roadways shown by Figure 3b in industrial parks around Hanoi and HCMC because of the outflow of goods on narrow 4-lane highways that are not constructed to handle higher loads and deteriorate rapidly due to

overuse (Donski-Lesiuk, 2023). Cabotage rules also restrict use of regional ports by foreign carriers putting greater pressure on roadways to transport goods within the country (Conventus Law, 2015). This roadway congestion is coupled with logistics delays at customs offices, and within Hai Phong and Vung Tau ports the have not received the necessary investment to digitalize operations, procure smart-port technology and build deep-sea terminals (Minh, 2016). This means trucks transport less goods, more outputs stay in warehouses, and goods sit in ports longer, all of which increases the business costs. Dang and Yeo (2018) identified that logistics costs account for 20% of Vietnam's GDP and cite it as the most detrimental to Vietnam's competitiveness. Reflective of this is in 2019 WEF Competitive index Vietnam ranked 104 (out of 141) in road connectivity, 103 in road quality, and 83 in efficiency of seaport services. This results in a stagnation in productivity that is crucial for Vietnam's continued growth (Figure 3c). Given that the state's robust capacity to overcome these obstacles, this report concludes that it is a transitory problem that the state will address in the next decade.

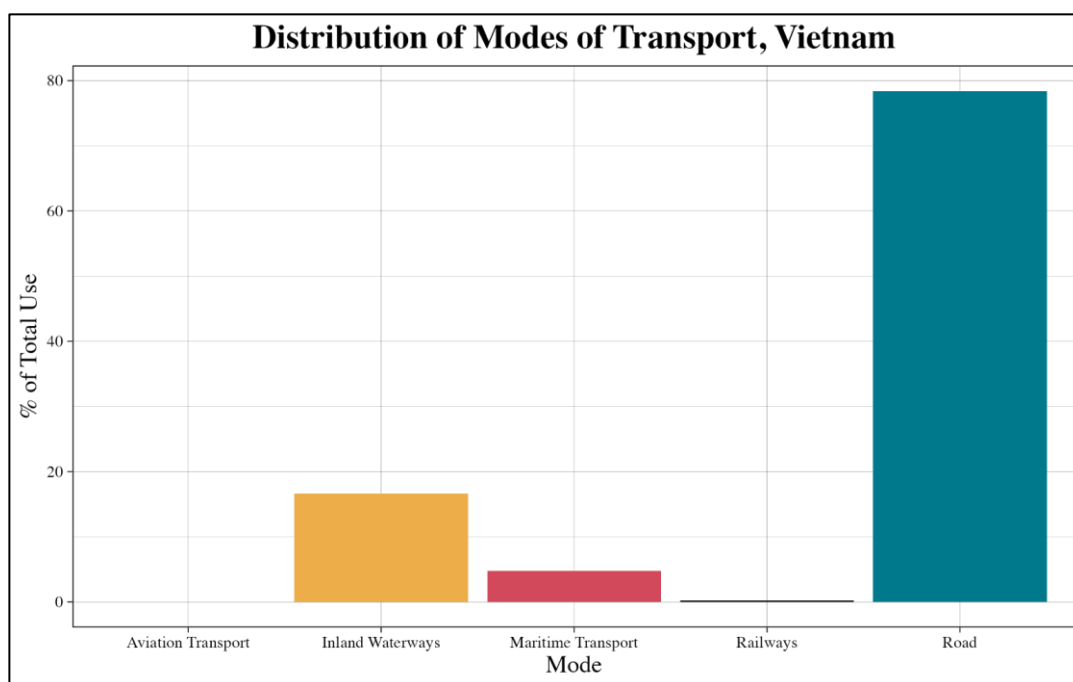


Figure 3b: Distribution of Modes of Transport, Vietnam

Source: Asian Transport Outlook

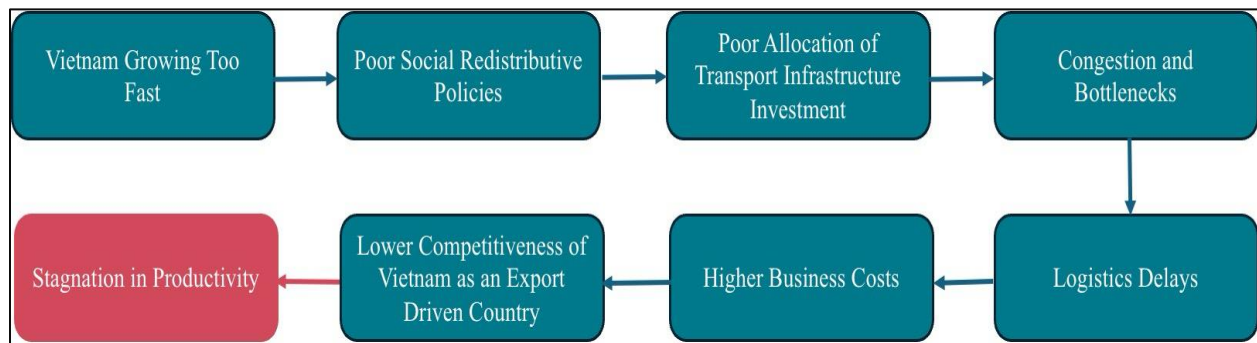


Figure 3c: Vietnam Syndrome

4. Therapeutics: Policy Recommendations

4.1 Public-Private Partnerships

Vietnam is recommended to forge public-private partnerships with foreign firms to integrate existing and planned transport networks to address congestion and logistics delays in the medium term. In similar spirit to the Chiapas case investigated by the Growth Lab, the problem is not lack of infrastructure, but poorly utilized infrastructure (Ricardo et al. 2015). This is exemplified by the over-reliance on roadways to transport goods throughout the country. These recommendations are meant to complement Vietnam's existing \$65 million transport long-term masterplan to expand Vietnam's existing roadways, construct high-speed rail, build deepwater ports, and improve international airports (Vo, 2023). The recommendation is split into two segments: one that focuses on **promoting a multimodal transport network to alleviate road congestion** and the second that aims to **digitalize and integrate technology in ports to reduce logistics delays caused by lack of integration**.

4.1.1 Promoting a Multimodal Transport Network

Promoting a multimodal transport network entails transitioning away from roadways into other forms of transportation to alleviate road congestion. This report recommends focusing on promoting rail as a meaningful alternative. Currently Vietnam transports less than 1% of its good via railways despite having a supportive flat and coastal geography (Communications, 2017). Moreover, Vietnam's low-end peers, Cambodia (8.29%) (BATRAK, 2025) and Bangladesh (10%) (Hossain & Ahsan, 2021) perform better than Vietnam. To accomplish this transition Vietnam should **retrofit passenger trains for freight use and construct a dedicated freight corridor parallel to the high-speed North-South lines**.

During off-peak season Japan has been using its Shinkansen trains to transport cargo in retrofitted freight cars. This model has been successful in addressing the labour shortages in its logistics and road transport sector and could be replicated easily in Vietnam. Although almost all train services in Vietnam are dedicated to passenger use, rail only accounted for 0.2% of passenger traffic in 2019 and has only been decreasing year after year (GSO, 2018). Since the trains are often empty, it provides a unique opportunity to address road congestion in the short term. To incentivize and inform transport-intensive industries to switch to rail, the state could provide preferential corporate

tax rates (<20%) to those firms (Pwc, 2025) who use rail and a percentage-based tax credit on expenses for using rail transport. These freight cars would be **installed in the Saigon-Hanoi railway with dedicated loading and unloading bays at key stations; Bien Hoa, Di An, Nam Dinh and Hai Phong** (Appendix 3). This solution is politically feasible because it addresses the urgent need to reduce road congestion with a low-cost solution and does not interfere with the state's current transport plan. Historically, Vietnam has also undertaken programs to achieve development goals, as evidenced by its preferential corporate tax rates for Vietnamese-based R&D projects (Pwc, 2025).

In the long-term, Vietnam should not only focus on upgrading and expanding existing rail infrastructure but also building a dedicated freight corridor parallel to the proposed high-speed North-South lines to industrial parks. An existing model is the Eastern-Western Dedicated Freight Corridors in India which is projected to transport an additional 10% of goods in India via rail increasing the rail transport share to 40% (India Briefing, 2024). These corridors were also constructed and financed via PPPs making it an even more attractive model for Vietnam to adopt (India Briefing, 2024). This solution is considered political feasible because Vietnam is investing in rail now to reduce the road passenger use, so this will support its aim of reducing road congestion.

4.1.2 Digitalization and Integration of Technology

Ports in and around Vietnams industrial hubs namely, Hai Phong and Vung Tau lag in their technological adoption compared to peer countries. Malaysia and Singapore have the most sophisticated port technologies in the world allowed them to gain a competitive edge and be the preferential hub for port users in Southeast Asia (Minh, 2016). Contrastingly, port in Vietnam primarily handle port operations manually through paperwork and phone calls (Banomyong et al. 2015). Since ports are such an important node of the supply chain and Vietnam's growth is primarily export-driven, improving the efficiency of Vietnam's ports is key in fostering future growth. The solution then is to **develop and adopt an Integrated Terminal Operations System (ITOS)** like Singapore's CITOS (Wen et al. 1992) that **integrates Vietnam's developed digital customs clearance processes** with corresponding **smart-port technologies** to reduce administrative delays and harmonize port operations with an information-driven approach. This solution would be adopted at **Hai Phong and Vung Tau**, Vietnam's busiest ports to have the largest impact.

An ITOS serves to track movement of goods in, inside, and out of the port and improve the scheduling of logistics and shipments from inline through a centralized data-sharing platform (Wen et al. 1992). This is accessible by all port users, including truck drivers, port operators, and shippers. This would reduce the reliance on paperwork to track container movement throughout the port and reduce delays in loading and unloading ships. Despite having designed a digital customs clearance portal (VNACCS/VCIS), at these ports, documentation is required to be submitted both online and on paper because these ports do not have an ITOS (Minh, 2016). To reduce congestion at customs offices integrating VNACCS/VCIS into the ITOS will help reduce the administrative burden on truckers. Time-delays and bottlenecks also occur at gates because of manual inspection processes (Minh, 2016); installation of Radio Frequency Identification (RFID) of goods and automated gate systems will also cut down on these delays. Lastly, in these ports, cranes are operated manually to load and unload these ships (Minh, 2016). Adopting remote-controlled cranes and gantries will help improve loading/unloading times and reduce the chance of human error.

We know these solutions are politically feasible because there exists a precedent with Vietnam's newer Long An and Da Nang ports having port-wide management systems (LEC). The issue lies in the on-ground adoption of these systems and resistance to technology adoption by port workers. To ensure this solution is effective, the PPP needs to take the form of Build-Operate-Transfer, to allow for technology partners to interface with port operators and provide workforce training.

4.1.3 Potential Partnerships and PPP Models

In table 4.1.3a we outline the potential partnerships and corresponding PPP model that would be the most reasonable and provide a brief justification for each PPP model. Port partners were selected based on whether they've worked with the Port of Singapore and Port of Tanjung Pelepas (Malaysia) to develop their smart-port and informational technology systems. Railway partners were selected based on expertise and previous experience developing in Vietnam. Note that these are suggested firms and that it is up to the discretion of the Ministry of Transport and Ministry of Planning and Investment to accept bids from the companies they deem suitable for the projects proposed in this report. The Saigon Newport Corp (SNP), Vietnam Logistics Business Association (VLA) and Vietnam Maritime Corporation (VIMC) would also be consulted in selection of partners and implementation.

Suggested Partner	Service Provided	PPP Model	Model Justification
Siemens	Installation of RFID and automated gate-systems.	Build-Transfer-Lease	Allows Siemens to take on initial costs while port operators pay in the long term. This allows for immediate use of automated-gate systems.
Kalamar	Development of ITOS, integration of VCIS/VNACCS, and workforce training.	Build-Operate-Transfer	Kalamar would build and operate the system in the short run. Allows for immediate use, and for Kalamar partners to interface with port operators and integrate customs clearance.
ZPMC	Installation of remote-controlled cranes and gantries.	Build-Transfer-Lease	Allows ZPMC to take on initial costs and port operators pay in the long term. This allows for immediate use of the services provided.
China Railway Engineering Corporation (CREC)	Retrofitting passenger cars to freight cars.	Build-Transfer-Lease	Allows CREC to take on initial costs and state pays in the long term.

			This allows for immediate use of the services provided.
Japan Freight Railway Company (JFRC)	Building North-South Freight Corridor.	Build-Own-Operate	JFRC to finance, build and operate the North-South Freight Corridor.

Table 4.1.3a: Suggested Partners

5. Conclusion

Vietnam is a rising tiger in the forest of global exports. The country has experienced high and stable growth in its GDP since the introduction of Đổi Mới and the first law of foreign investment in 1986. It is also sufficiently diversified and in fact, more complex than the trend suggests, given its GDP per capita. However, Vietnam's reigning position is in threat, given its rapidly aging population, short demographic window of opportunity, falling relative productivity across sectors and stagnating gross capital formation. In other words, Vietnam's bullish growth is entirely dependent on the "magical" total factor productivity, whose plateauing will bring growth to a standstill. It is thereby imperative keep the TFP from plateauing, and finding the answer to the question - **How can Vietnam sustain its current growth rate in the long run?**

The binding constraint hindering Vietnam's sustainable growth is its **transport infrastructure**. Specifically, the country is wrestling with severe inefficiencies in the two modes of transportation it relies on the most – **roadways and waterways**. Chronic road congestion delays freight from factories to ports, while over-capacity and procedural bottlenecks at seaports hamper export reliability and raise logistics costs. To navigate these inadequacies, caused primarily by poor social redistributive policies and infrastructure allocation investment, Vietnam must take immediate action.

To address the challenges, we recommend: 1) forging public-private partnerships with foreign firms to integrate existing and planned transport networks to address congestion and logistics delays in the medium term; 2) adopting a multimodal transport network entails transitioning away from roadways into other forms of transportation to alleviate road congestion; 3) developing and adopt an Integrated Terminal Operations System (ITOS) that integrates Vietnam's developed digital customs clearance processes with corresponding smart-port technologies.

By collaborating with proven private-sector partners who have delivered similar solutions in peer markets, Vietnam can overcome its infrastructure bottlenecks and secure the productivity gains essential for long-term growth.

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Appendices

Appendix 1: Economic Growth History

Appendix 1.1: Growth Trajectory

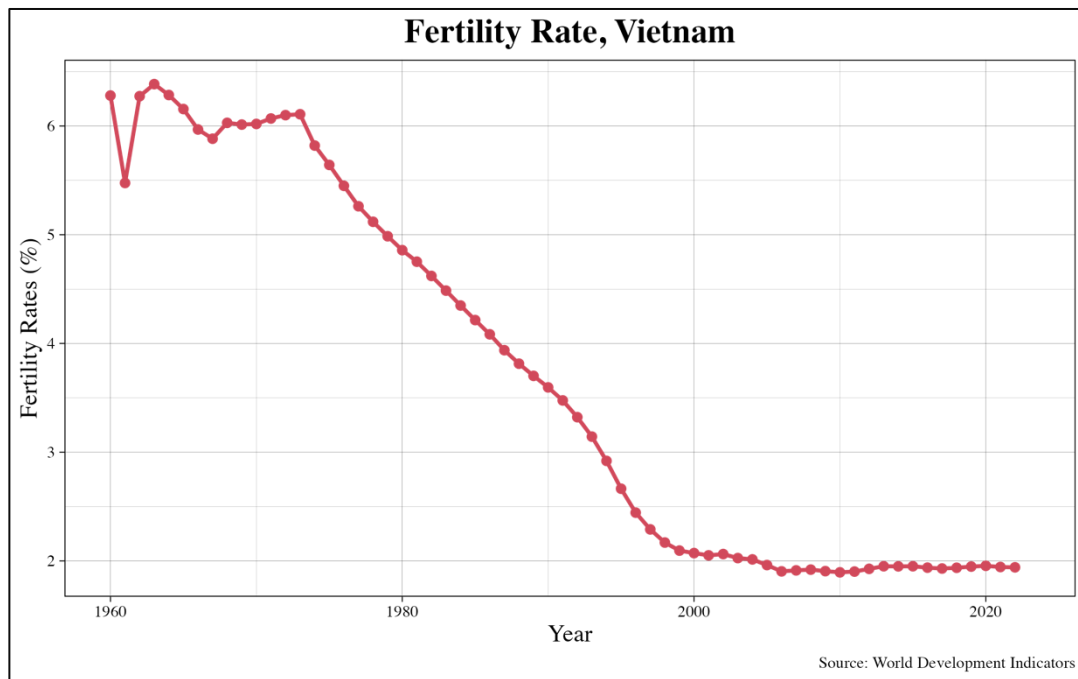


Figure 1.1k: Fertility Rate, Vietnam

Source: World Development Indicators

Appendix 2: Growth Diagnostics

Appendix 2.1: Transport Infrastructure

Appendix 2.2: Human Capital

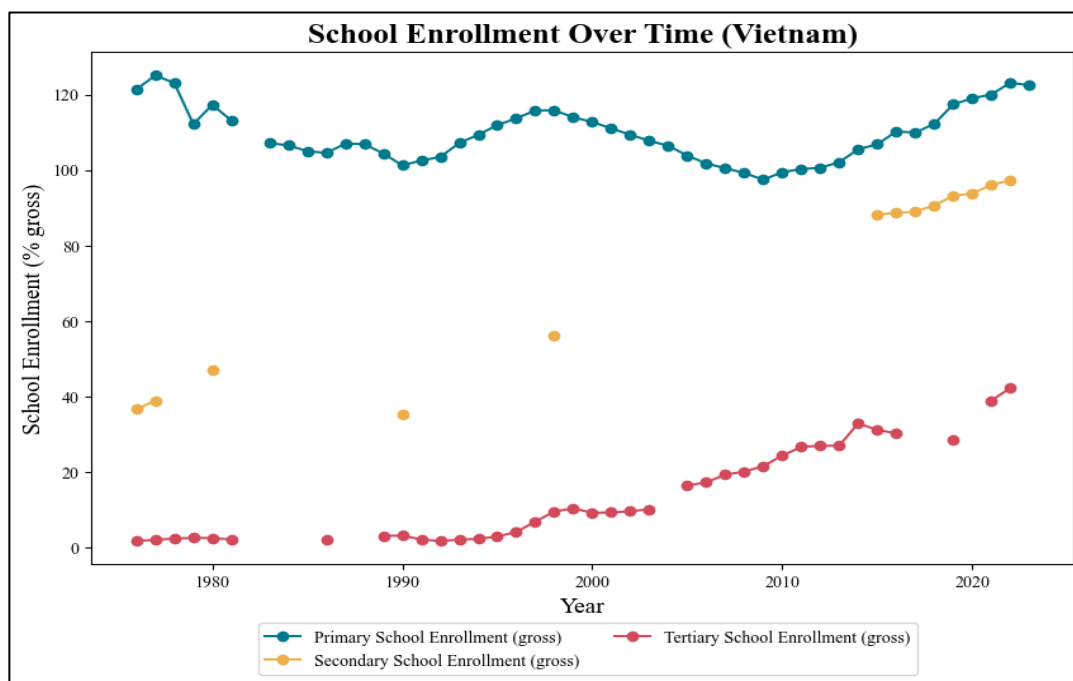


Figure 2.2a: School Enrollment Over Time, Vietnam

Source: World Development Indicators

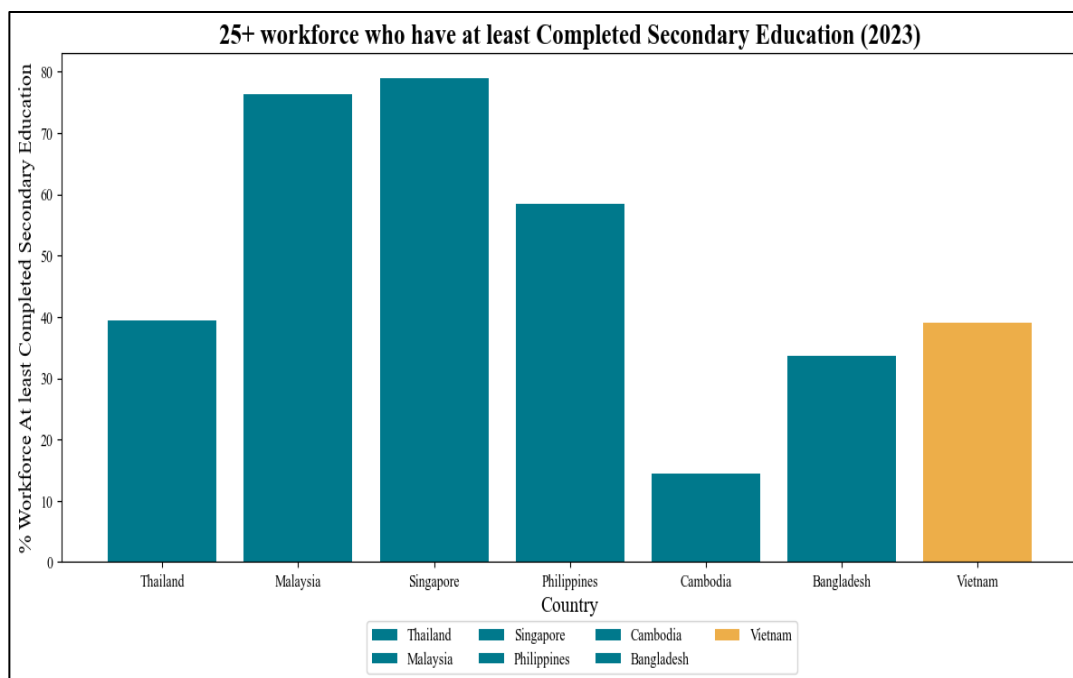


Figure 2.2b: 25+ Workforce with Atleast Secondary Education, Vietnam and Peers
Source: World Development Indicators

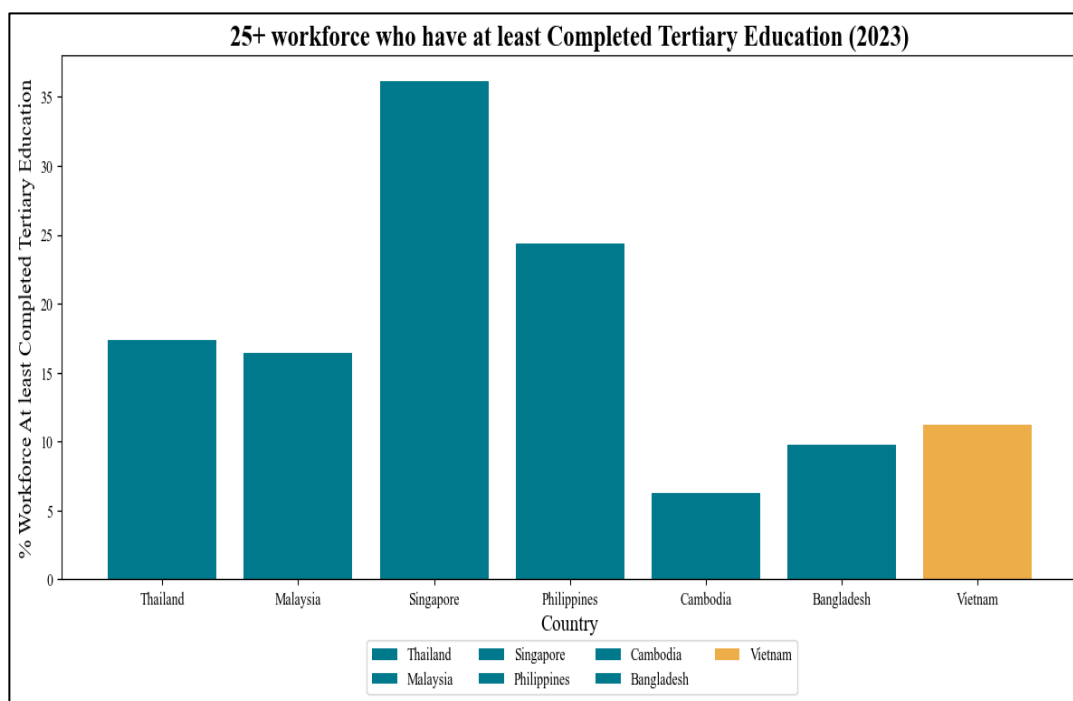


Figure 2.2c: 25+ Workforce with Atleast Tertiary Education, Vietnam and Peers
Source: World Development Indicators

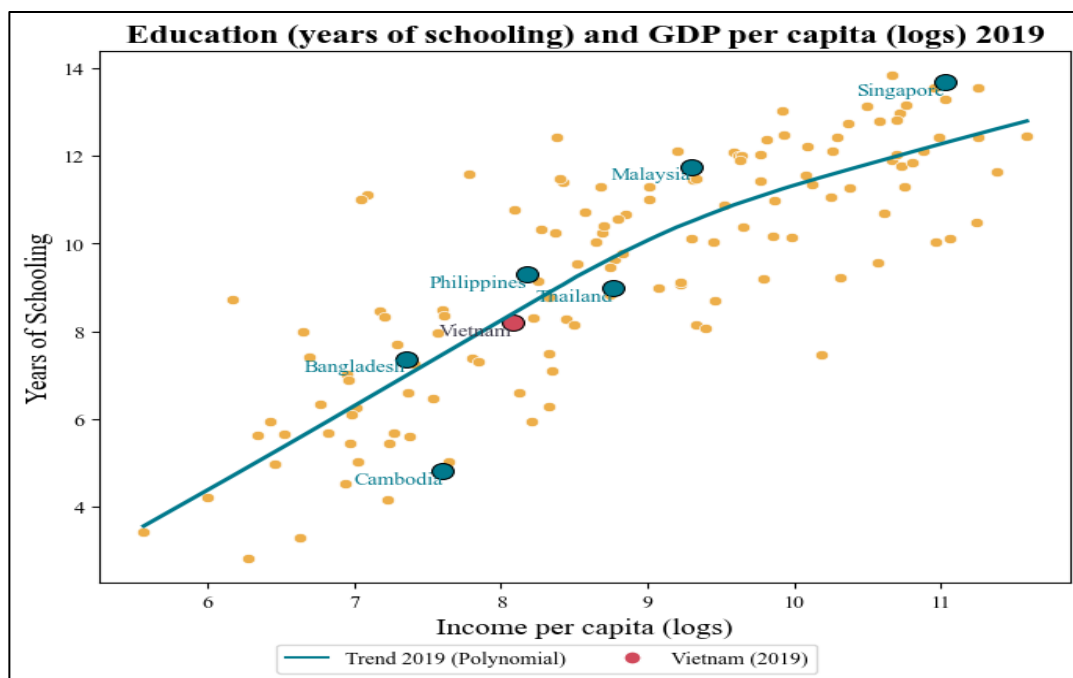


Figure 2.2d: Education and GDPPC

Source: World Development Indicators

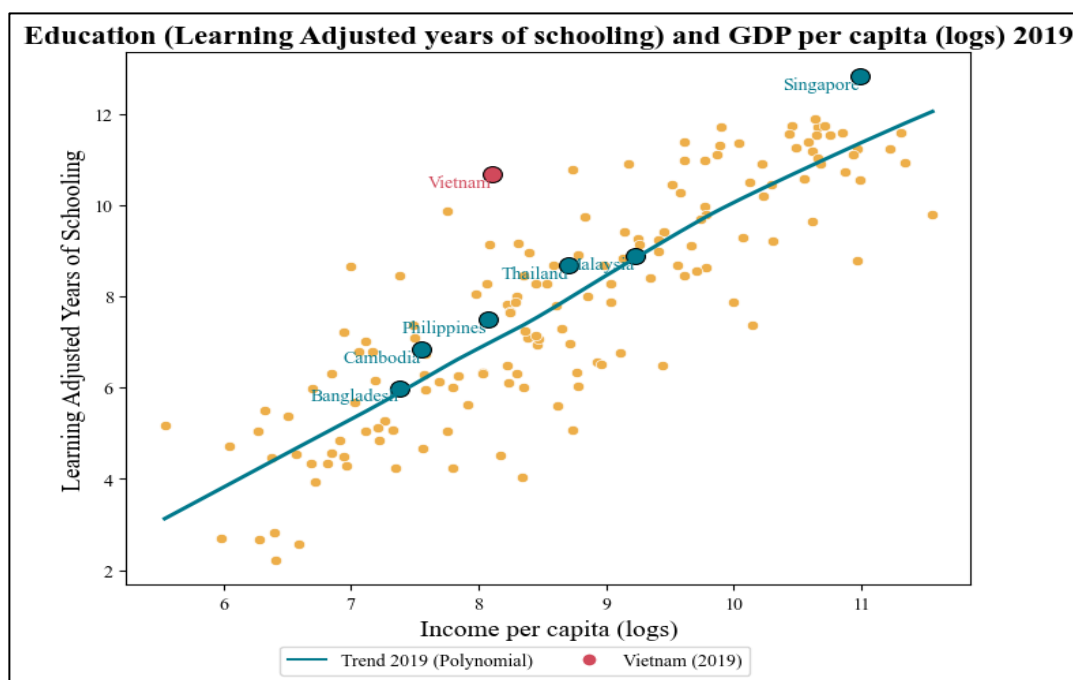


Figure 2.2e: Learning Adjusted Years of Schooling and GDPPC

Source: World Development Indicators

Structure of the Labor Force by Technical Qualifications in Quarter 2, 2017

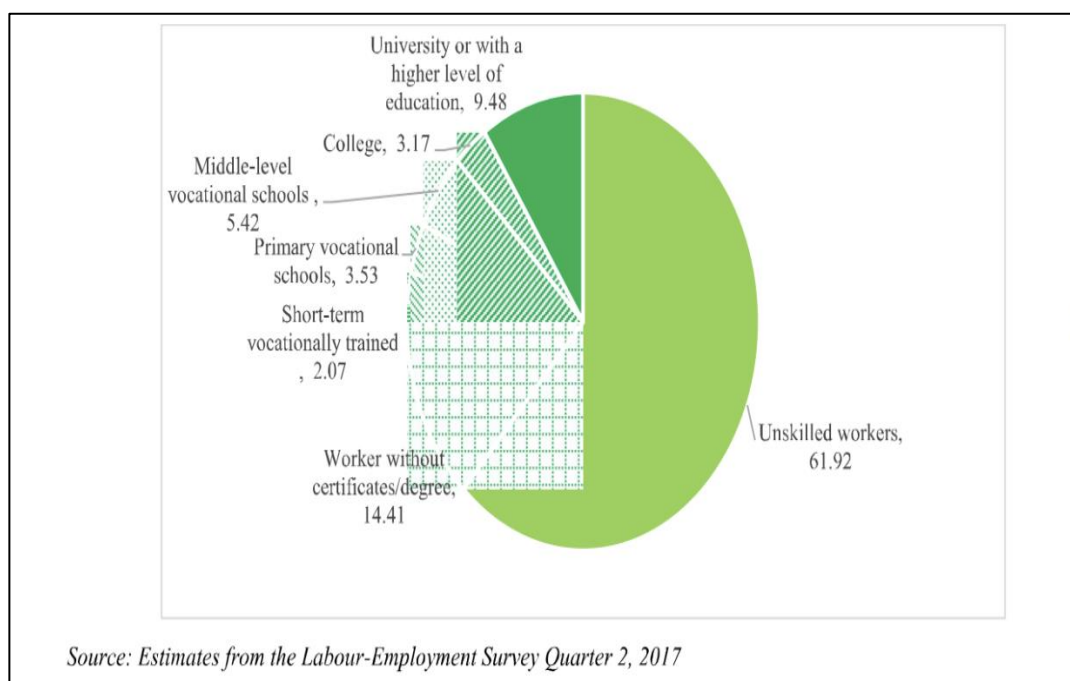


Figure 2.2f: Structure of Labour Force

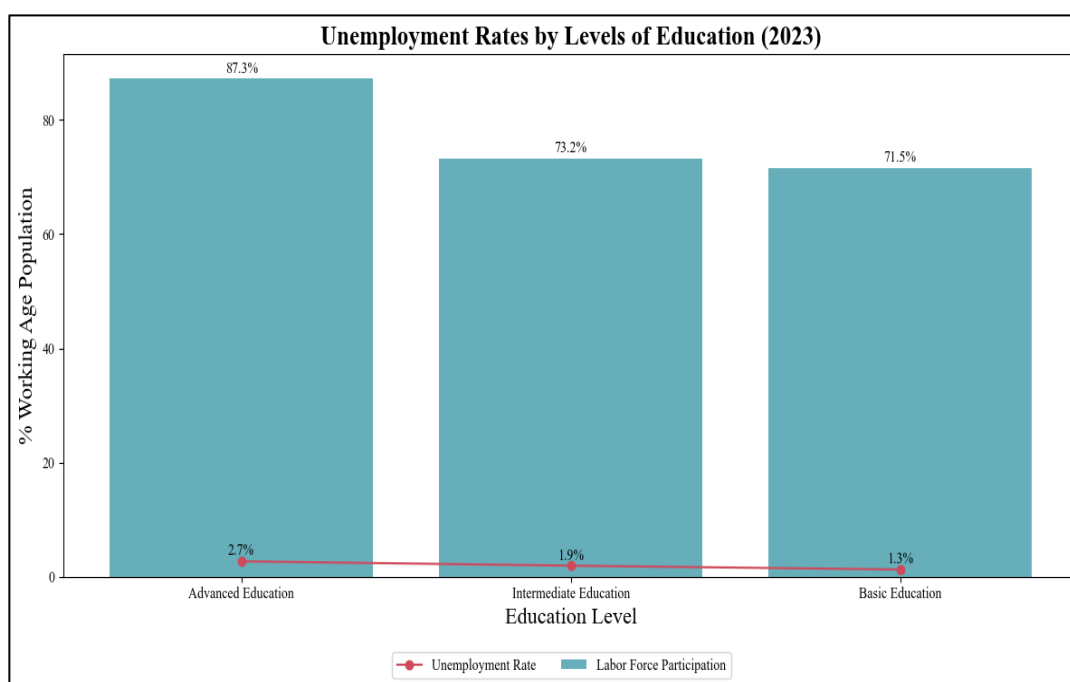


Figure 2.2g: Unemployment Rates by Level of Education

Source: World Development Indicators

Baseline Estimates using OLS

Variable	1998	2004	2006	2008	2010	2012	2014
School year	0.025** (0.005)	0.063** (0.002)	0.068** (0.002)	0.076** (0.004)	0.066** (0.002)	0.057** (0.002)	0.052** (0.002)
Exp	0.016** (0.005)	0.034** (0.002)	0.037** (0.002)	0.041** (0.003)	0.039** (0.003)	0.039** (0.002)	0.043** (0.002)
Exp ²	−0.001** (0.000)	−0.001** (0.000)	−0.001** (0.000)	−0.001** (0.000)	−0.001** (0.000)	−0.001** (0.000)	−0.001** (0.000)
Sex ($M = 1$)	0.194** (0.031)	0.155** (0.018)	0.160** (0.010)	0.204** (0.019)	0.203** (0.015)	0.204** (0.013)	0.137** (0.013)
Urban	0.384** (0.049)	0.188** (0.042)	0.174** (0.041)	0.165** (0.037)	0.205** (0.021)	0.127** (0.019)	0.154** (0.017)
Constant	−0.098 (0.085)	0.259** (0.031)	0.489** (0.026)	0.740** (0.054)	1.334** (0.044)	1.856** (0.037)	2.056** (0.035)
Observations	3,449	5,657	5,955	6,124	7,166	7,581	7,697
Prob > F	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
R-squared	0.220	0.309	0.321	0.315	0.321	0.310	0.306

Robust SEs are in parentheses, corrected for sample weights and sample clusters, statistically significant at 10 per cent (+), at 5 per cent (*), and at 1 per cent (**); all models control for 8 region dummies; dependent variable is hourly wage in log, which is measured in VND 1,000 (and for all tables hereafter).

Table 2.2a: Baseline Estimates using OLS

Source: Tran and Pham, 2019

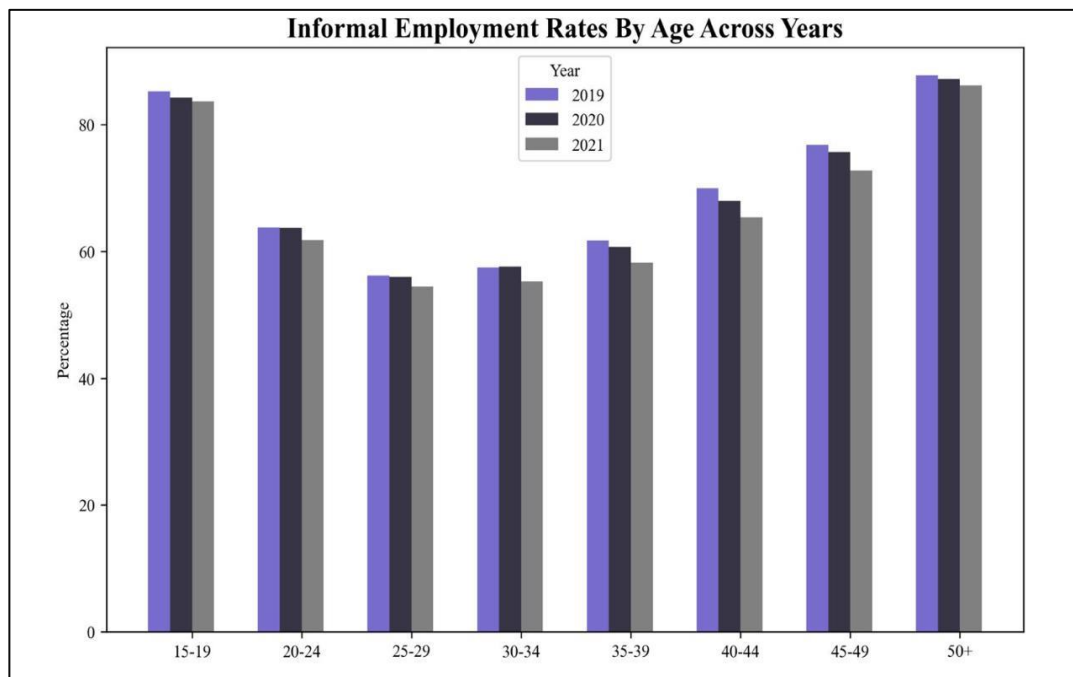


Figure 2.2h: Informal Employment

Source: World Development Indicators

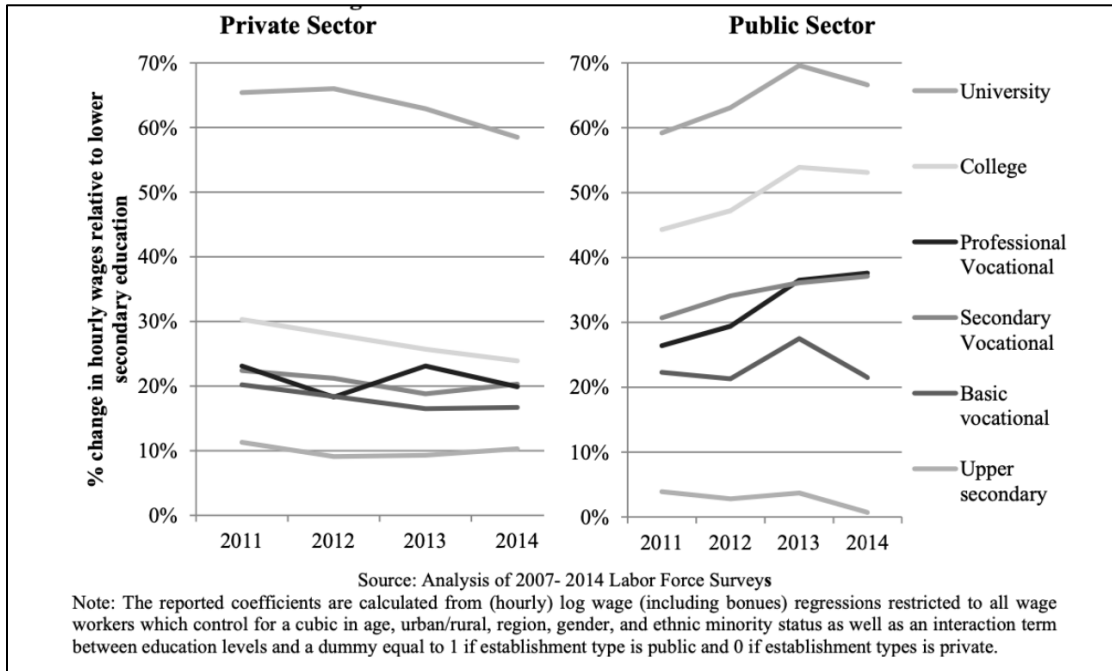


Figure 2.2i: Returns to Education Over Time

Source: Analysis of LFS

Explanatory variables	OLS	Simultaneous Quantile Regression Estimator				
	Estimator	10th Quantile	25th Quantile	50th Quantile	75th Quantile	90th Quantile
Mismatch	-0.29*** (0.021)	-0.55*** (0.044)	-0.31*** (0.021)	-0.22*** (0.017)	-0.19*** (0.022)	-0.11*** (0.027)
Age	0.06*** (0.003)	0.07*** (0.003)	0.06*** (0.003)	0.05*** (0.002)	0.05*** (0.003)	0.05*** (0.003)
Age squared	-0.00*** (0.000)	-0.00*** (0.000)	-0.00*** (0.000)	-0.00*** (0.000)	-0.00*** (0.000)	-0.00*** (0.000)
Gender	0.16*** (0.012)	0.06*** (0.014)	0.10*** (0.010)	0.15*** (0.010)	0.19*** (0.011)	0.20*** (0.019)
Public sector	-0.26*** (0.016)	-0.16*** (0.016)	-0.17*** (0.012)	-0.22*** (0.015)	-0.29*** (0.018)	-0.37*** (0.027)
SOEs	-0.02 (0.022)	0.01 (0.027)	0.02 (0.020)	-0.02 (0.021)	-0.05** (0.026)	-0.09** (0.039)
FDI	0.20*** (0.031)	0.28*** (0.035)	0.25*** (0.029)	0.20*** (0.024)	0.23*** (0.037)	0.22*** (0.053)
Others	0.10 (0.151)	-0.22 (0.396)	-0.25** (0.110)	-0.16 (0.186)	-0.01 (0.189)	0.20 (0.243)
Red River	0.12*** (0.019)	0.06*** (0.024)	0.06*** (0.016)	0.05*** (0.016)	0.06*** (0.019)	0.08** (0.030)
North Central Coast	-0.02 (0.020)	-0.05** (0.022)	-0.05*** (0.015)	-0.06*** (0.013)	-0.06*** (0.016)	-0.02 (0.025)
Highlands	0.06*** (0.023)	0.05* (0.027)	0.03* (0.019)	-0.01 (0.018)	-0.00 (0.025)	0.01 (0.027)
South East	0.24*** (0.021)	0.16*** (0.026)	0.16*** (0.017)	0.16*** (0.019)	0.19*** (0.023)	0.22*** (0.036)
Mekong Delta	0.03 (0.021)	-0.00 (0.027)	-0.02 (0.016)	-0.06*** (0.015)	-0.08*** (0.019)	-0.06** (0.029)
Urban region	0.12*** (0.014)	0.07*** (0.016)	0.08*** (0.011)	0.08*** (0.011)	0.09*** (0.013)	0.10*** (0.017)
Log PCI	0.66*** (0.138)	0.44*** (0.152)	0.24* (0.121)	0.42*** (0.122)	0.49*** (0.151)	0.55*** (0.206)
Constant	5.61*** (0.569)	6.03*** (0.630)	7.10*** (0.499)	6.61*** (0.502)	6.59*** (0.623)	6.61*** (0.842)
R2/Pseudo R2	0.33	0.20	0.17	0.18	0.19	0.23
Observations	10,802	10,802	10,802	10,802	10,802	10,802

Notes: * significant at 10%; ** significant at 5%; *** significant at 1%. Bootstrapped standard errors (100 replications) in parentheses. The omitted categories in the dummy variable analyses are matched education, female sex, wage employment in the private sector, the midlands and northern mountains, and rural region. SOE: state owned enterprises; foreign direct investment (FDI).

Table 2.2b: The Impact of Educational Mismatch on Wage Earnings

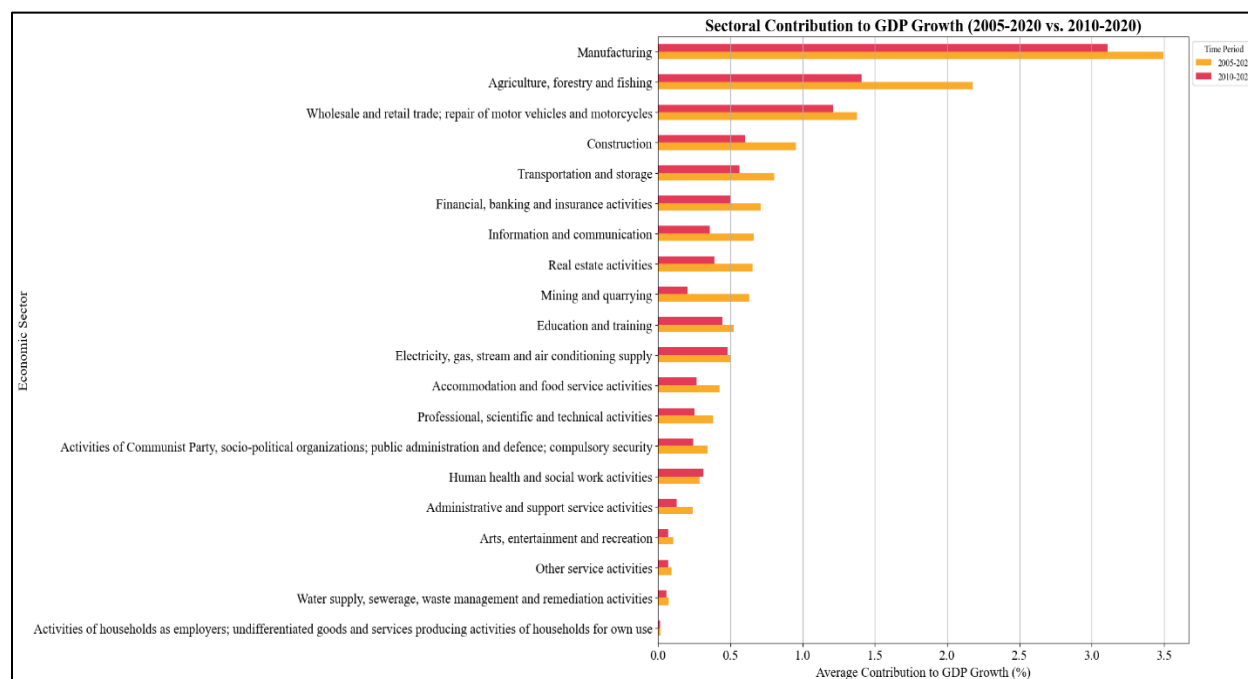


Figure 2.2j: Sectoral Contribution to GDP Growth

Source: Own Computation Using GSO Data

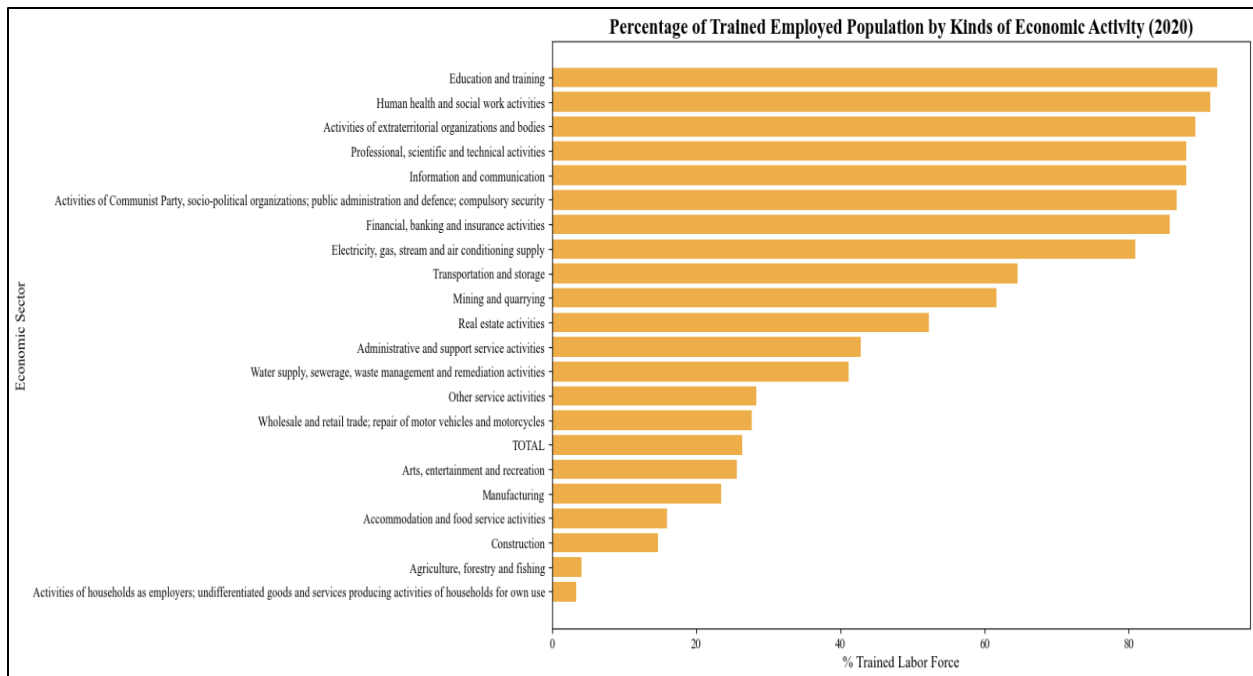


Figure 2.2k: Percentage of Trained Employees by Economic Activity

Source: Own Computation Using GSO Data

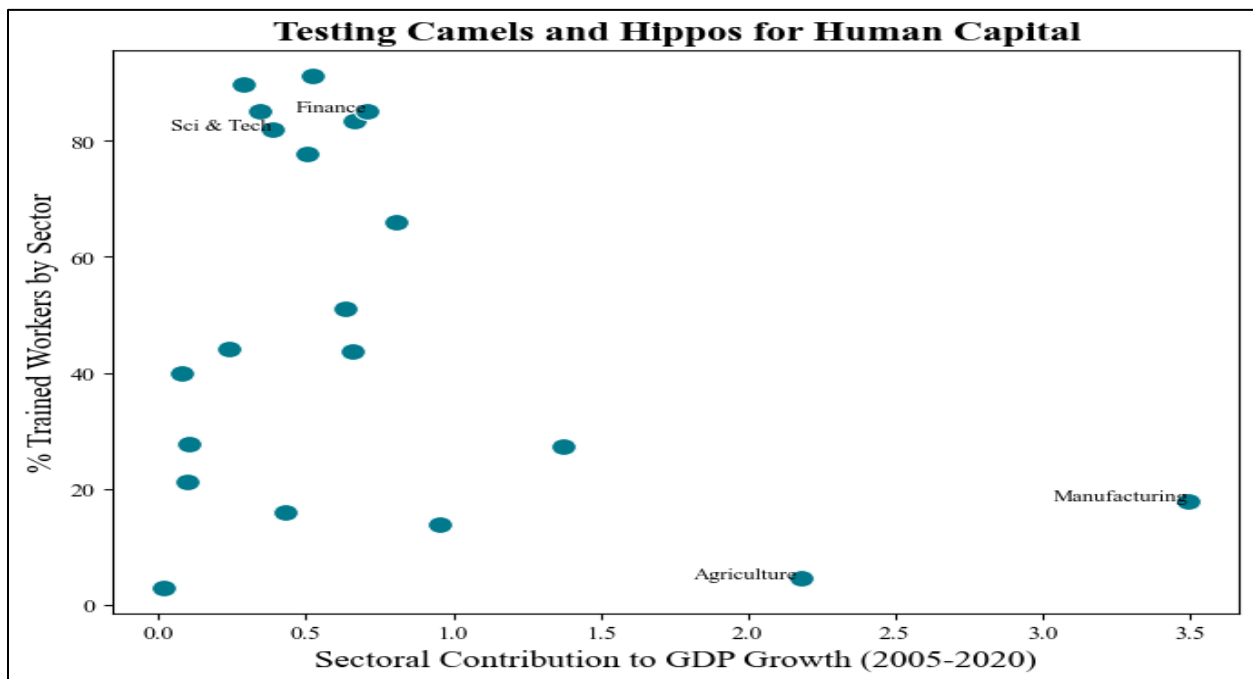


Figure 2.2l: Camels and Hippos, Human Capital

Source: Own Computation Using GSO Data

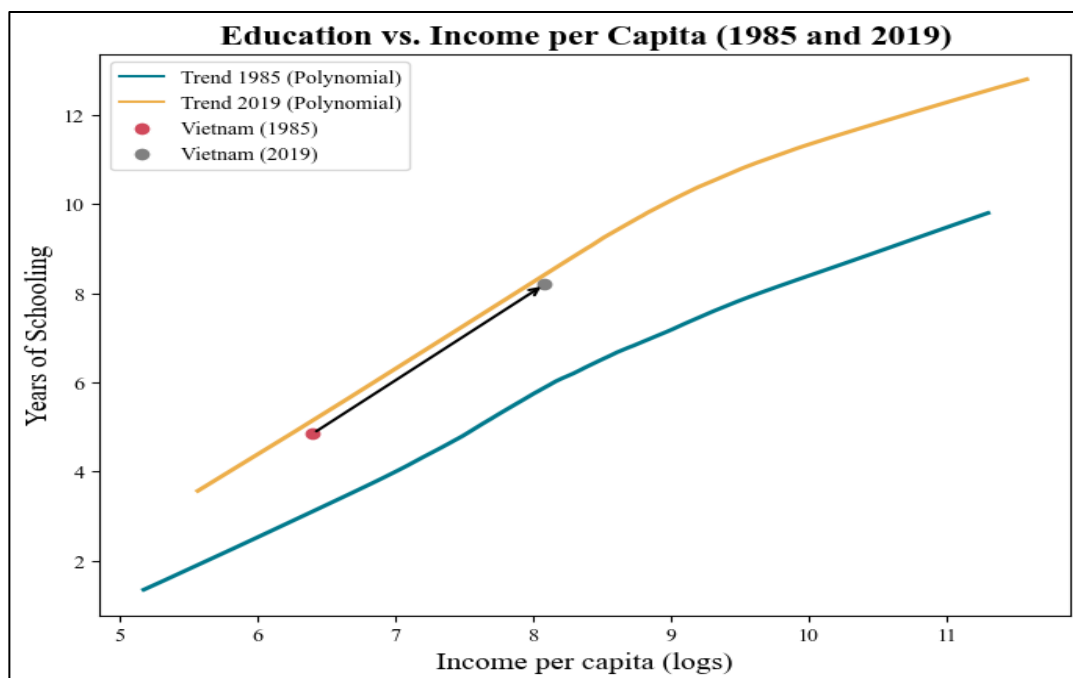


Figure 2.2m: Education v/s Income Per Capita, Vietnam

Source: World Development Indicators

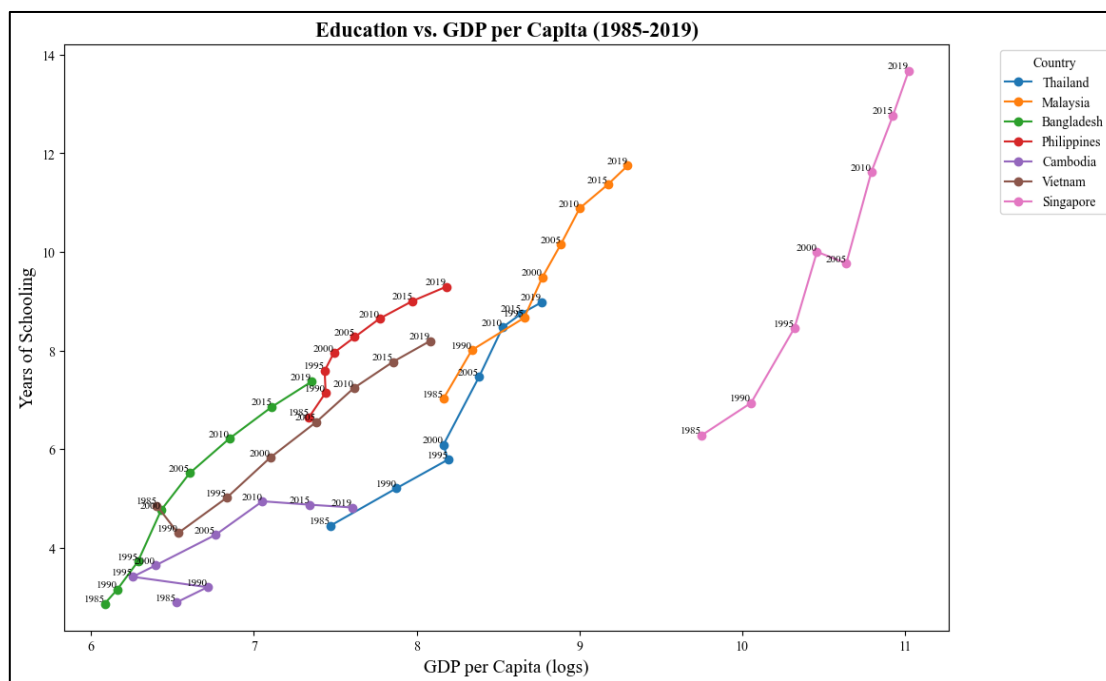


Figure 2.2n: Movements-in-Movements (Years of Schooling), Vietnam

Source: World Development Indicators

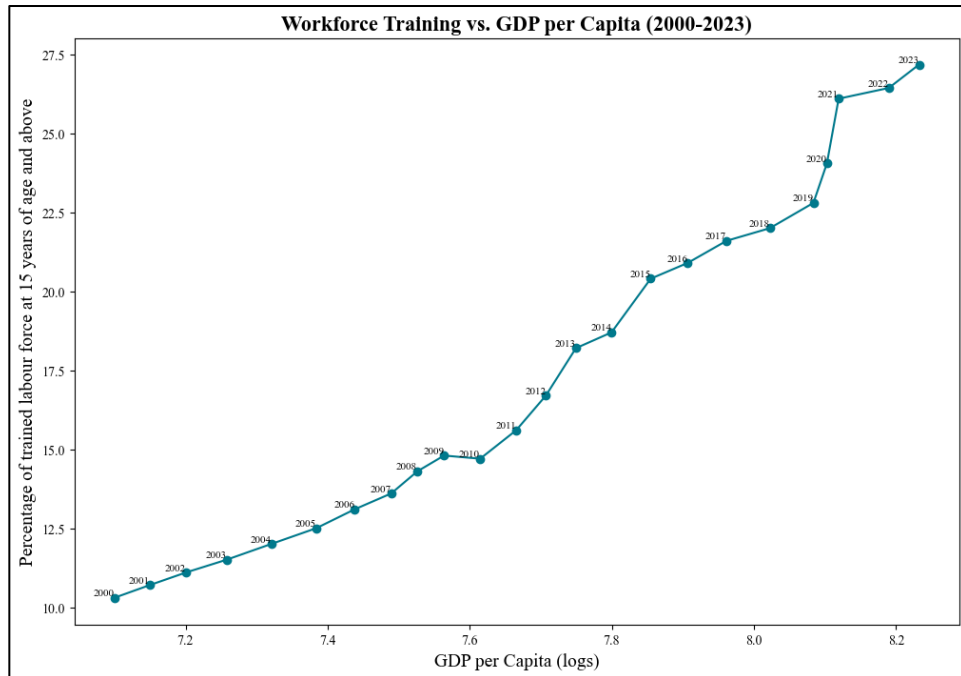


Figure 2.2o: Workforce Training v/s GDPPC, Vietnam

Source: General Statistics Office

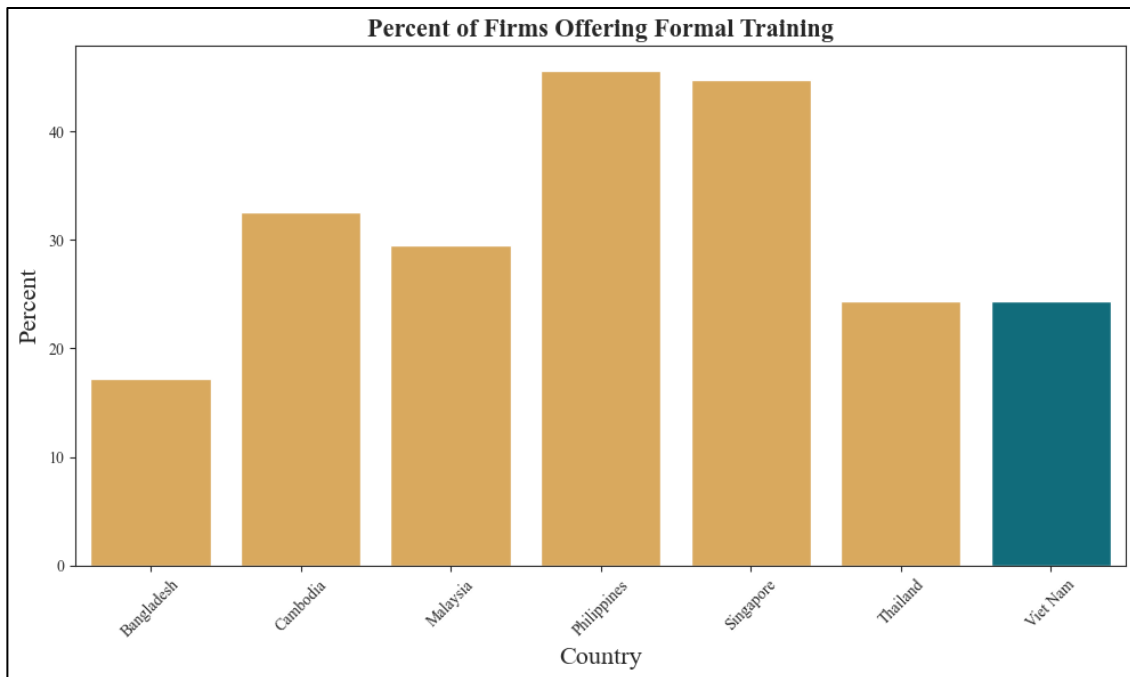


Figure 2.2p: Percentage of Firms Offering Formal Training, Vietnam and Peers

Source: World Bank Enterprise Survey

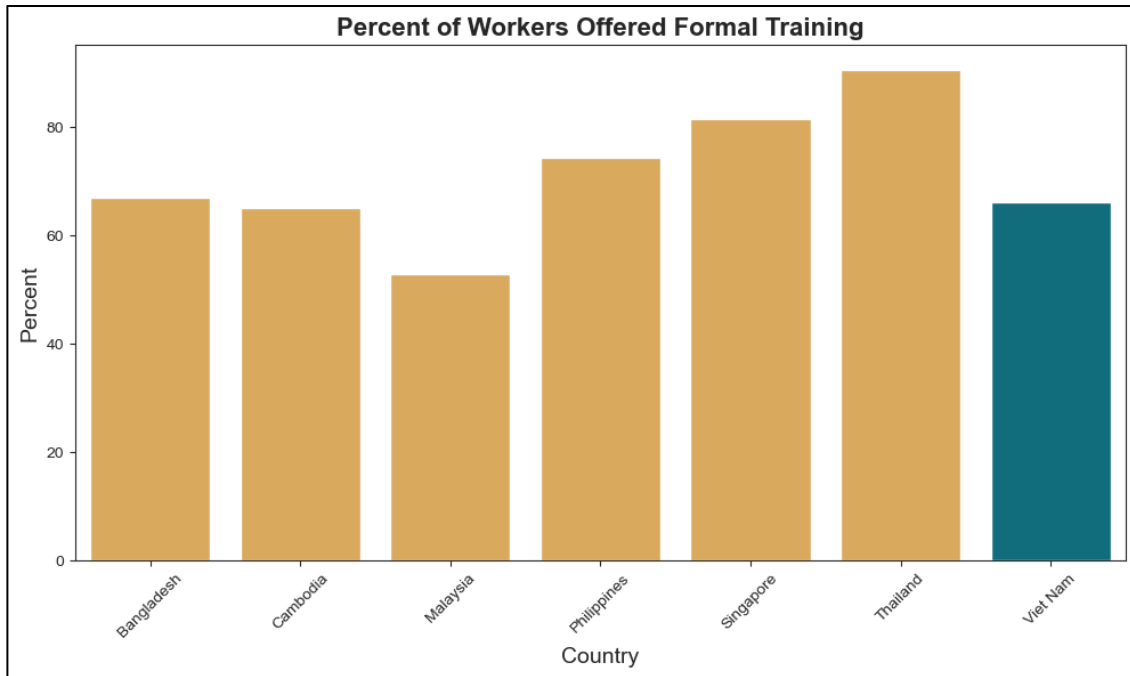


Fig 2.2q: Percentage of Workers Offered Formal Training, Vietnam and Peers

Source: World Bank Enterprise Survey

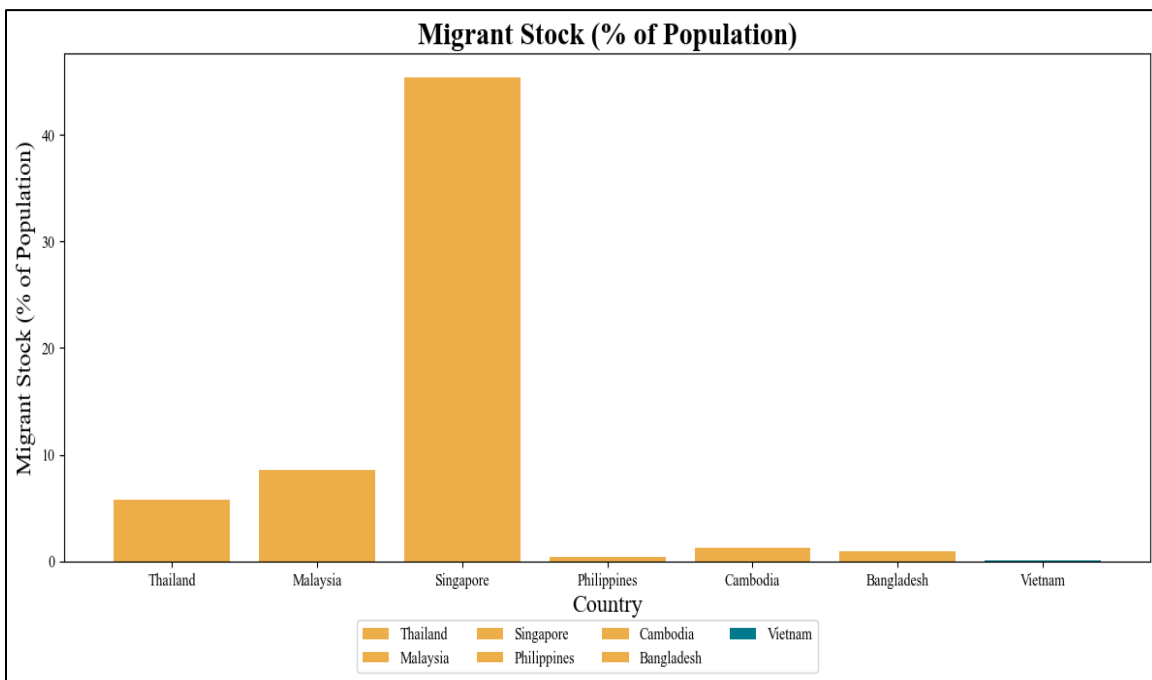


Figure 2.2r: Migrant Stock as A Percentage of Population, Vietnam and Peers

Source: World Development Indicators

Appendix 2.3: Access to Finance

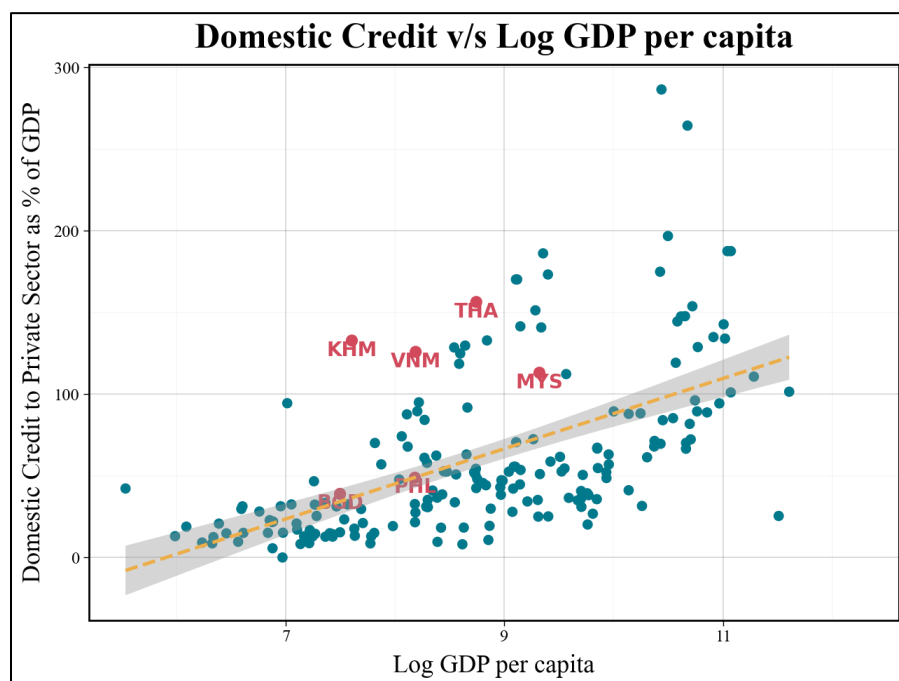


Figure 2.3a: Domestic Credit v/s Log GDPPC

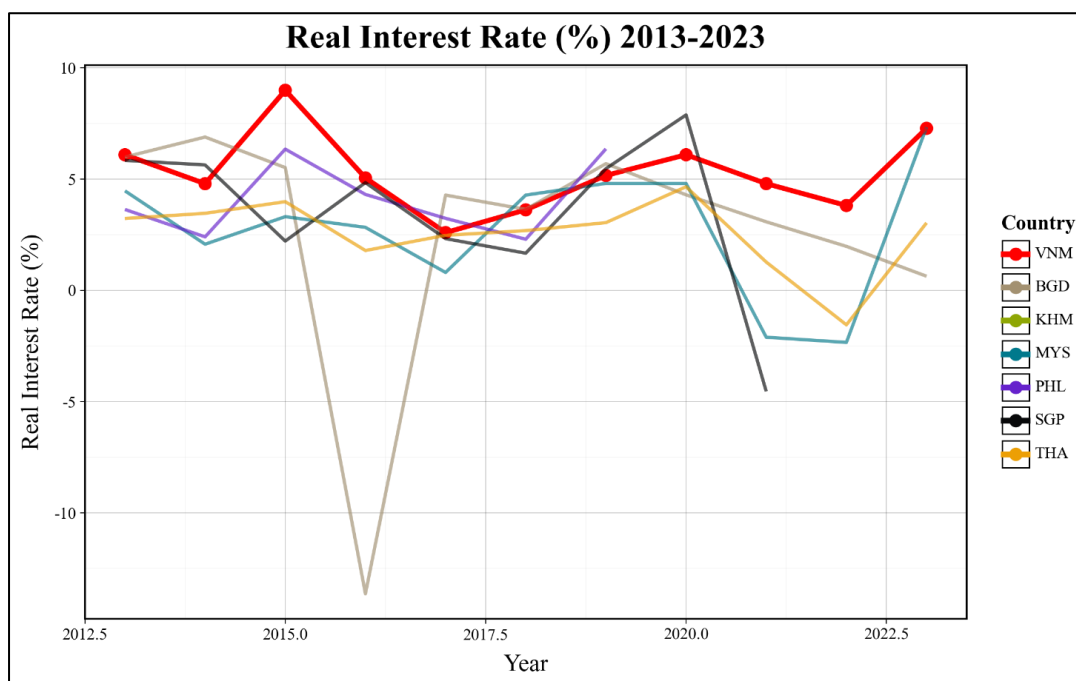


Figure 2.3b: Real Interest Rate, Vietnam and Peers

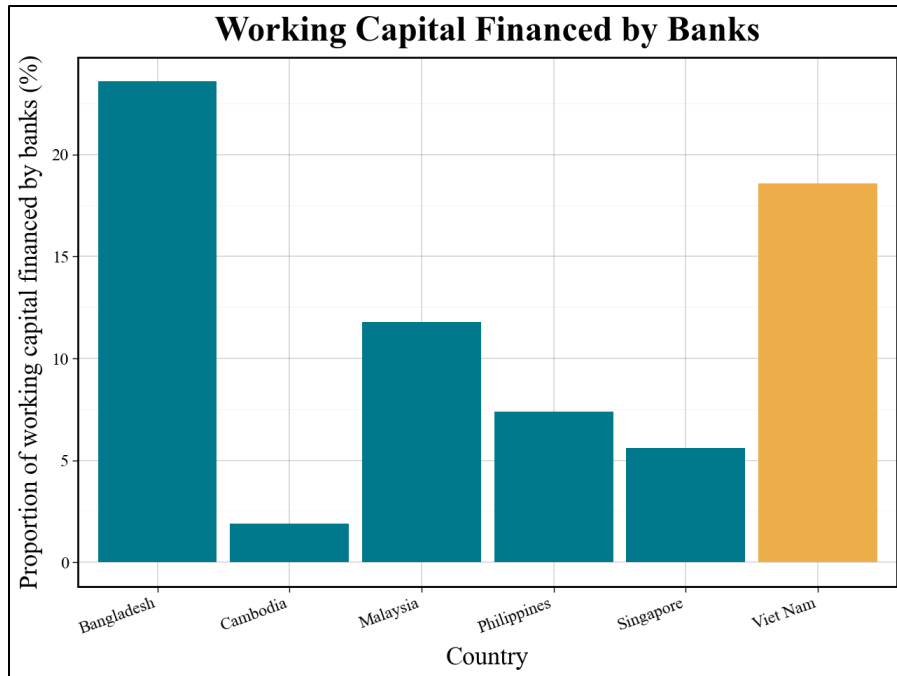


Figure 2.3c: Working Capital Financed by Banks, Vietnam and Peers

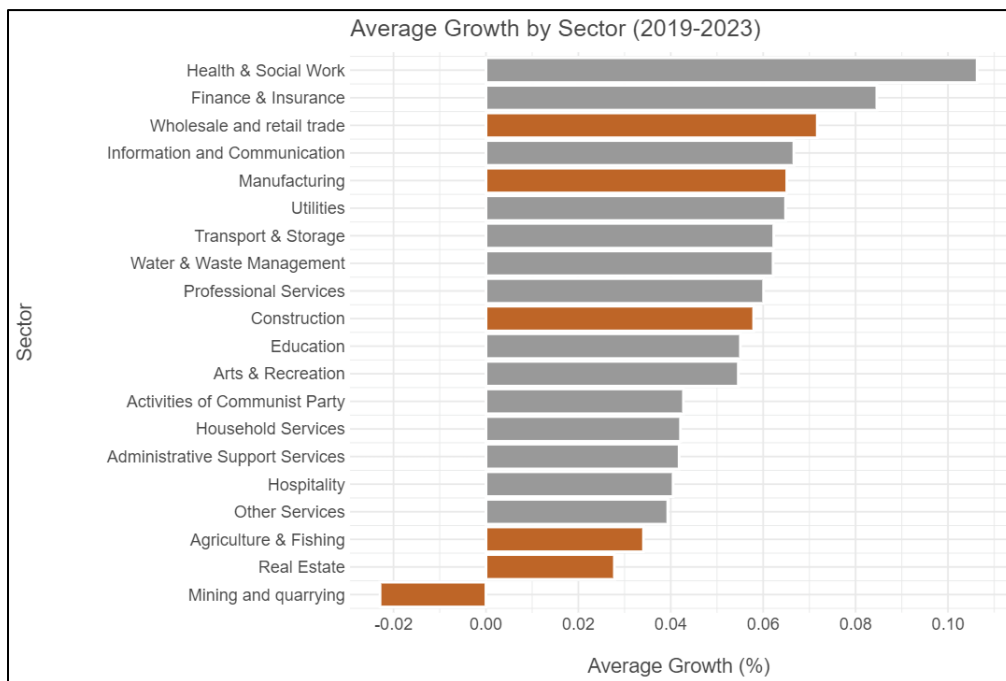
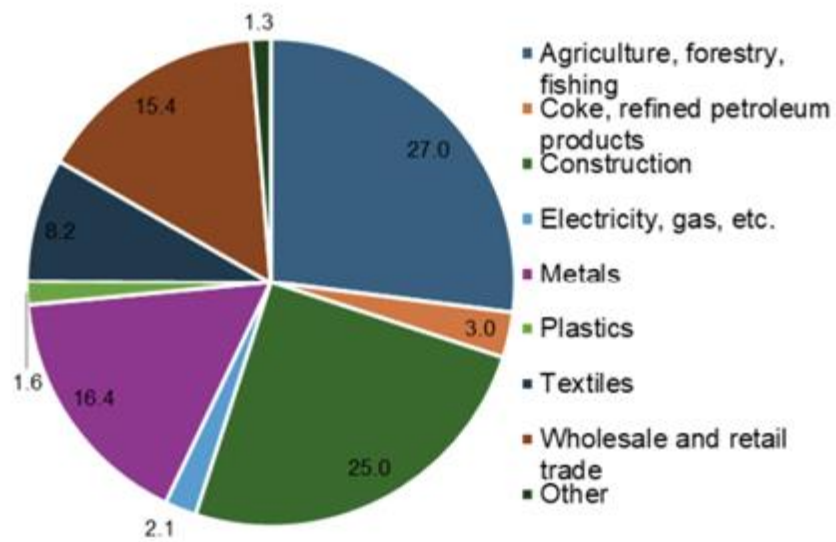


Figure 2.3d: Average Growth by Sector, Vietnam

Figure A3.4. Trade Finance Usage by Sector



Source: WTO and IFC (2024), AMRO staff calculations.

Figure 2.3e: Trade Finance Usage by Sector

Source: World Trade Organisation

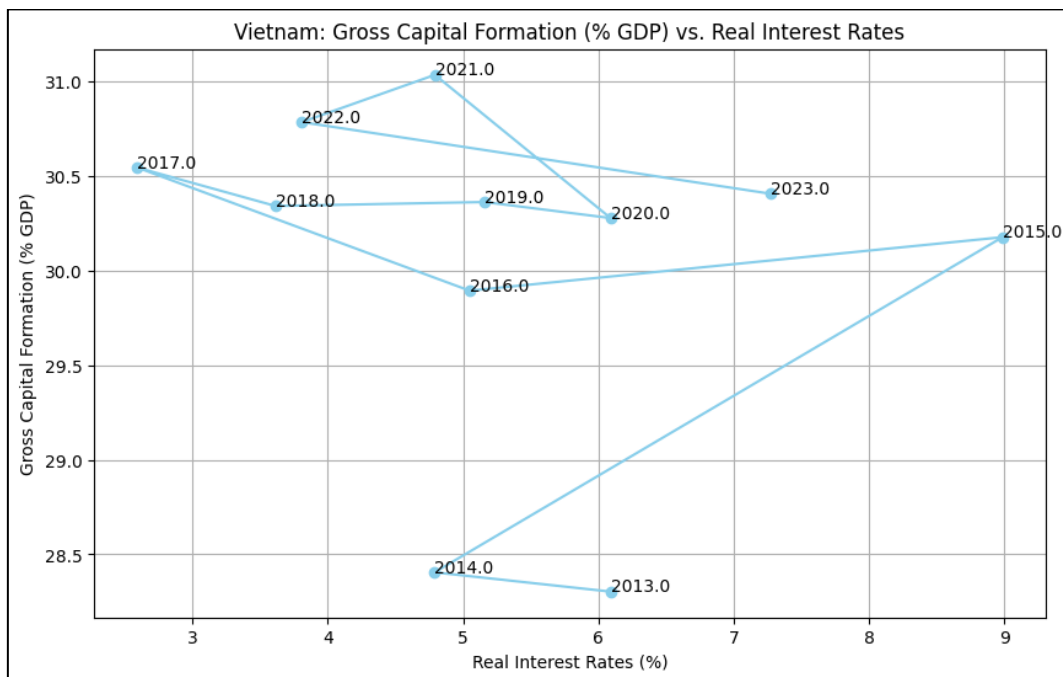


Figure 2.3f: Movements-in-Movements (Gross Capital Formation), Vietnam

Source: World Bank

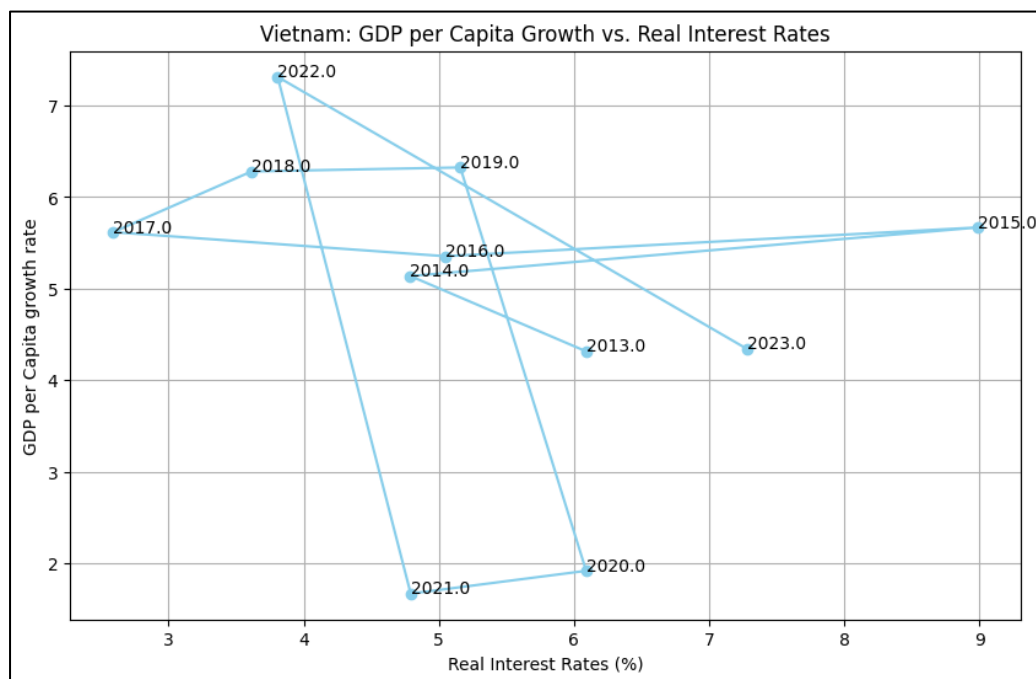


Figure 2.3g: Movements-in-Movements (GDPPC), Vietnam

Source: World Bank

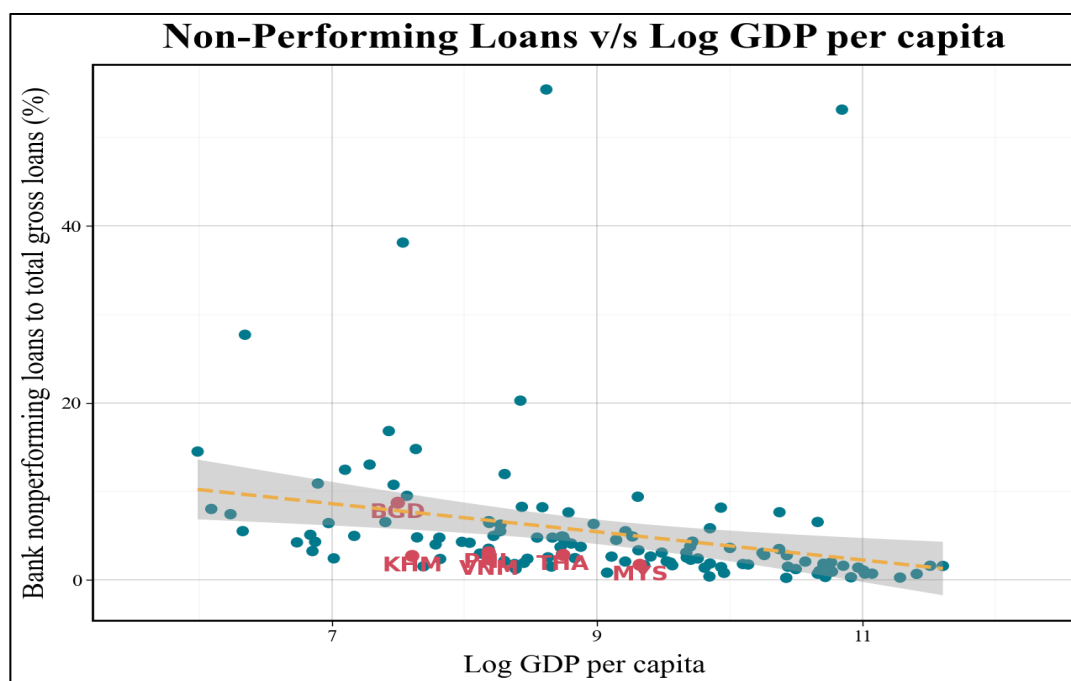


Figure 2.3h: NPL v/s GDPPC

Date	Government Bonds (LCY billions)	Corporate Bonds (LCY billions)	Total (LCY billions)
Mar-2022	723,210.79	0.00	723,210.79
Jun-2022	493,632.20	0.00	493,632.20
Sep-2022	356,135.78	0.00	356,135.78
Dec-2022	215,465.85	0.00	215,465.85
Mar-2023	278,514.94	0.00	278,514.94
Jun-2023	401,350.21	0.00	401,350.21
Sep-2023	378,081.19	0.00	378,081.19
Dec-2023	498,613.29	0.00	498,613.29
Mar-2024	561,724.90	0.00	561,724.90
Jun-2024	670,542.00	0.00	670,542.00
Sep-2024	751,207.05	0.00	751,207.05
Dec-2024	845,838.63	1.00	845,839.63

Table 2.3a: Bonds

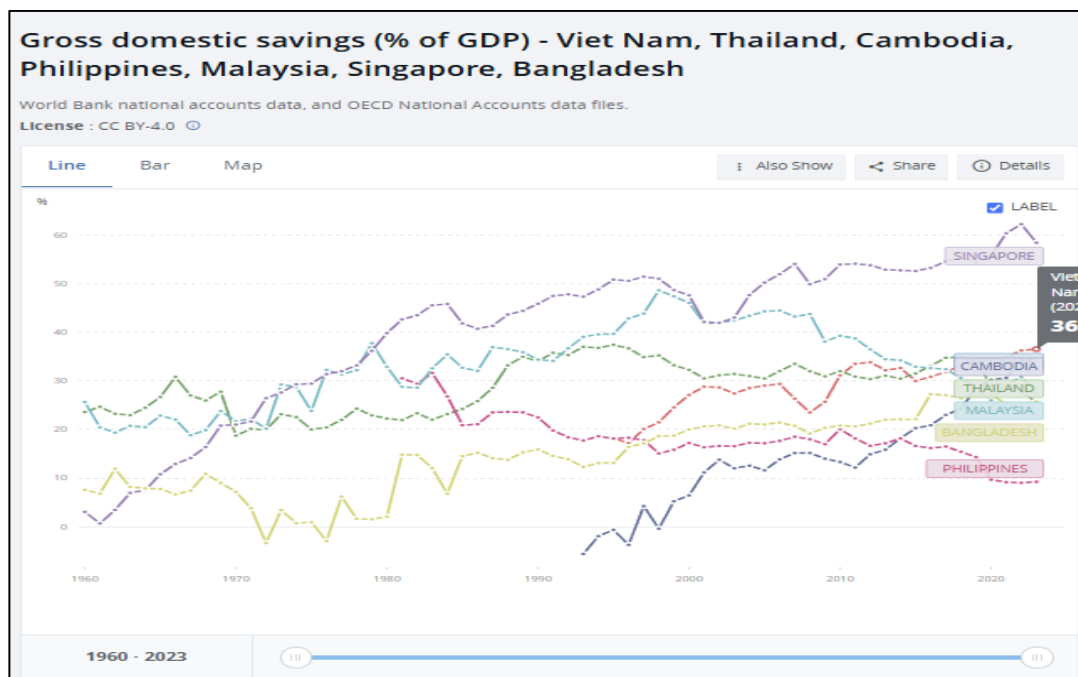


Figure 2.3i: Gross Domestic Savings (% of GDP), Vietnam and Peers

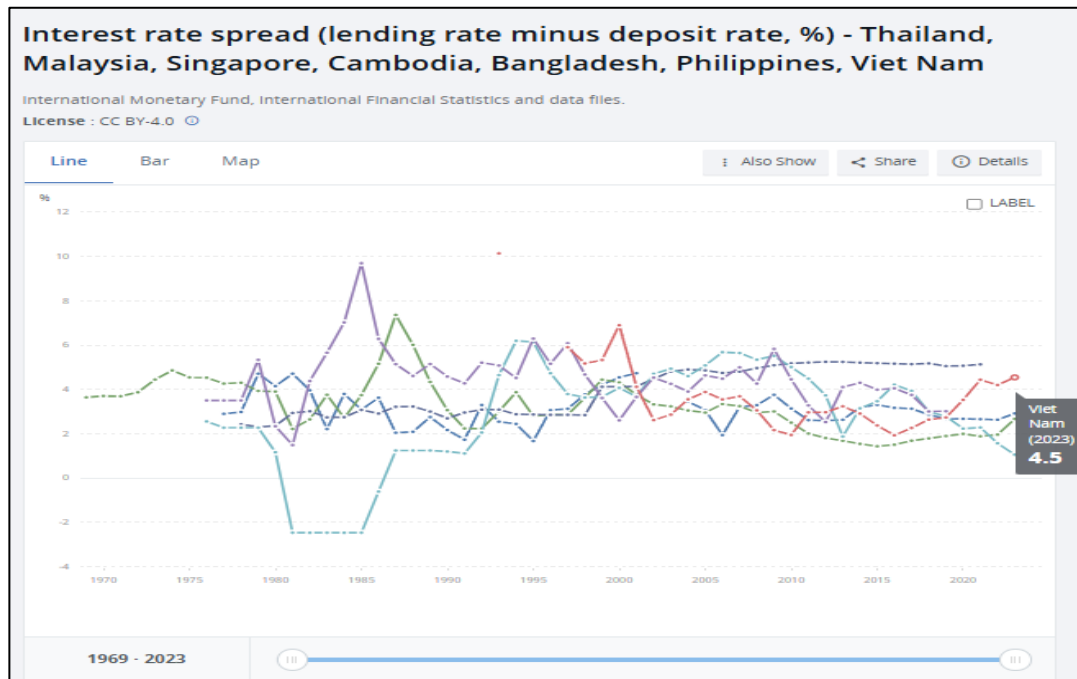


Figure 2.3j: Interest Rate Spread, Vietnam and Peers

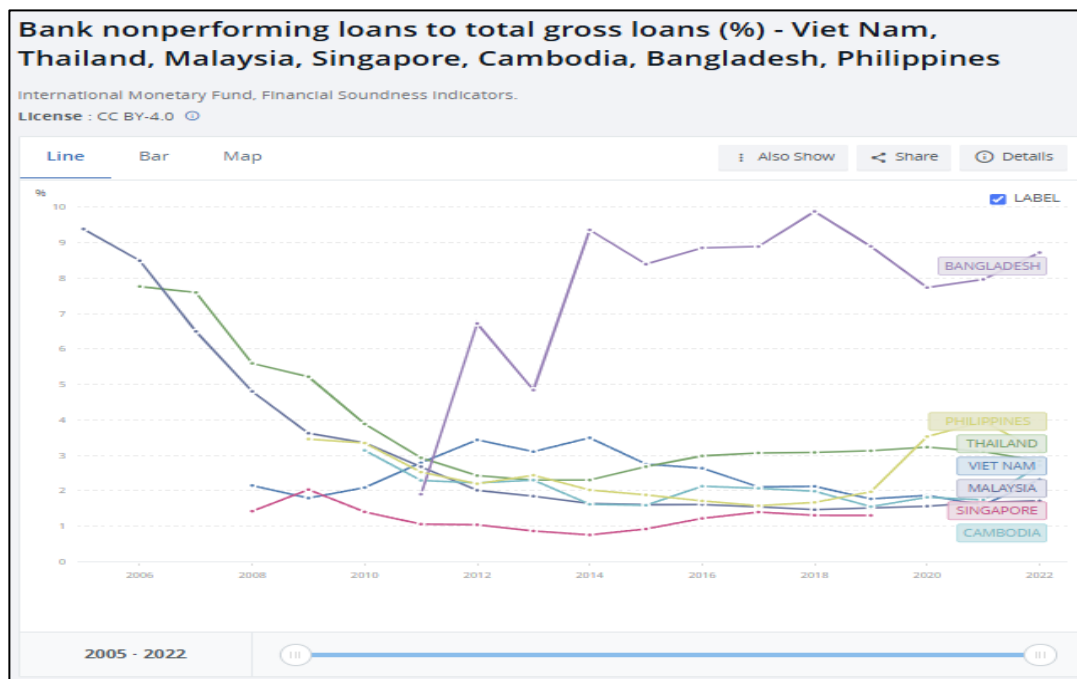


Figure 2.3k: Bank non-performing loans, Vietnam and Peers

Appendix 2.4: Macroeconomic Risks

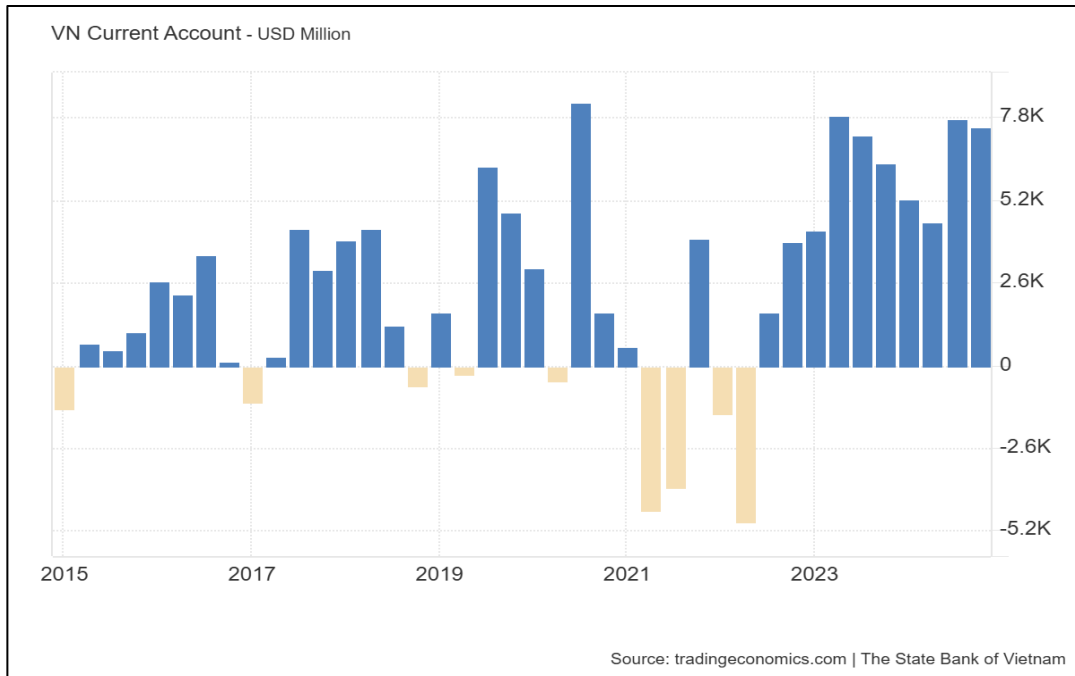


Figure 2.4a: Current Account, Vietnam

Source: The State Bank of Vietnam

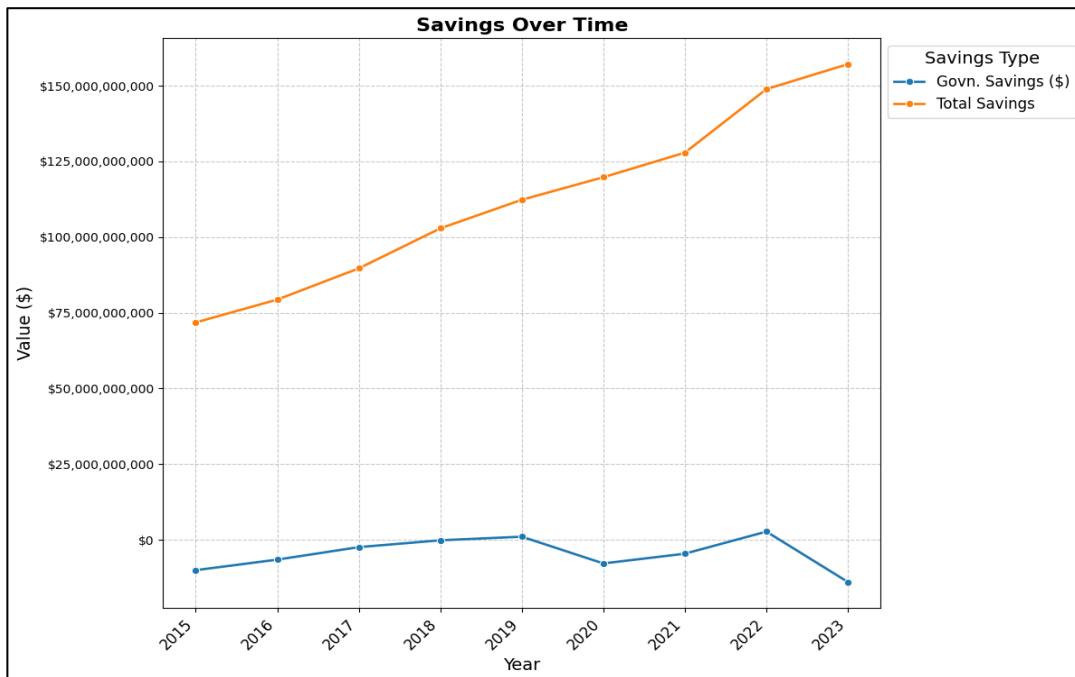


Figure 2.4b: Savings Over Time, Vietnam

Source: Trading Economics

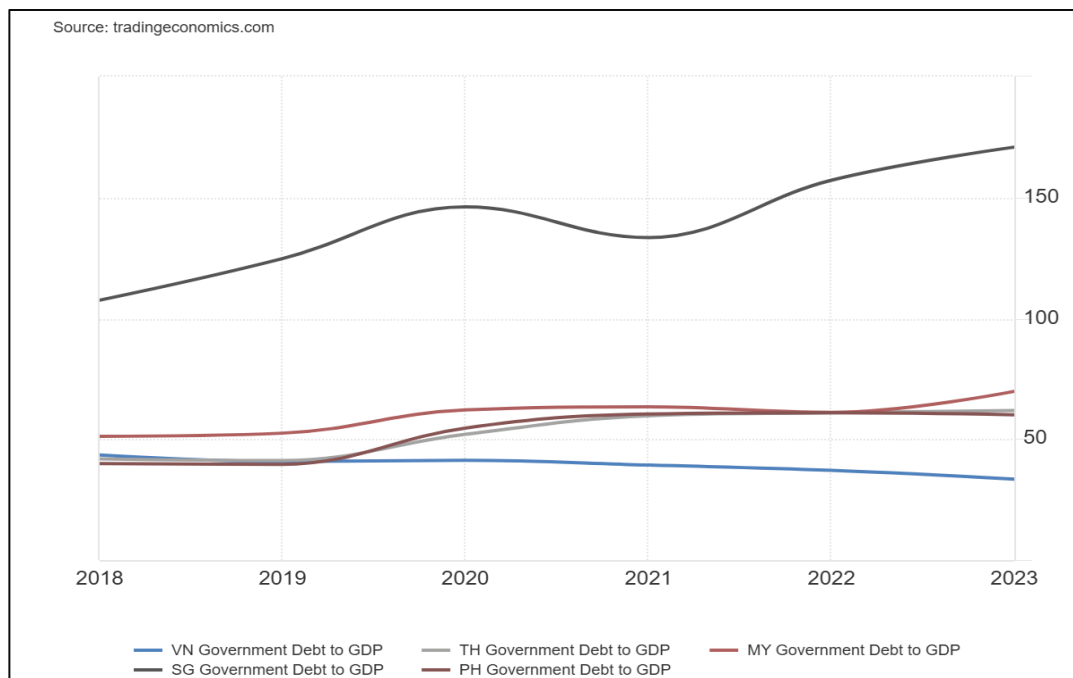


Figure 2.4c: Debt to GDP Ratio, Vietnam and Peers

Source; Trading Economics

Appendix 2.5: Appropriation

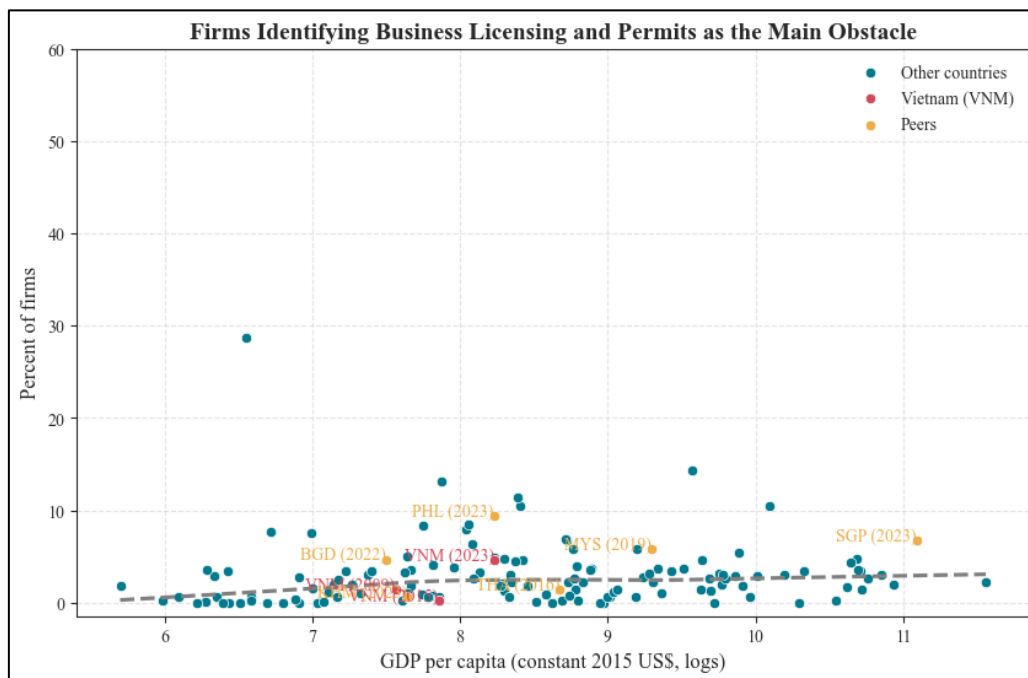


Figure 2.5a: Business Licensing and Permits as Main Obstacle

Source: World Bank Enterprise Survey and World Development Indicators

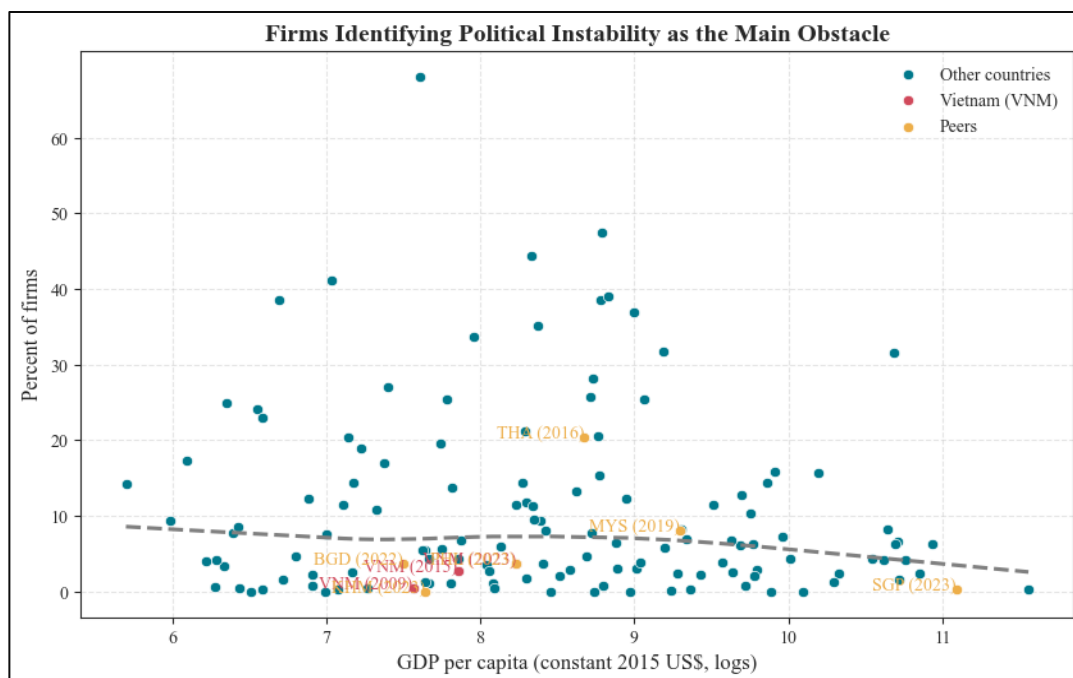


Figure 2.5b: Political Instability as Main Obstacle

Source: World Bank Enterprise Survey and World Development Indicators

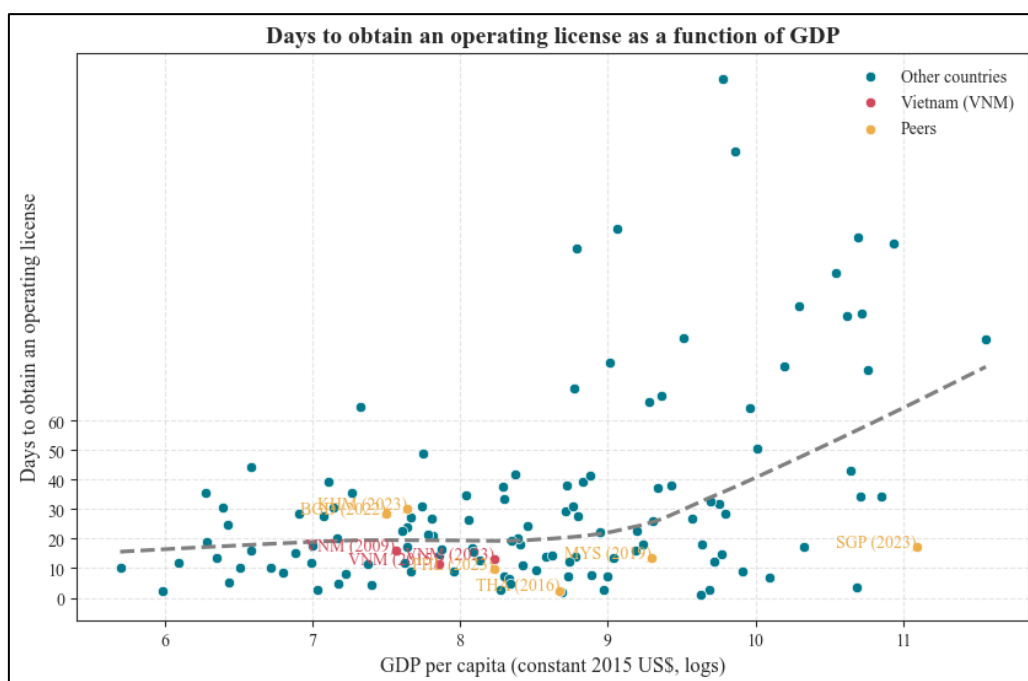


Figure 2.5c: Days to Obtain Operating License

Source: World Bank Enterprise Survey and World Development Indicators

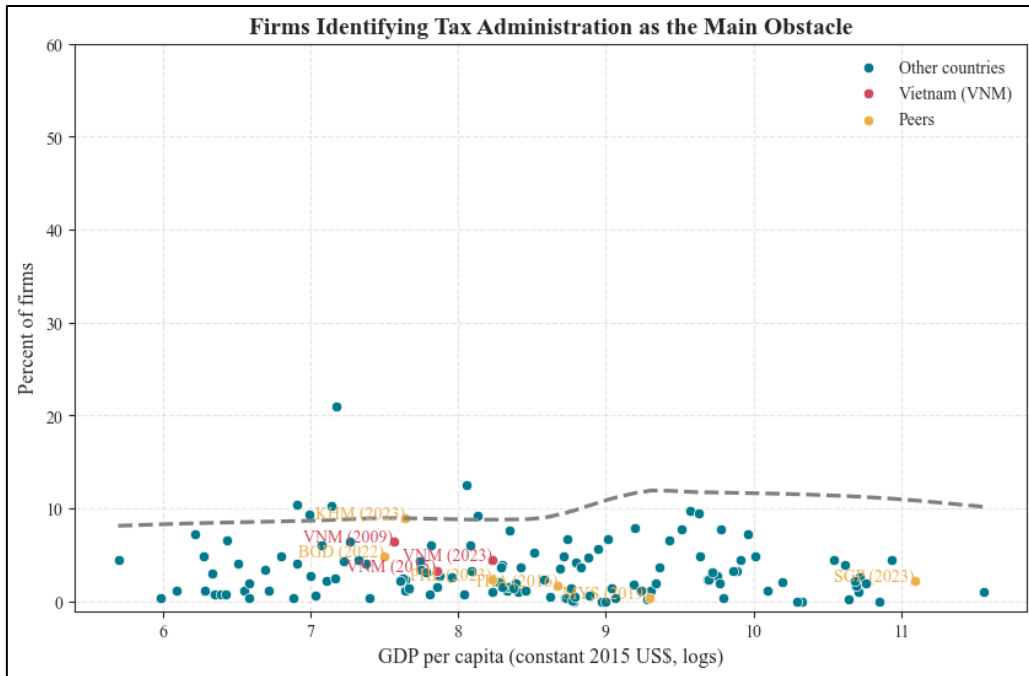


Figure 2.5d: Tax Administration as Main Obstacle

Source: World Bank Enterprise Survey and World Development Indicators

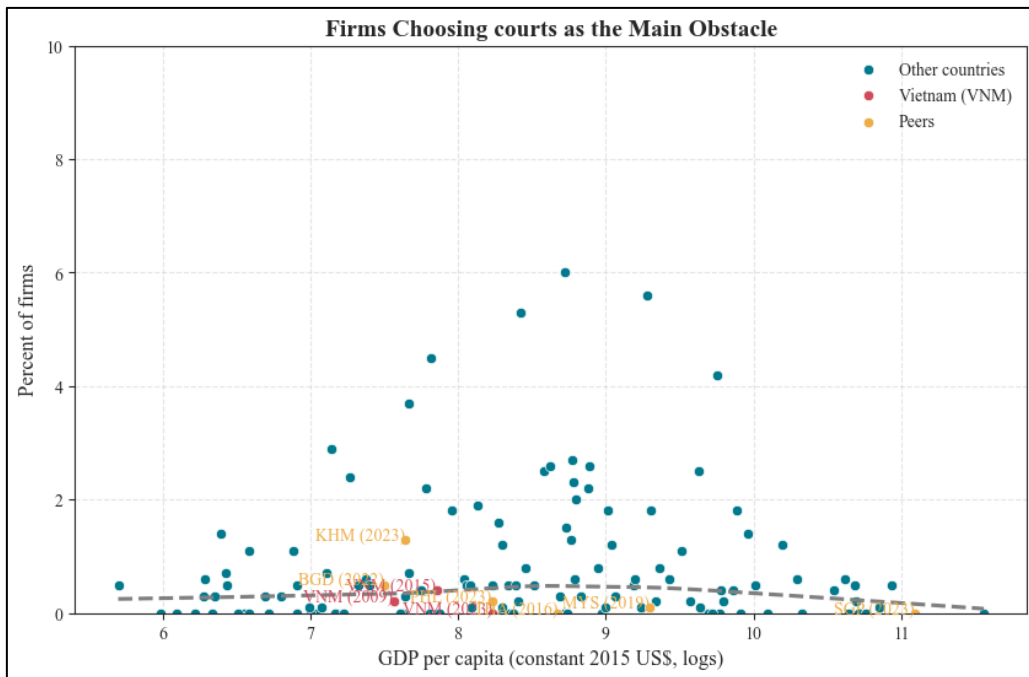


Figure 2.5e: Courts as Main Obstacle

Source: World Bank Enterprise Survey and World Development Indicators

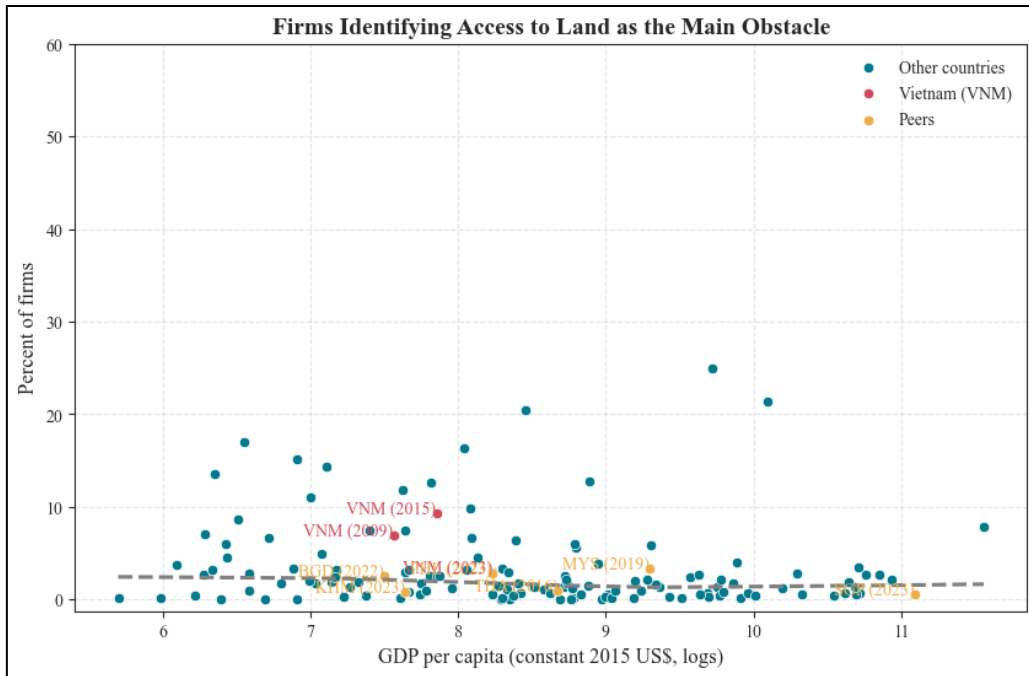


Figure 2.5f: Access to Land as Main Obstacle

Source: World Bank Enterprise Survey and World Development Indicators

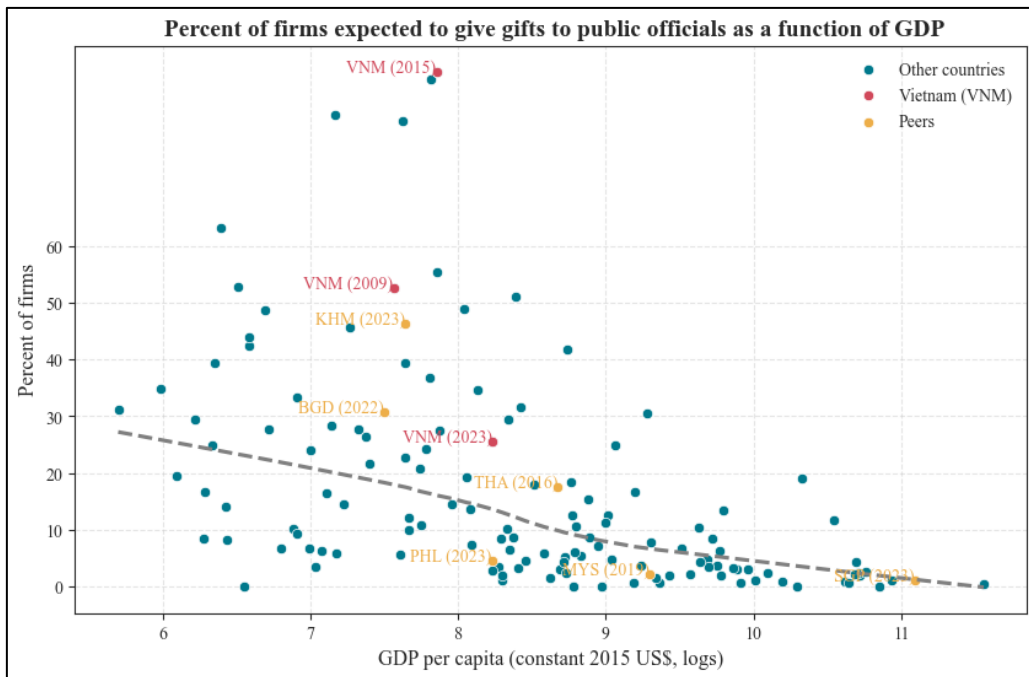


Figure 2.5g: Percentage of Firms Expected to Give Gifts to Public Officials

Source: World Bank Enterprise Survey and World Development Indicators

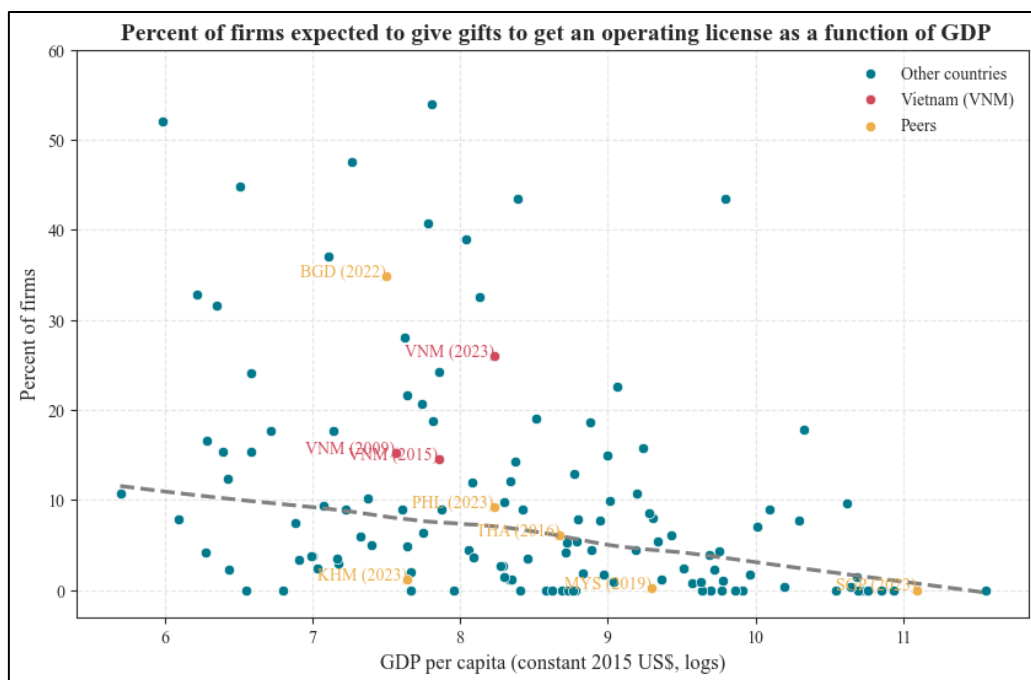


Figure 2.5h: Percentage of Firms Expected to Give Gifts to Get an Operating License

Source: World Bank Enterprise Survey and World Development Indicators

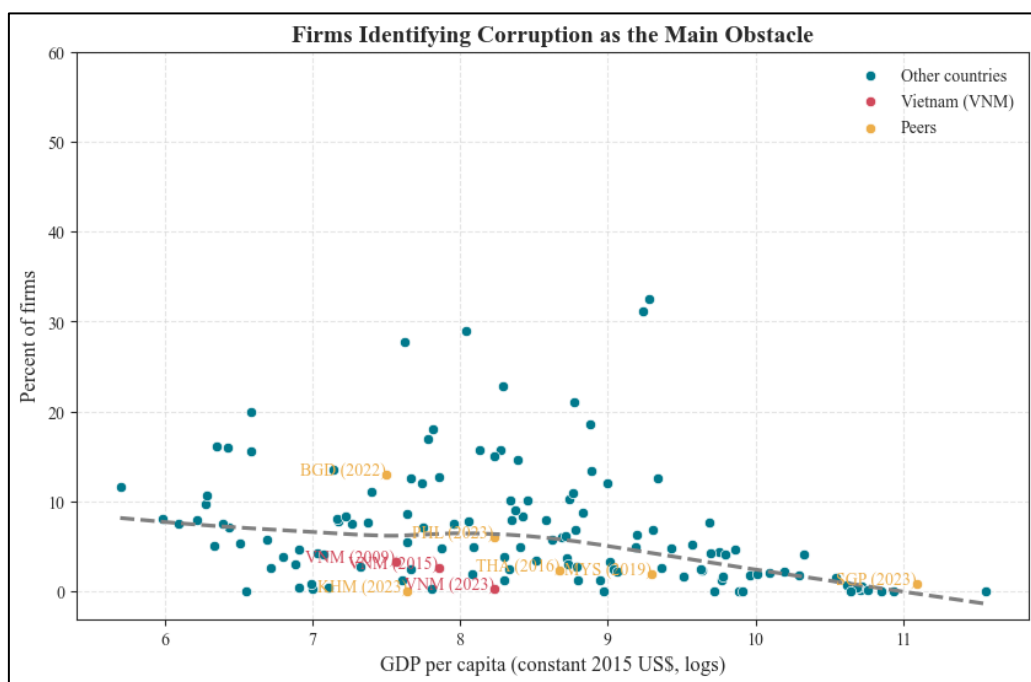


Figure 2.5i: Corruption as Main Obstacle

Source: World Bank Enterprise Survey and World Development Indicators

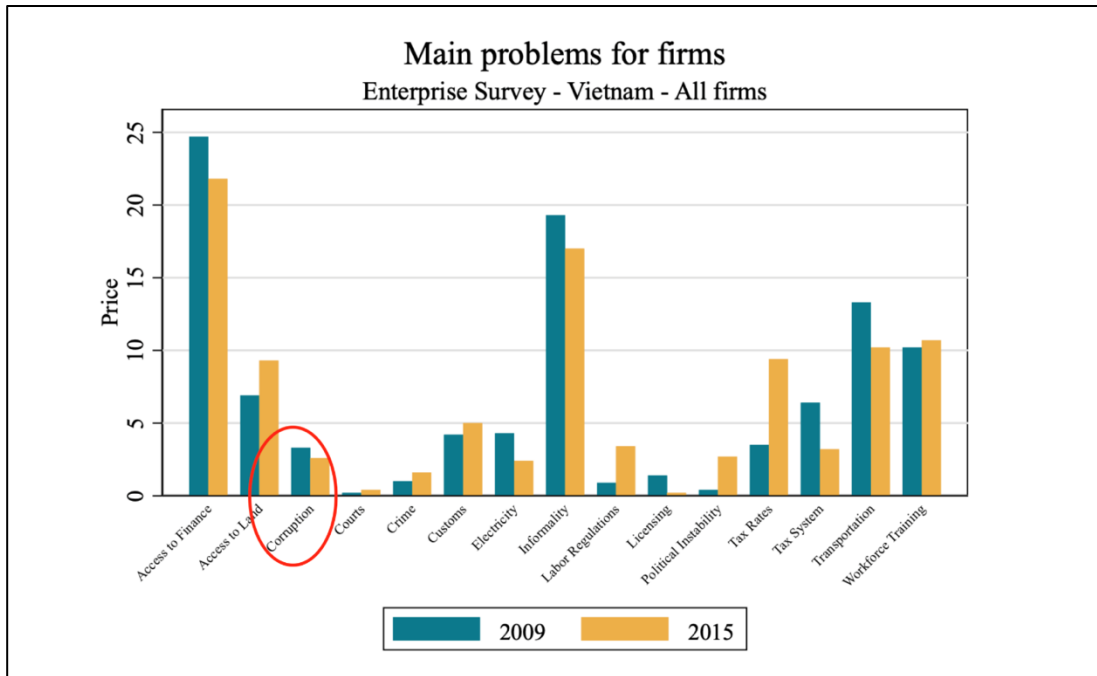


Figure 2.5j: Main Problems for All Firms

Source: World Bank Enterprise Survey

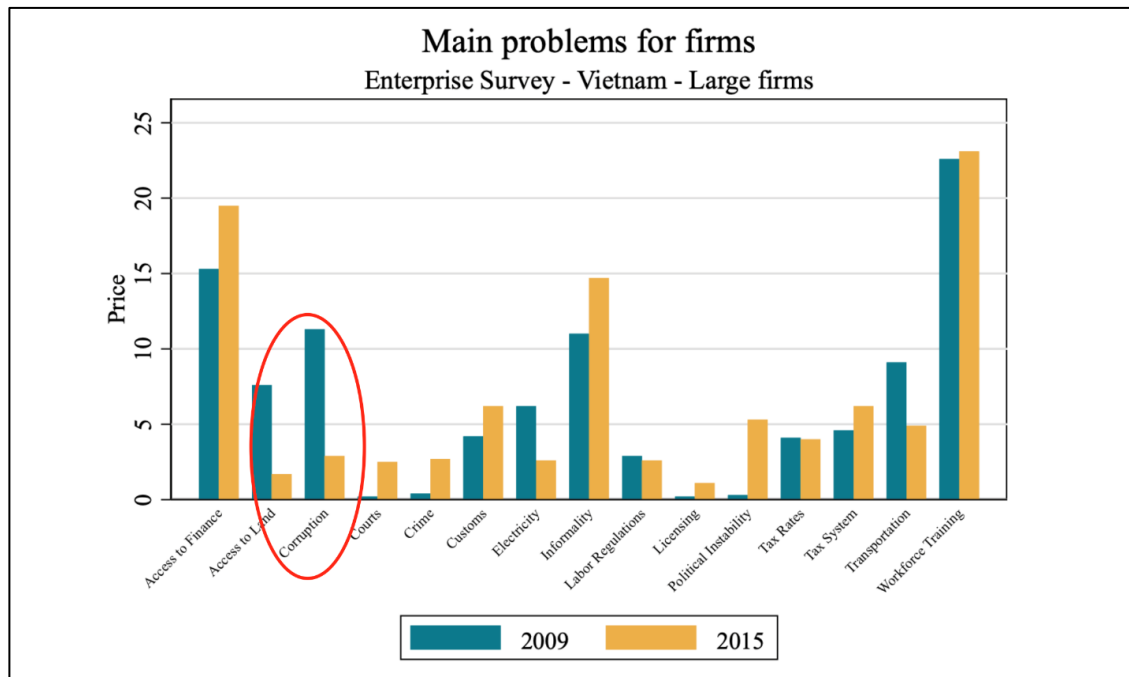


Figure 2.5k: Main Problems for Large Firms

Source: World Bank Enterprise Survey

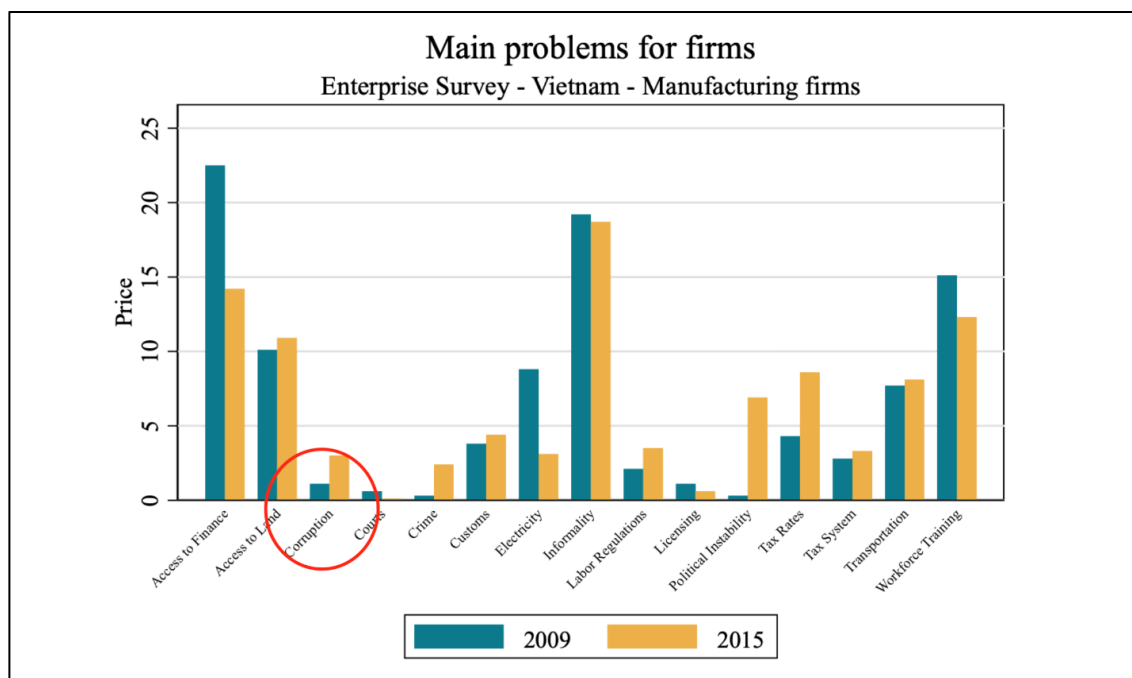


Figure 2.5l: Main Problems for Manufacturing Firms

Source: World Bank Enterprise Survey

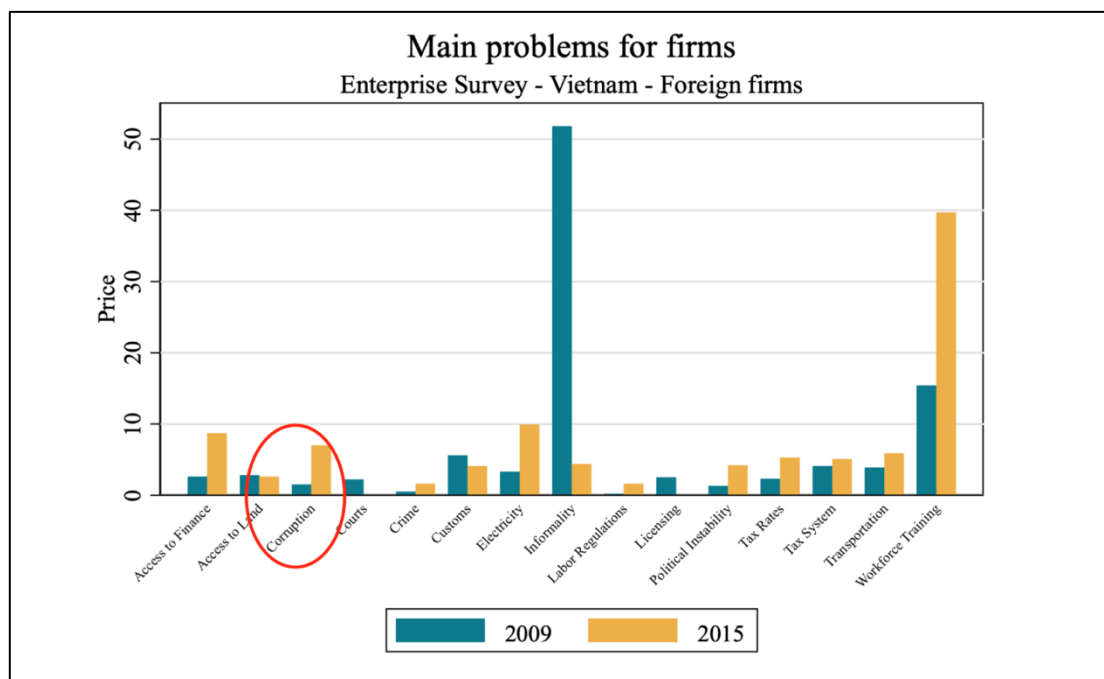


Figure 2.5m: Main Problems for Foreign Firms

Source: World Bank Enterprise Survey

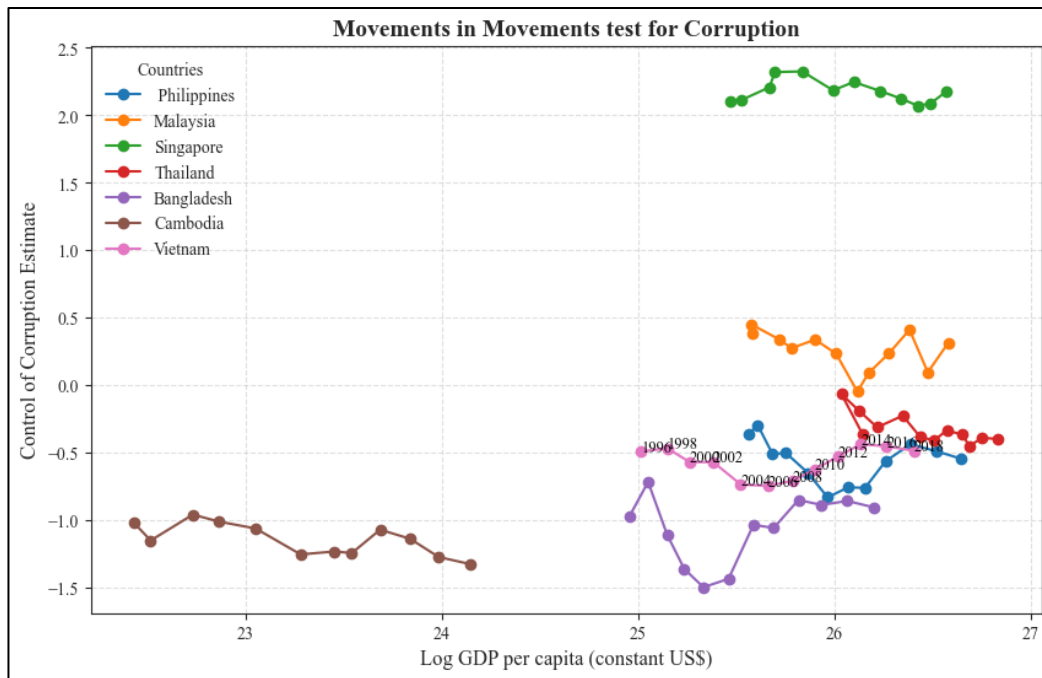


Figure 2.5n: Movements-in-Movements (Control of Corruption), Vietnam and Peers

Source: World Bank Governance Indicators

Appendix 3: Growth Syndrome

Alternate Causal Chains Explored

We explored the following causal chains apart from the one chosen for the report. The one chosen was most plausible, based on available data and headlines.

1. Vietnam growing too fast >> Poor allocation of transport infrastructure investment >> Congestion and Bottlenecks >> Stagnation in productivity
2. Vietnam growing too fast >> Poor allocation of transport infrastructure investment >> Congestion and Bottlenecks >> More labour hours spent in unproductive activities >> stagnation in individual labour productivity
3. Vietnam growing too fast >> Poor allocation of transport infrastructure investment >> Congestion and Bottlenecks >> Logistic Delays >> Higher Business Cost >> lower resources to invest in R&D >> Fewer innovative ideas >> stagnation in total factor productivity
4. Vietnam growing too fast >> Poor allocation of transport infrastructure investment >> Congestion and Bottlenecks >> Logistic Delays >> Higher Business Cost >> Less competitive products on export markets as prices for goods increases.

Appendix 4: Therapeutics

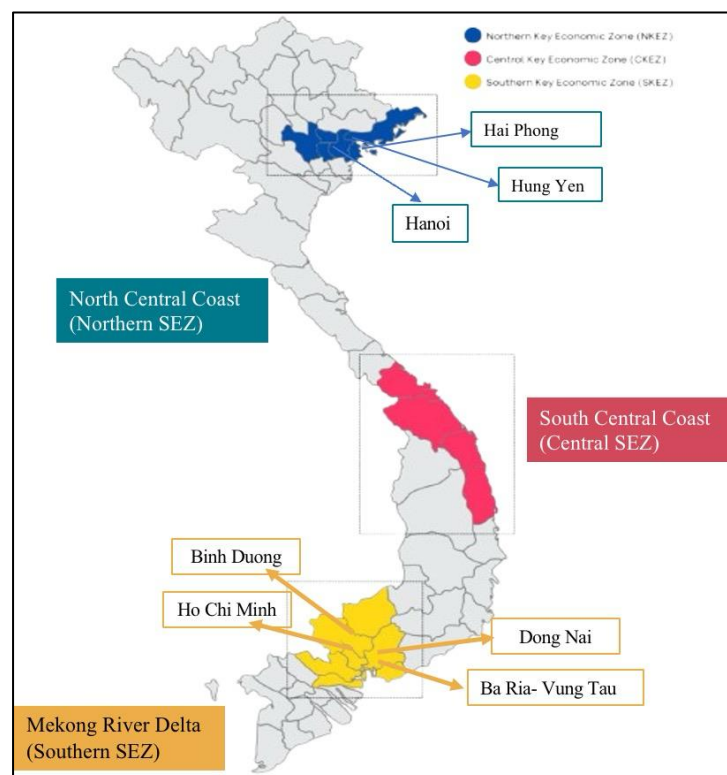


Figure 4a: Map of Key SEZs

Source: Asian Transport Outlook



Figure 4b: Map of Railway Network and Planned Rail

Source: Wikipedia, 2015

Appendix 5: Peer Selection

To find Vietnam's peers we conducted a clustered dendrogram analysis (CDA) with a several relevant variables listed in Table 1, followed by additional considerations to find a set of countries that minimized the variance between groups. We ran two models; one that included all the countries, and one that was restricted to East Asia and Pacific. The second analysis is restricted to the regional level (EAS: East Asia and Pacific and SAS: South Asia) to allow for consideration of unobserved regional characteristics. This section supplements aspirational peer selection with additional analysis to evidence similar growth trajectories.

Appendix 5.1: Cluster Analysis

We conducted a cluster analysis in Python which group sets of countries according to Euclidean distances from Vietnam based on several country characteristics (table 1) in 2022. A breakdown of the features selected for the clustering analysis is provided in Table A-1.

Variable	Justification
Real GDP (constant 2015 US\$)	Levels of output, and size of economy
Urban population (% total)	Disaggregation in urbanization due to FDI flows in Vietnam
Total Population (gross)	Size of country and economy
Economic Complexity Index	Account for product diversity and future growth
Labour Force Participation Rate (% of total working age population)	Captures the large informal activity
Female Labour Force Participation Rate (% of total working age population)	Captures potential legal and social differences, and full utilization of human capital
Polity IV (3) - Index of Democracy	Political differences effecting institutions relevant for growth
Exports of goods and services (% of GDP)	Capture's Vietnams export-oriented growth and trade openness
Agriculture, value-added (% of GDP)	Sectorial composition
Manufacturing, value-added (% of GDP)	Sectorial composition

Services, value-added (% of GDP)	Sectorial composition
Primary School Enrolment Ratio (% gross)	Education levels in Vietnam for human capital

Table 5.1a: Variables Selected for Clustering

The clustering dendrogram analysis was done on two samples. One was done a sample of all the countries in the world. The second was conducted on a subsample of the Asia Pacific and South Asia region. The analysis found the Euclidean distances between Vietnam and the two samples and clustered them into groups. The further group is from Vietnam, the more unlike the two are based on the features we chose. Since hierarchical clustering requires full information, we decided to drop the NA observations for each of the above features. The clustering model was restricted to 2022, because it is the most recent year provided and has the most complete information before dropping NAs. Below are the results of the world CDA and regional CDA.

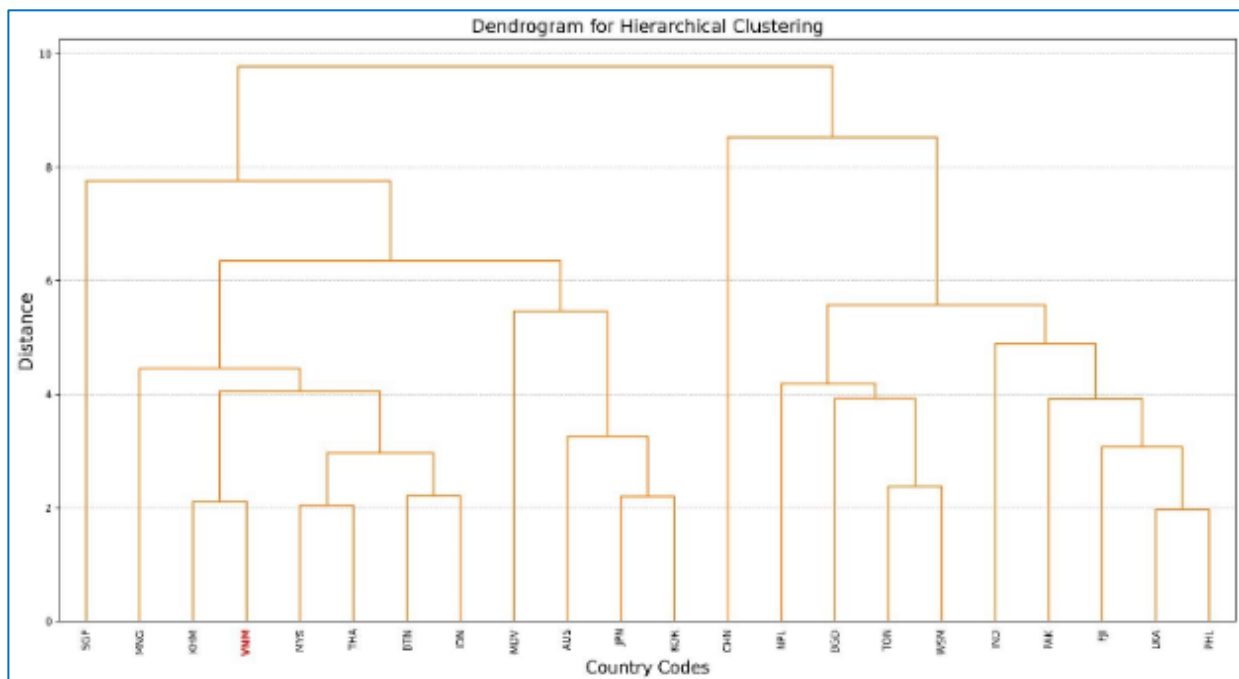


Figure 5.1a: Regional CDA

Appendix 5.2: Rejection by Product Space and Trade Composition:

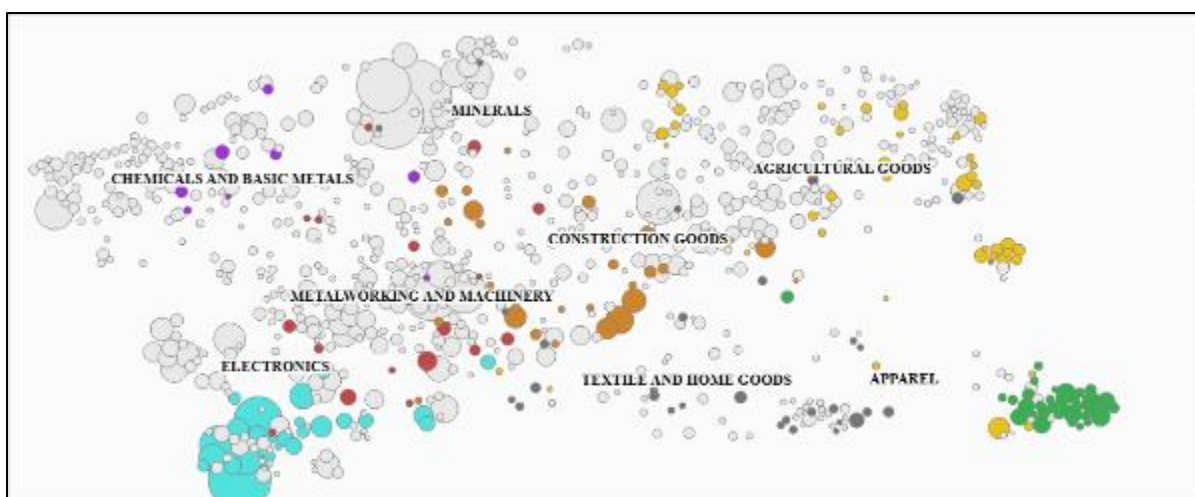
Although ECI and sectorial features are included it does not capture the opportunities for export diversification which can determine different paths for growth between countries. Trade composition is used to determine whether countries with similar product spaces are dependent on

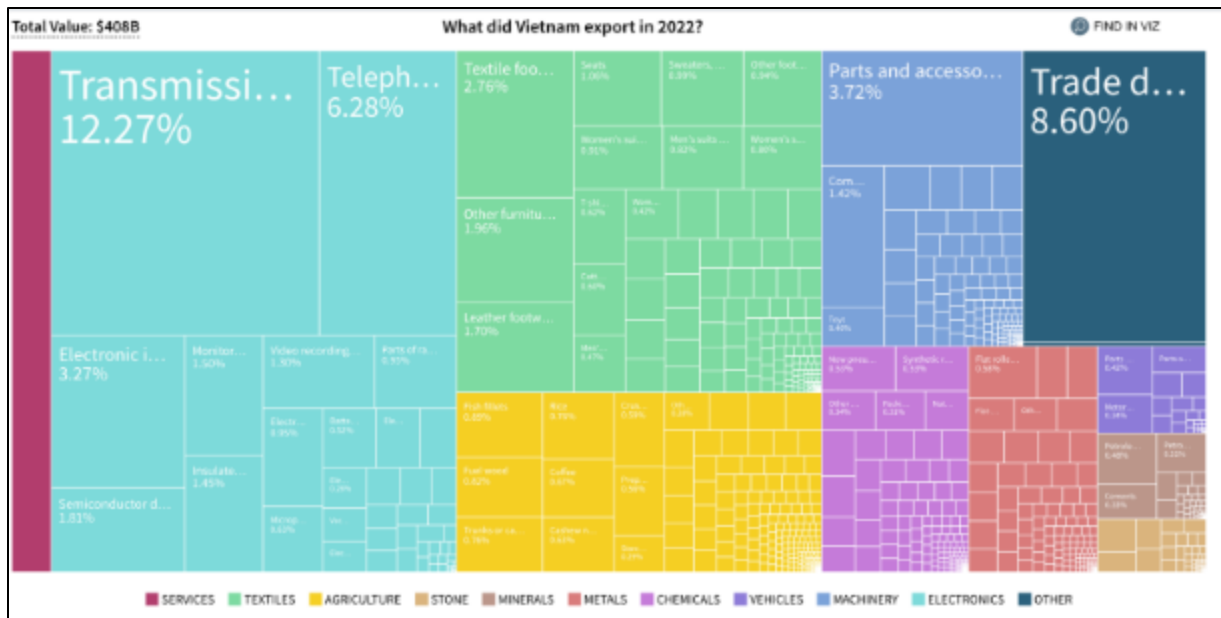
the export industries as Vietnam. If product space does not match up, then the fundamental economic structure is likely to explain major deviations in growth between countries. Below you will find product spaces for each of the matched countries based on the matching/clustering algorithm. Product Space is taken for latest year as growth diagnostics model is forward looking.

Product Space and Trade Composition of World CDA

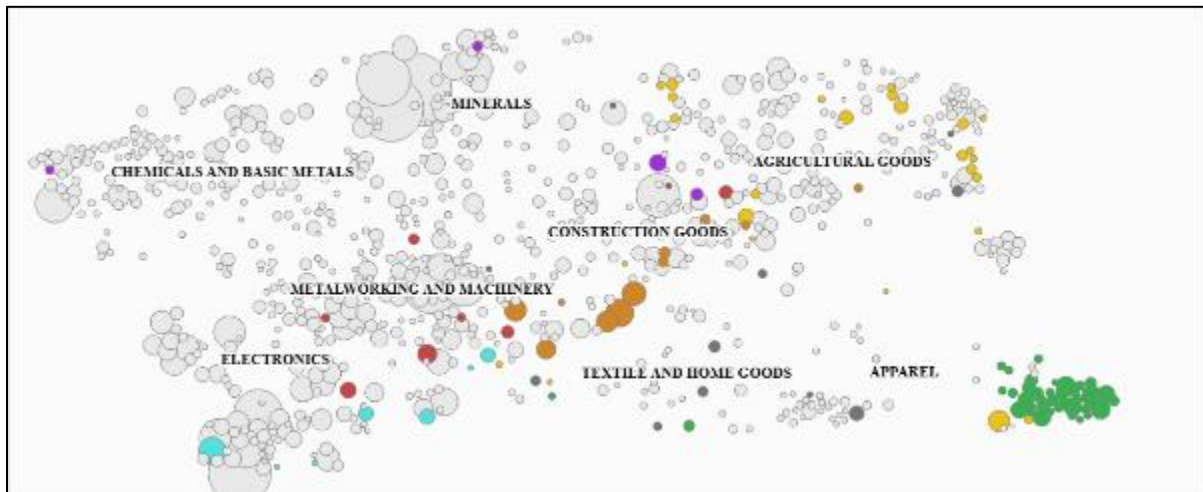
Vietnam

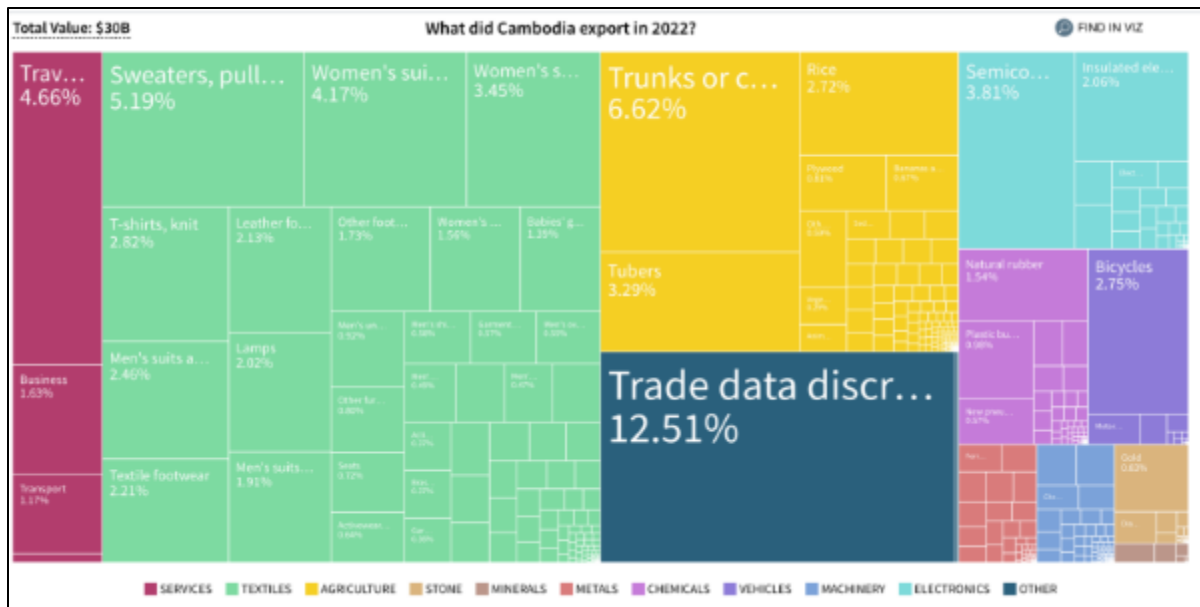
Vietnam's product space is highly diversified but concentrated in electronics and apparel. Like other Asia 'tigers', although they've moved into new product spaces, apparel makes up a constant chunk of its trade composition. Therefore, we consider concentration into these two areas key when selecting peers for Vietnam. Trends in product spaces and trend composition are important in our analysis because our growth question pertains to product diversification.



Cambodia

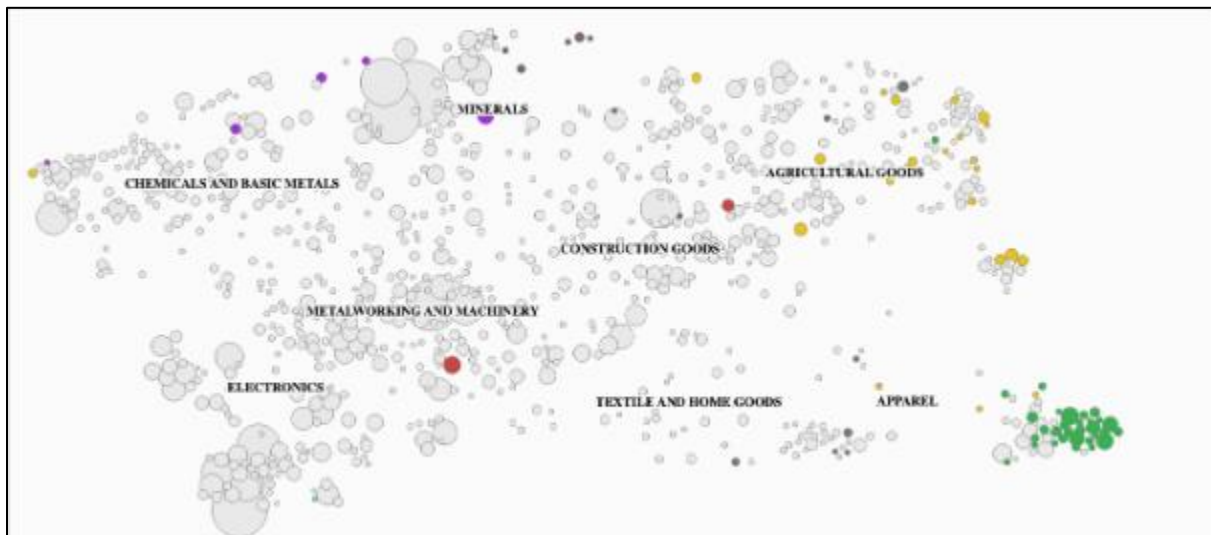
Cambodia's product space mirrors that of Vietnam's 10-20 years ago, where it is slowly shifting into more complex product spaces like electronics and construction goods. Cambodia serves then as a good lower-bound peer comparator.





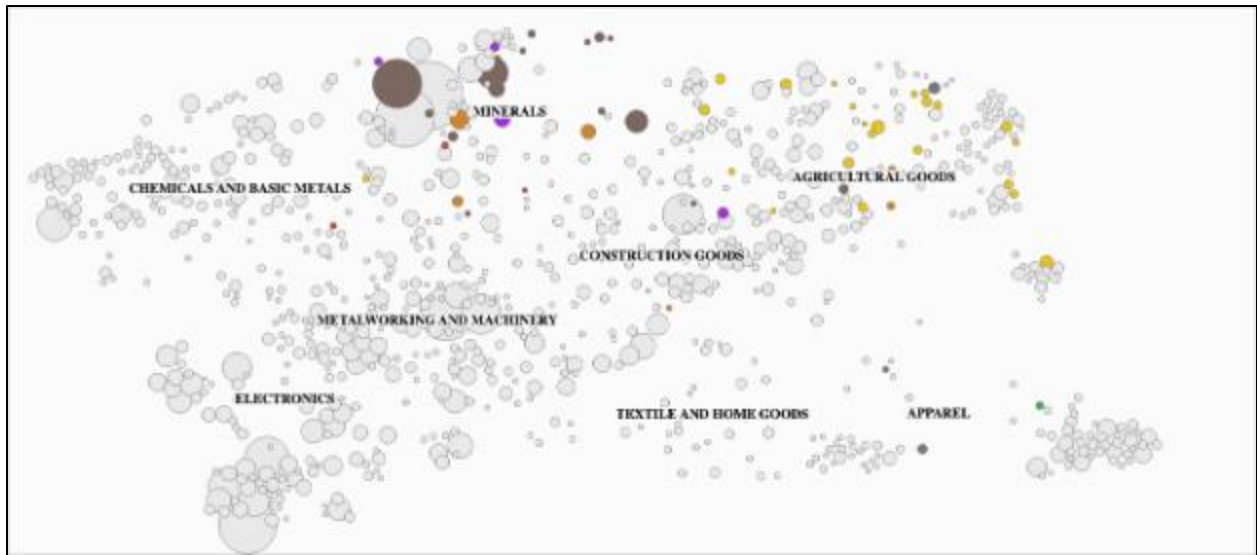
Madagascar

Madagascar has a similar concentration in apparel as Vietnam; however it has little to no product diversification into electronics and other industry goods. This signals that Madagascar does not serve as a good comparator for Vietnam's growth.



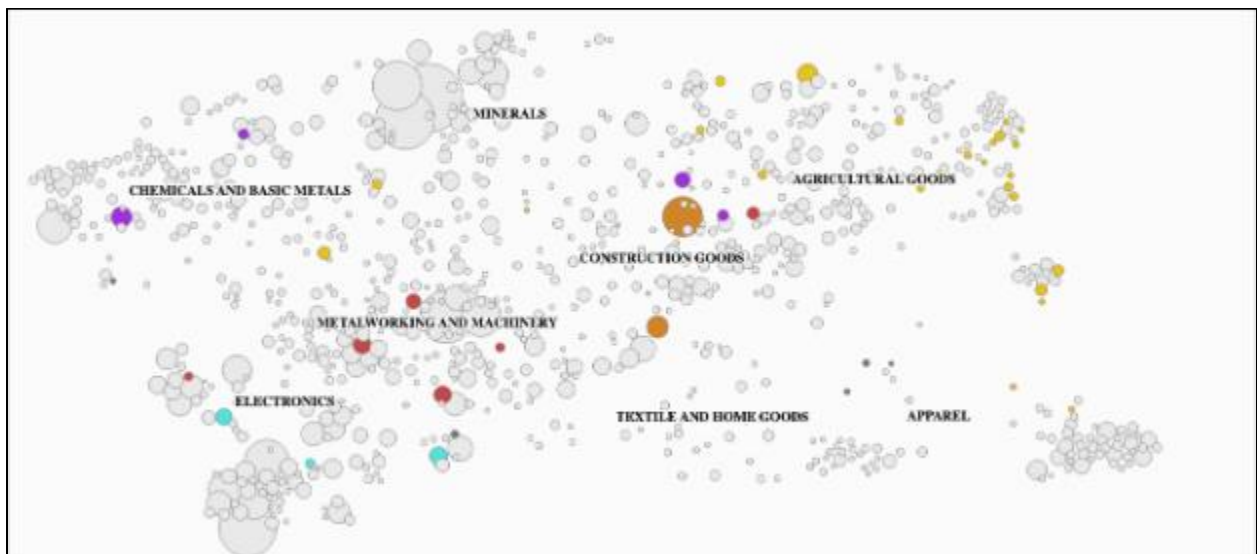
Mozambique

Mozambique product space is concentrated in minerals, an area in which Vietnam does not have a comparative advantage and therefore does not make up significant share of its product space. Mozambique then is not a good comparator of Vietnam's future growth.



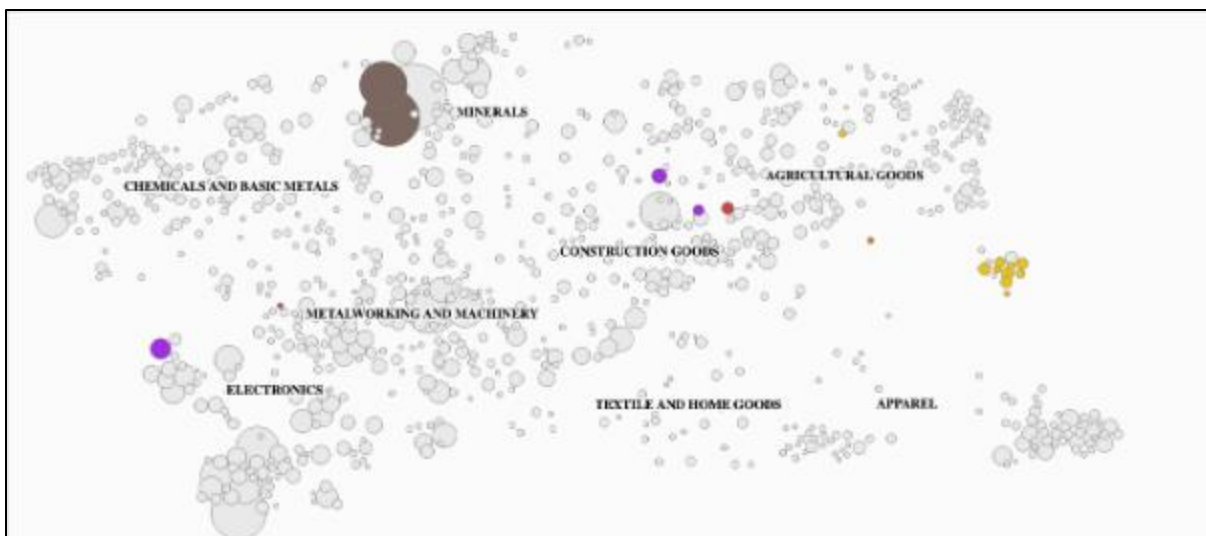
Tonga

Tonga has little concentration in industries of interest for Vietnam and does not serve as a good comparator.



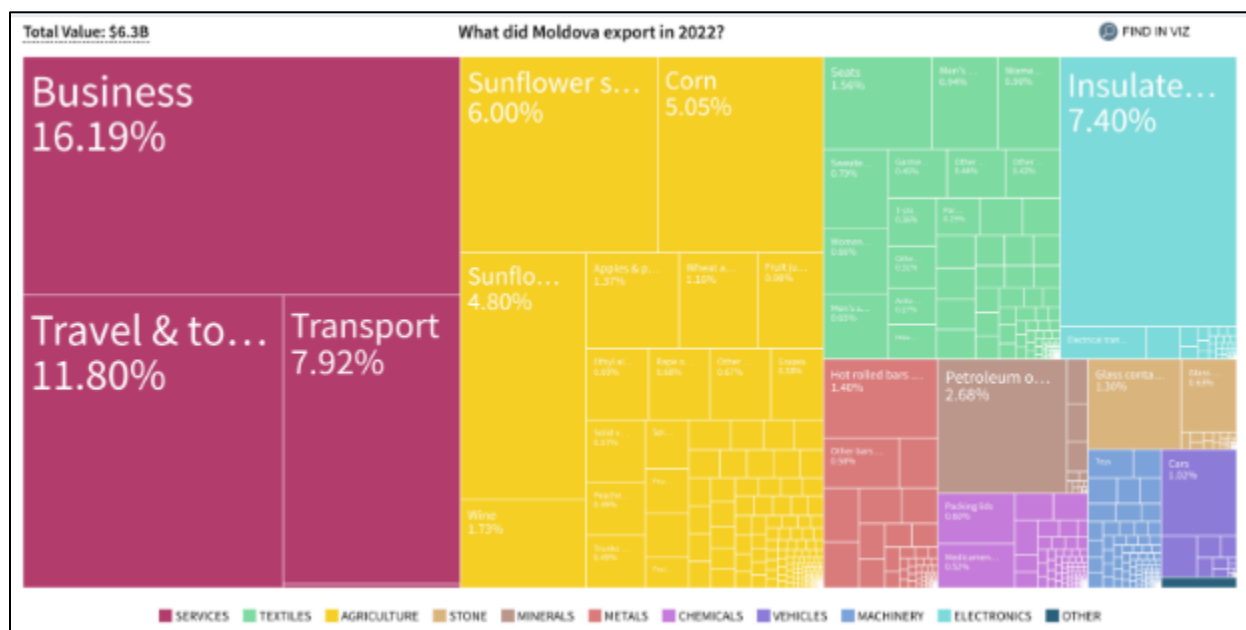
Samoa

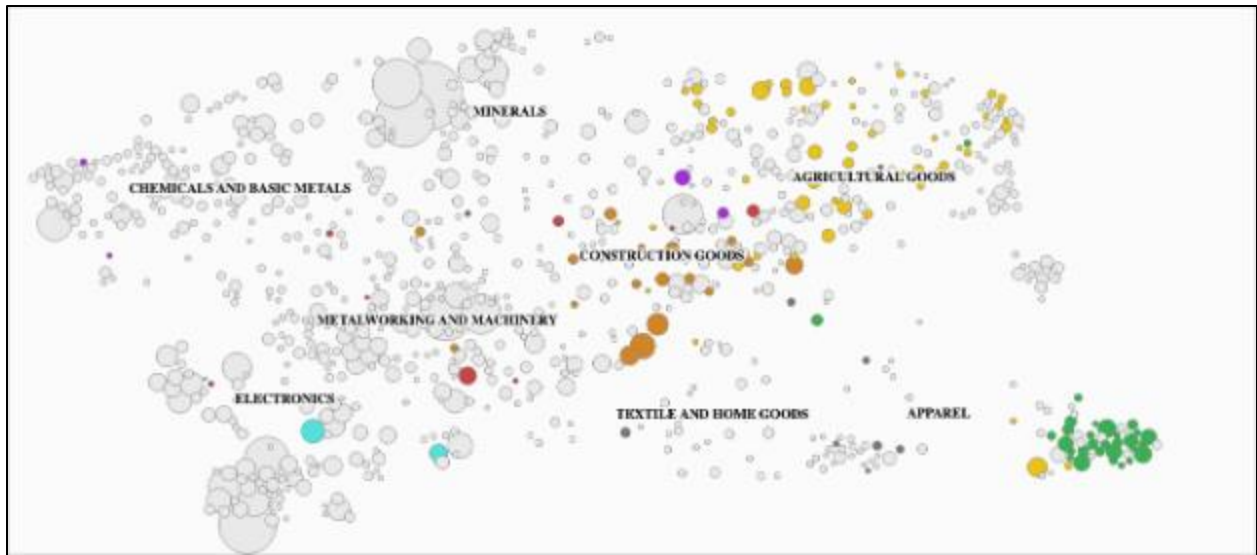
Samoa has a large concentration of its product space in electronics; however, this is misleading because it makes up a small share of its trade composition with most of its trade being driven by services. This is informative enough to conclude Samoa is on a different growth trajectory than Vietnam.



Moldova

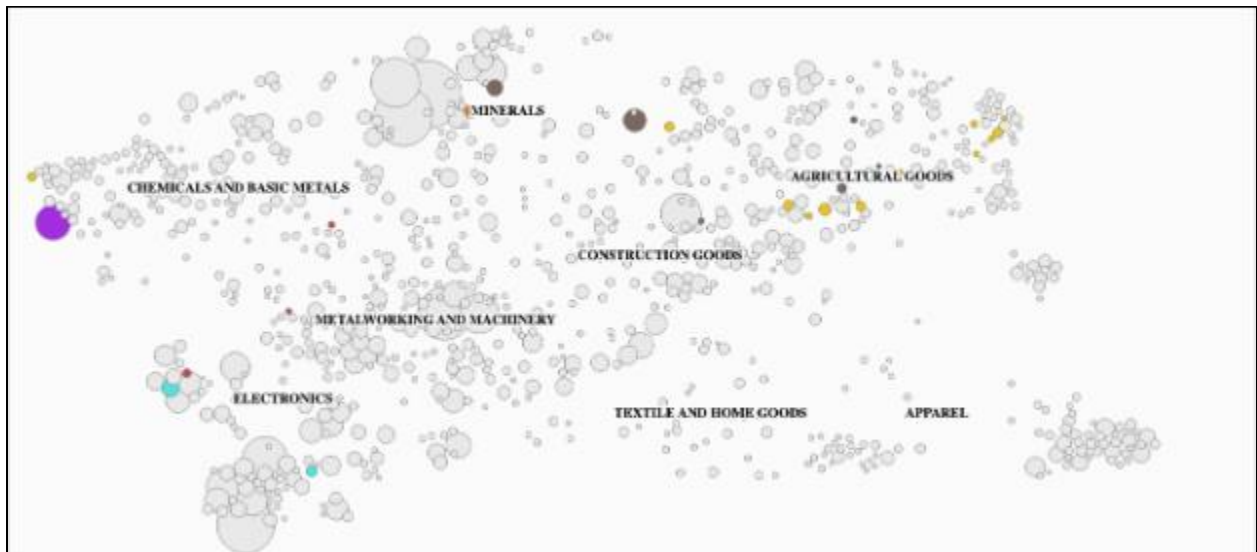
Moldova has a concentration in apparel and is slowly emerging in the electronics industry, but this picture is misleading because Moldova's exports are primarily composed of services and agricultural products. This put Moldova on a different growth path than Vietnam.





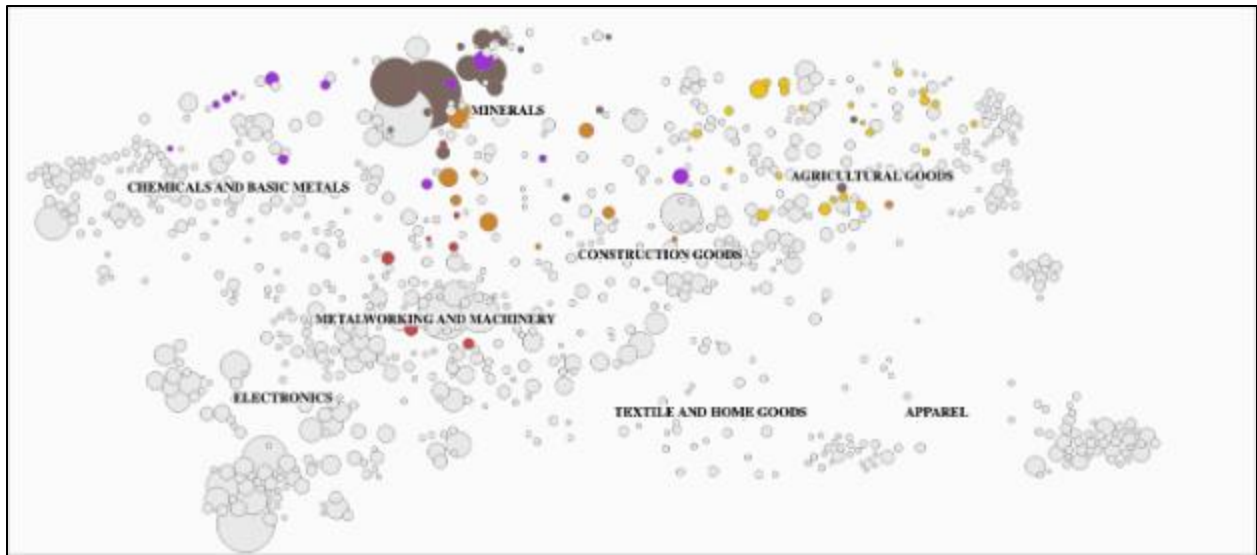
Bhutan

Bhutan has little concentration in the industries of interest for Vietnam and therefore is not a good comparator.



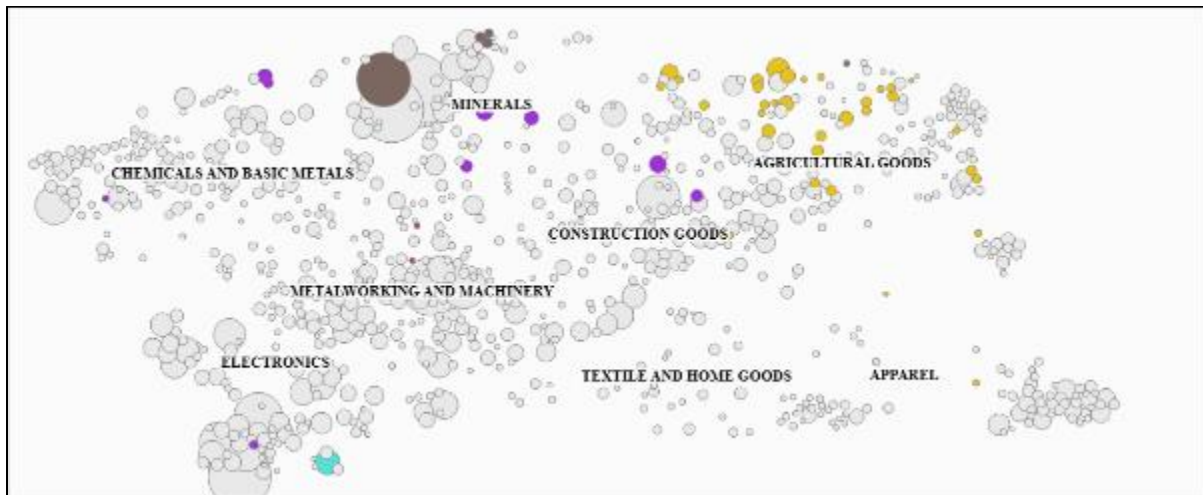
Kazakhstan

Kazakhstan has little concentration in the industries of interest for Vietnam and therefore is not a good comparator.



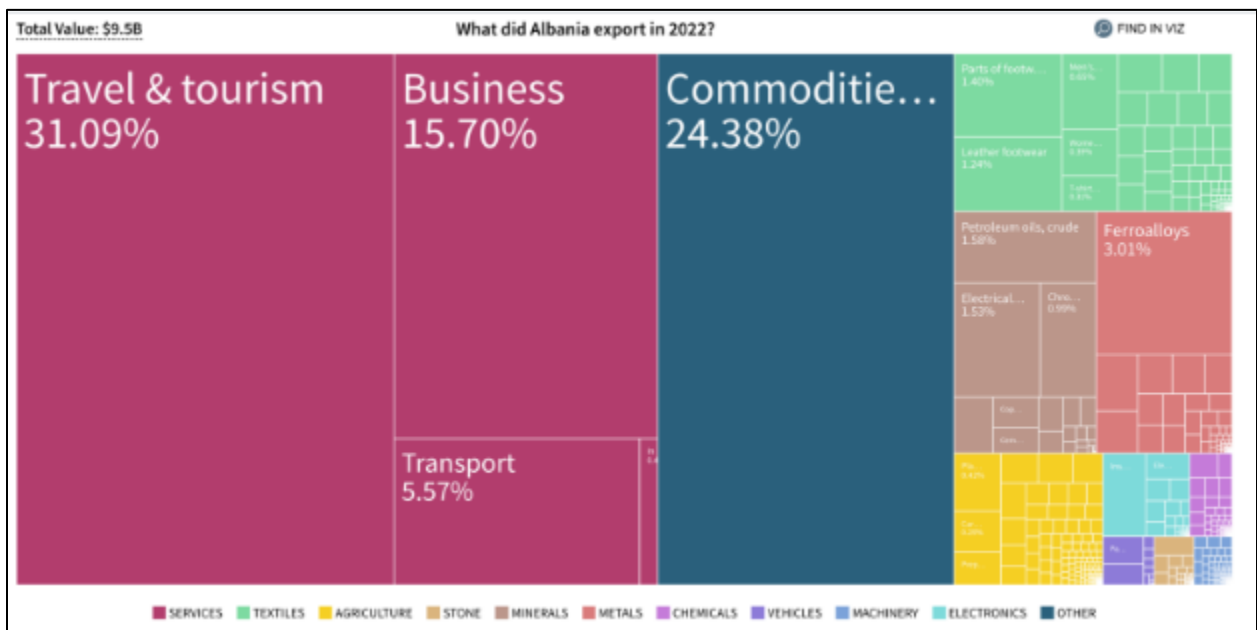
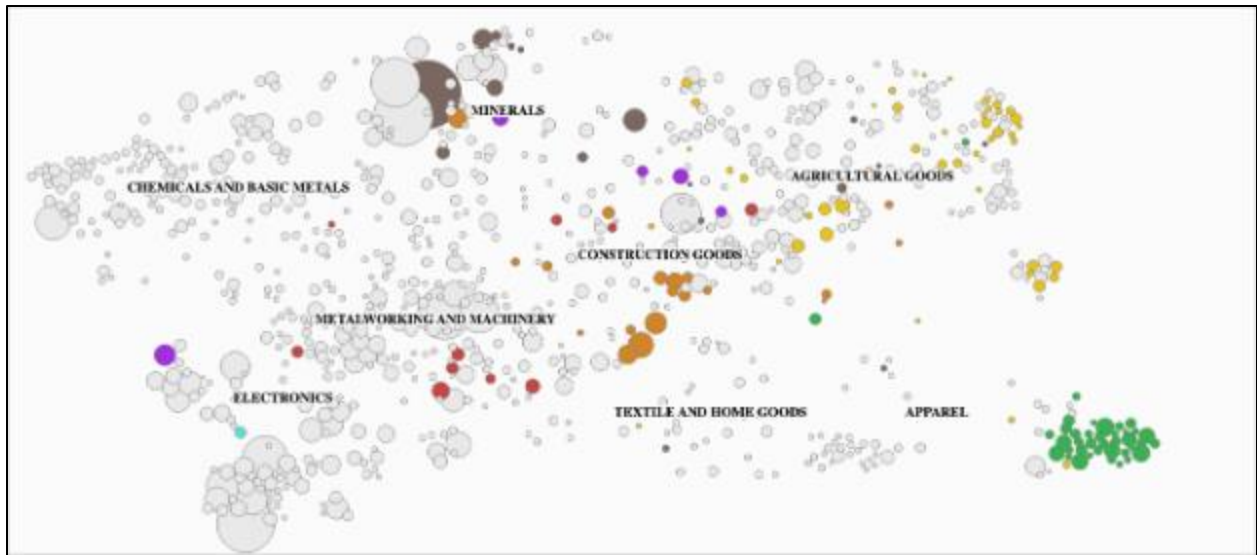
Bolivia

Bolivia has little concentration in the industries of interest for Vietnam and therefore is not a good comparator.



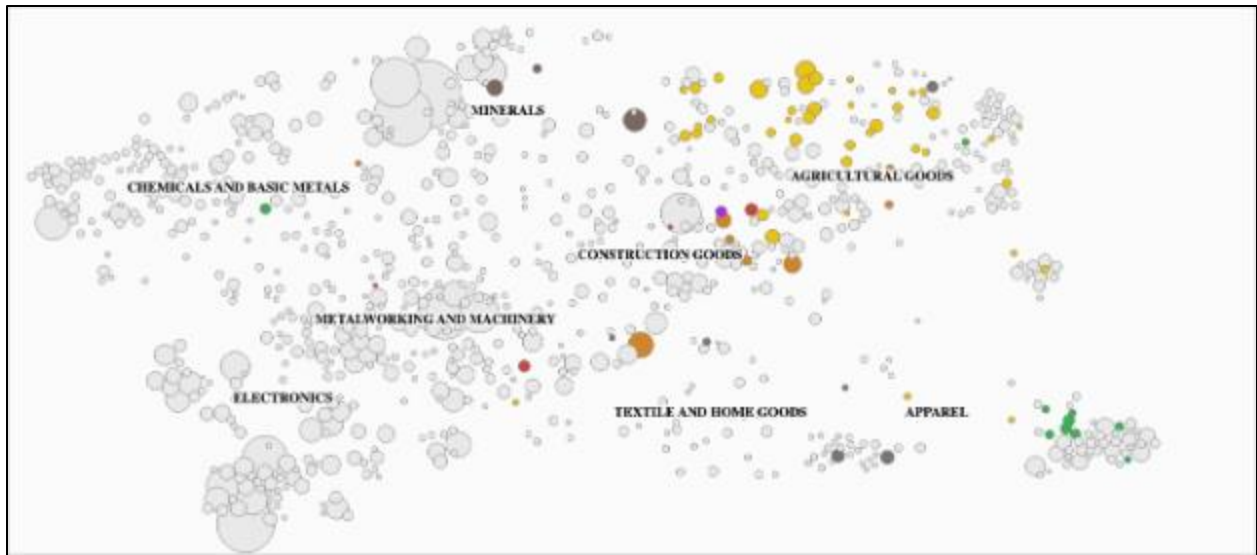
Albania

Albania has a concentration in apparel and has diversified into manufacturing, however Albania's exports are driven primarily by services and unspecified commodities. This means Albania and Vietnam are on different growth paths.



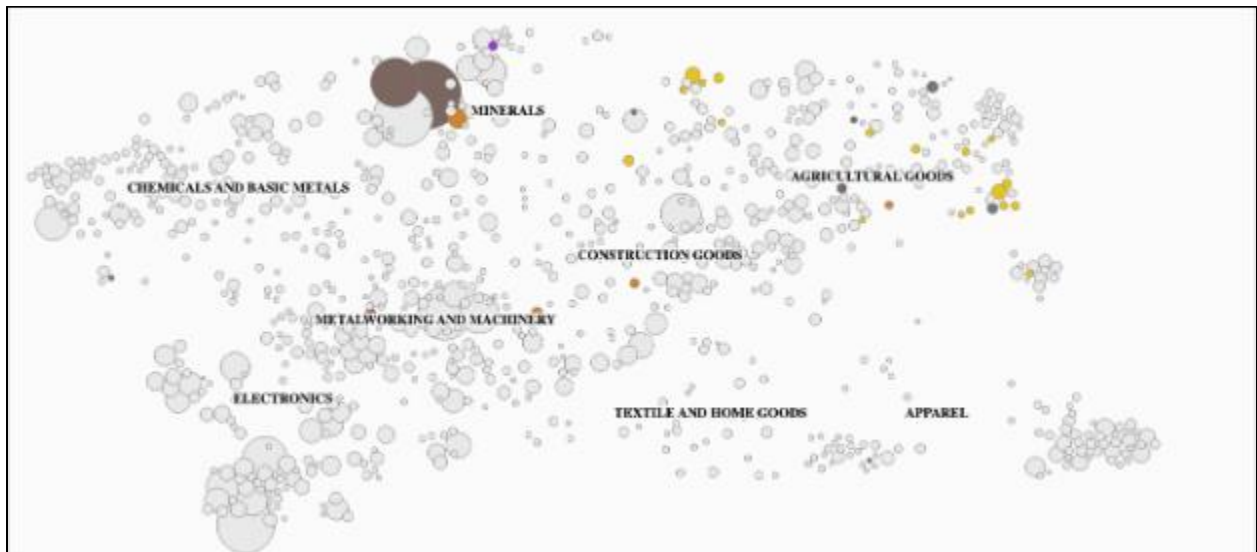
Paraguay

Paraguay has little concentration in the industries of interest for Vietnam and therefore is not a good comparator.



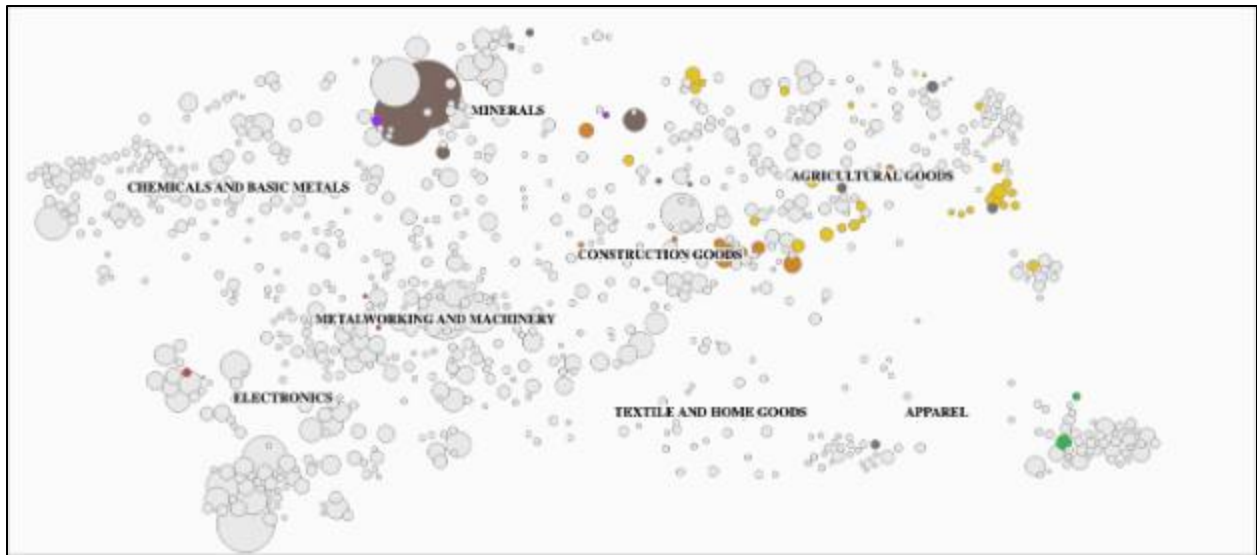
Cameroon

Cameroon has little concentration in the industries of interest for Vietnam and therefore is not a good comparator.



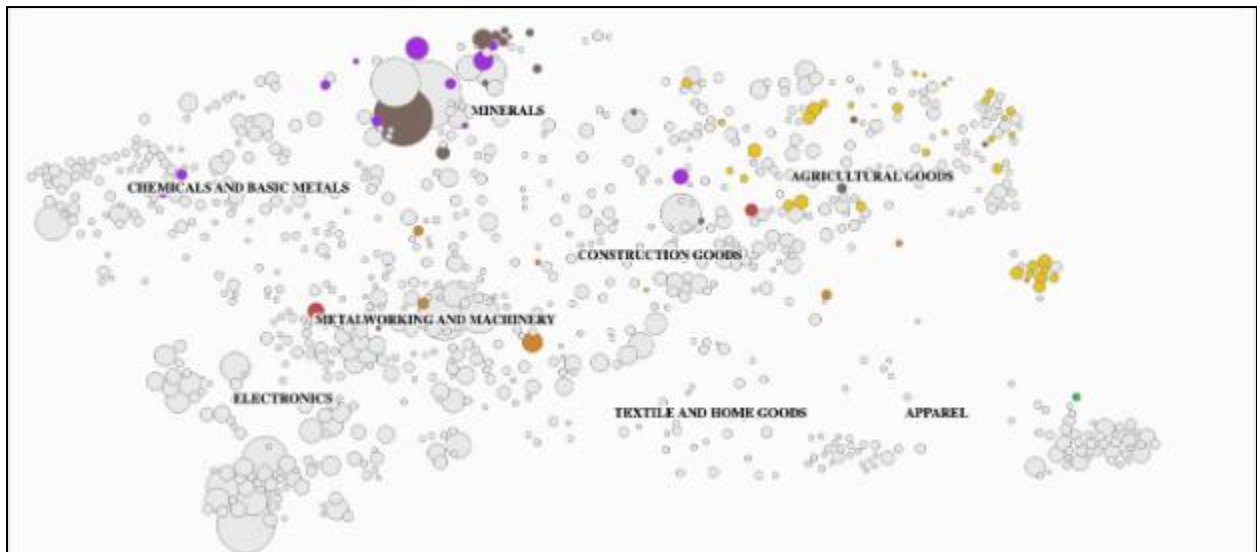
Cote d'Ivoire

Cote d'Ivoire has little concentration in the industries of interest for Vietnam and therefore is not a good comparator.



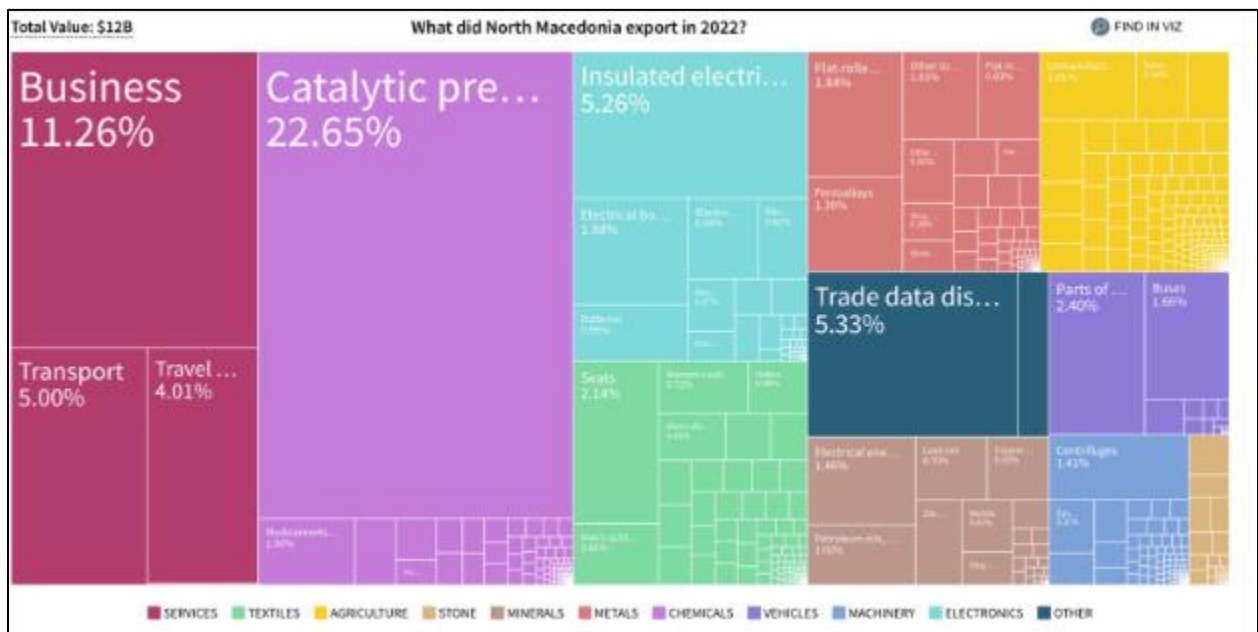
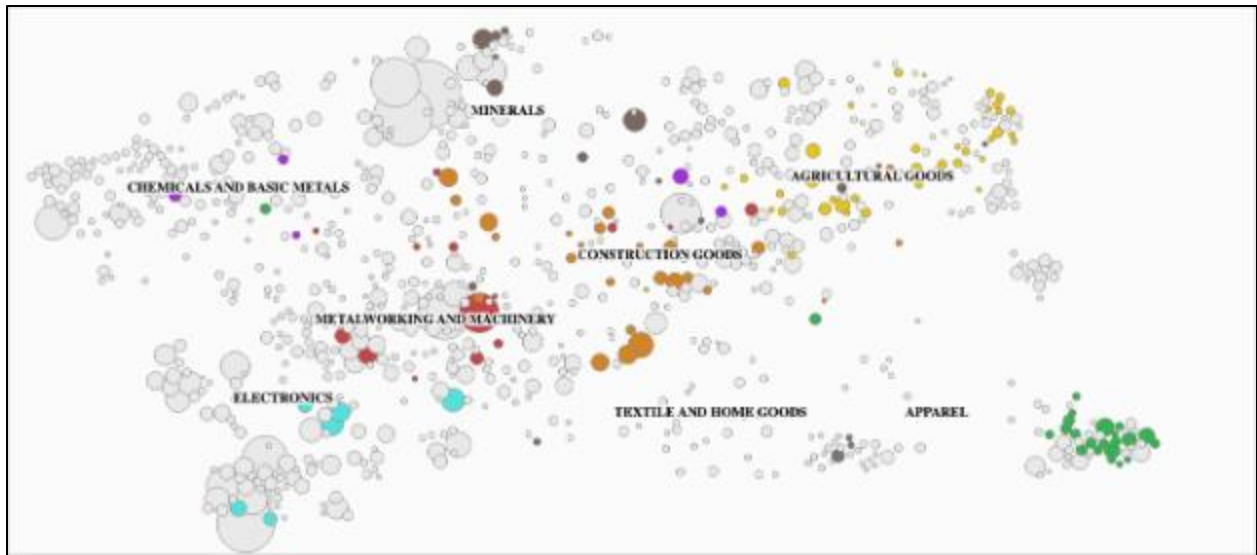
Namibia

Namibia has little concentration in the industries of interest for Vietnam and therefore is not a good comparator.



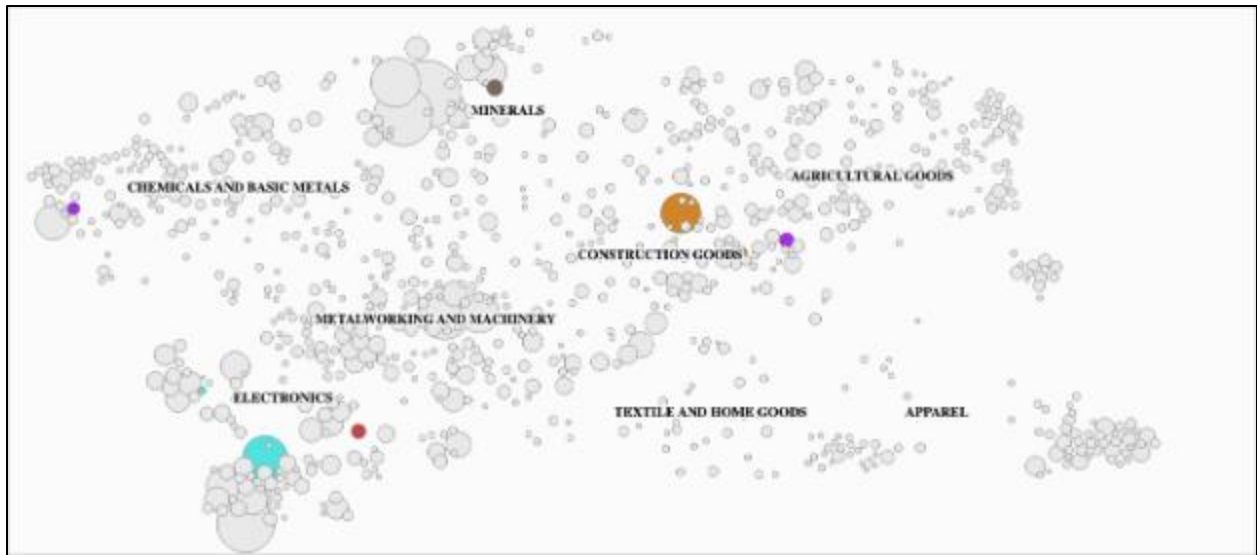
North Macedonia

North Macedonia has concentrations in apparel and electronics, but its exports are driven by services and its chemicals putting it on a different growth trajectory than Vietnam.



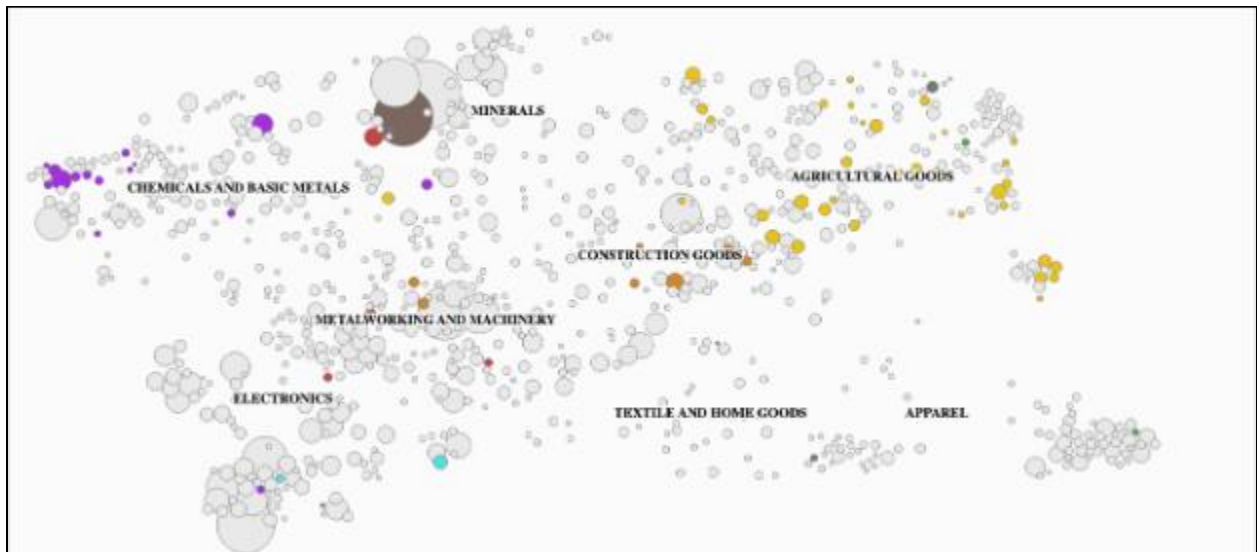
South Georgia

Although South Georgia has a budding concentration into electronics, it is not diversified into anything else. This means that South Georgia would be a poor comparator for Vietnam's growth.



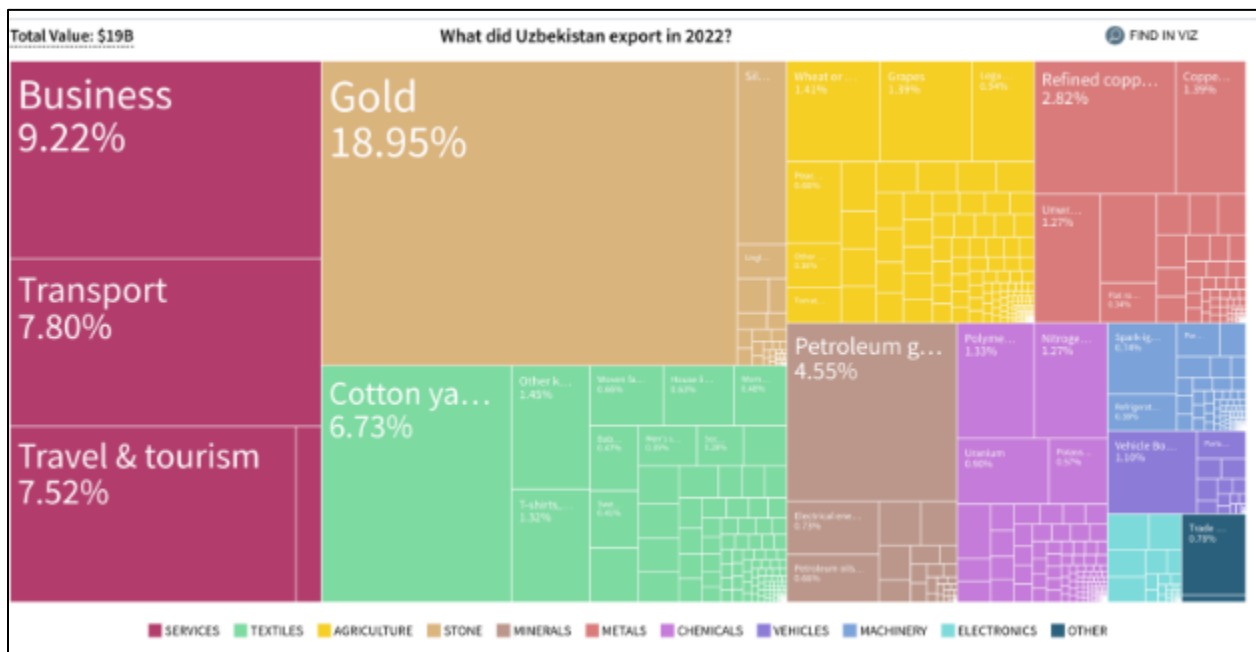
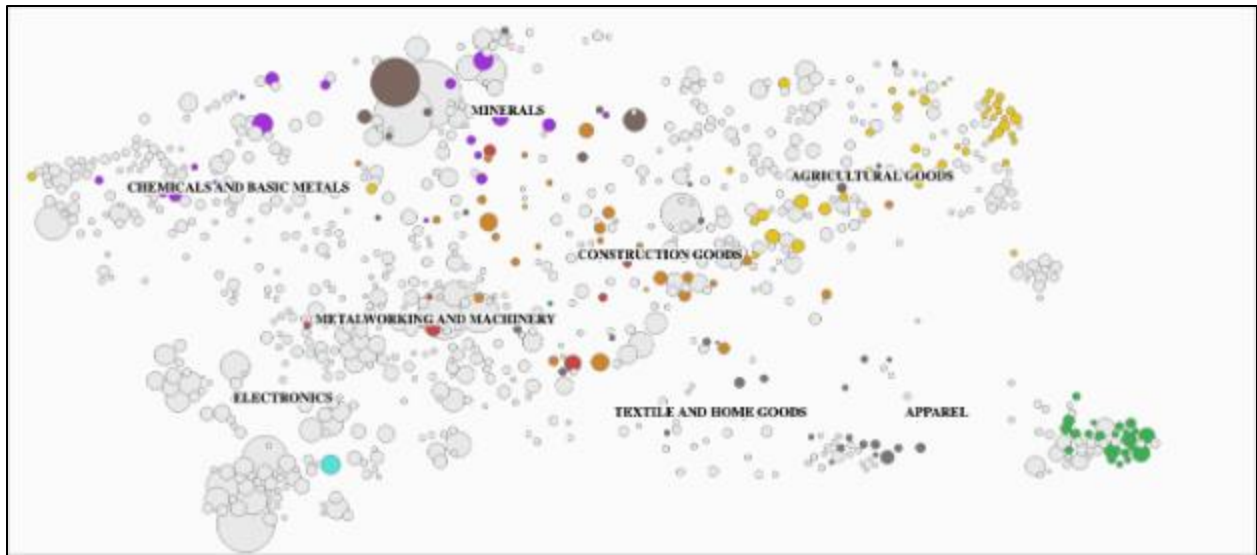
Belize

Belize has little concentration in the industries of interest for Vietnam and therefore is not a good comparator.



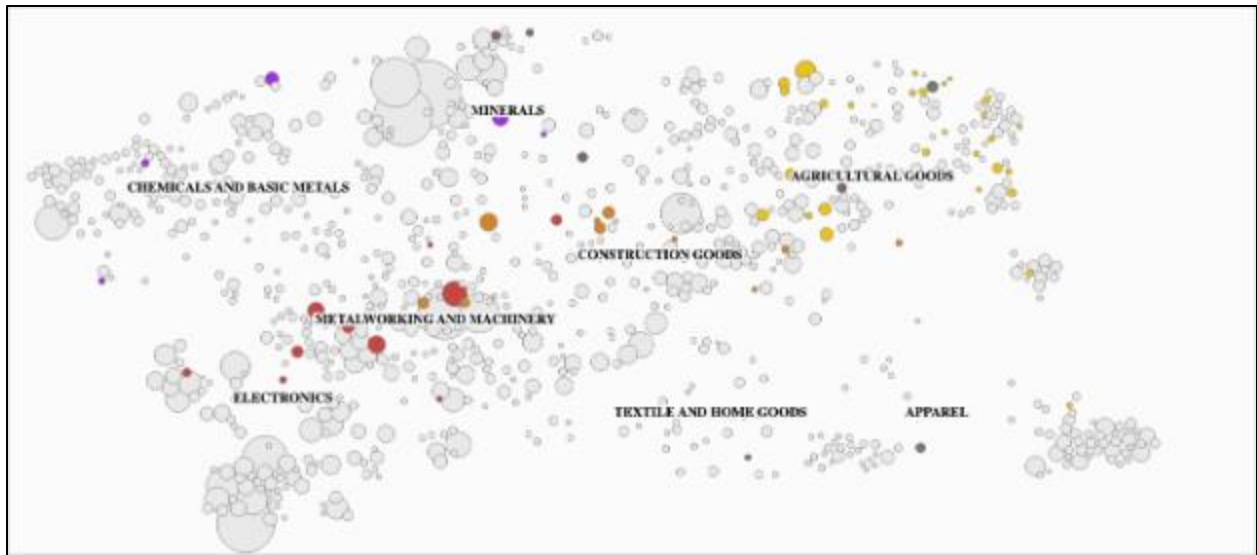
Uzbekistan

Uzbekistan has concentrations in apparel and electronics; however, its exports are characterized by services and gold. This is sufficient to reject Uzbekistan as a peer for Vietnam.



Burkina Faso

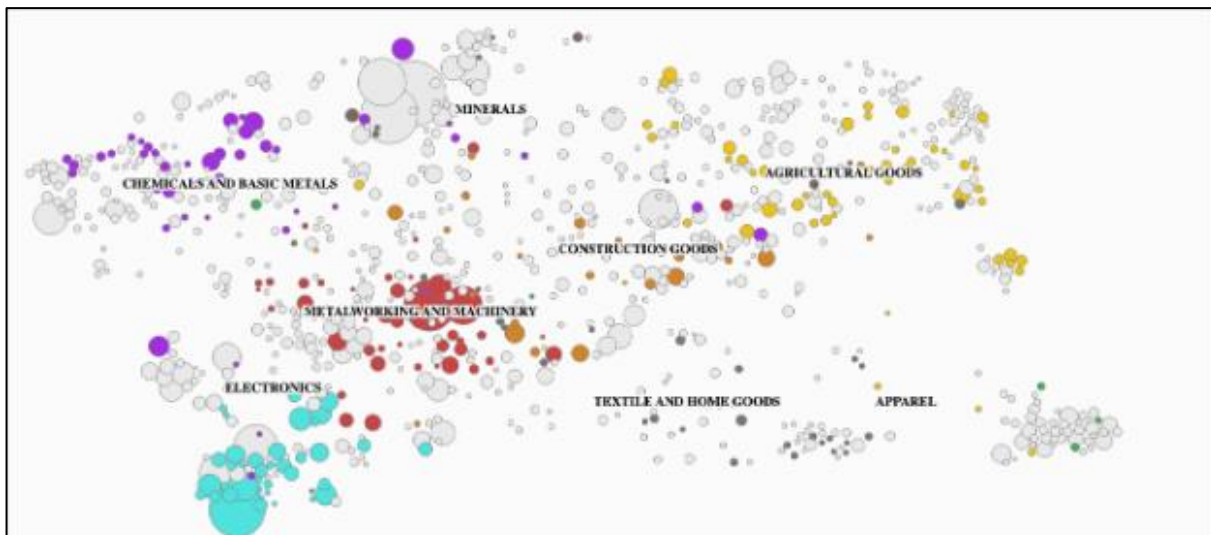
Burkina Faso has little concentration in the industries of interest for Vietnam and therefore is not a good comparator.

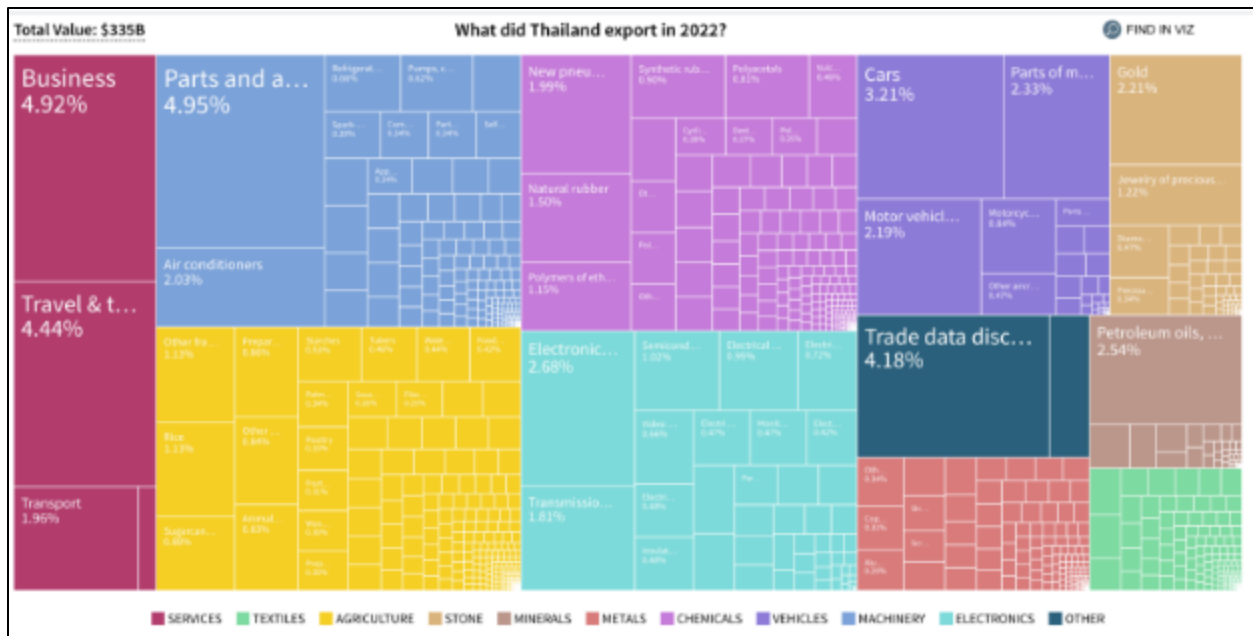


Product Space and Trade Composition of Regional CDA

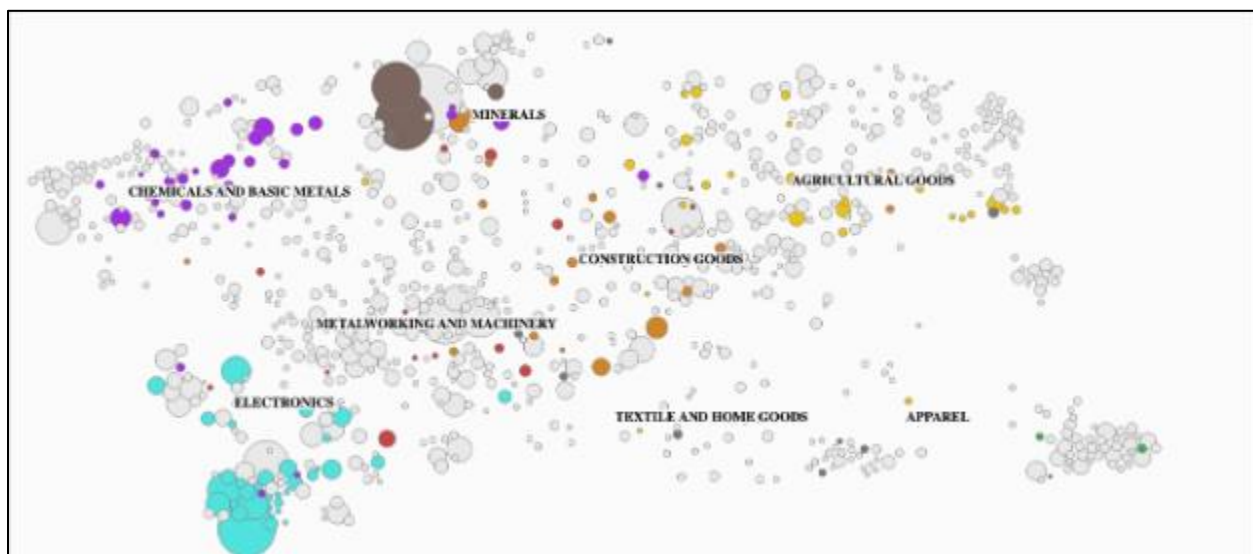
Thailand

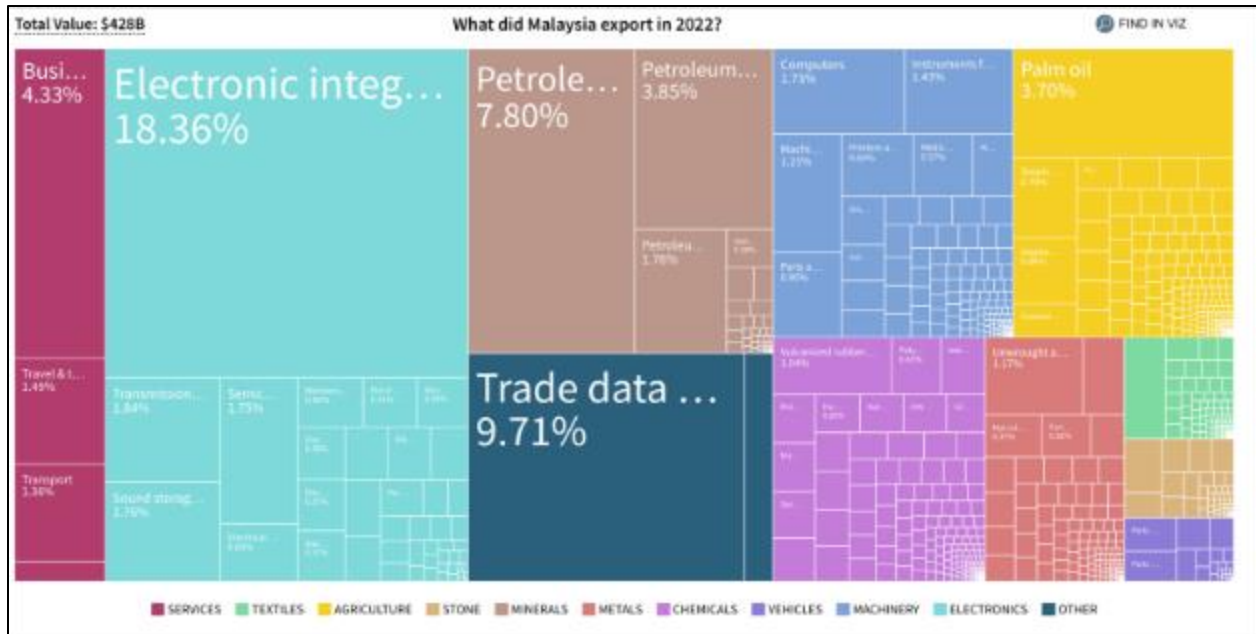
Thailand has a large concentration into electronics, which is one of two industries driving Vietnam's growth. It has larger concentrations of chemicals and metals, and no specialization in apparels. Looking at its trade composition, Thailand evenly exports chemicals, machinery, electronics and agricultural products. Since it is diversified and evenly composed, we cannot reject Thailand based on its product space and trade composition. More consideration needs to be given especially given its geography.



Malaysia

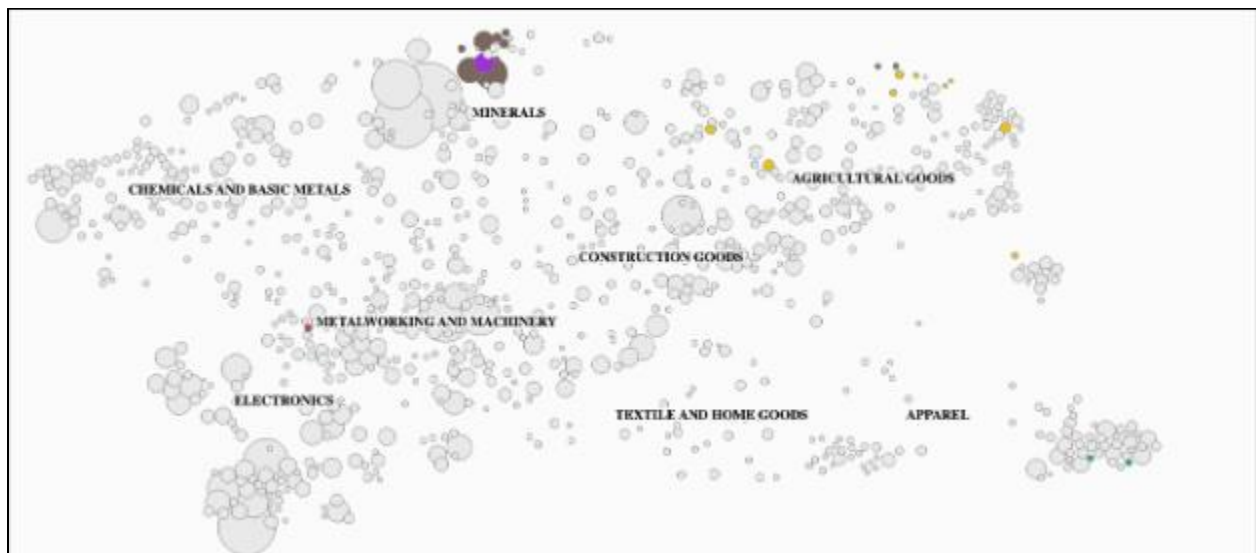
Malaysia is concentrated in electronics, minerals and chemicals. Its trade composition is primarily made up of electronics, minerals and machinery. Given its geography, further considerations must be given and it cannot be rejected based on its product space and trade composition.





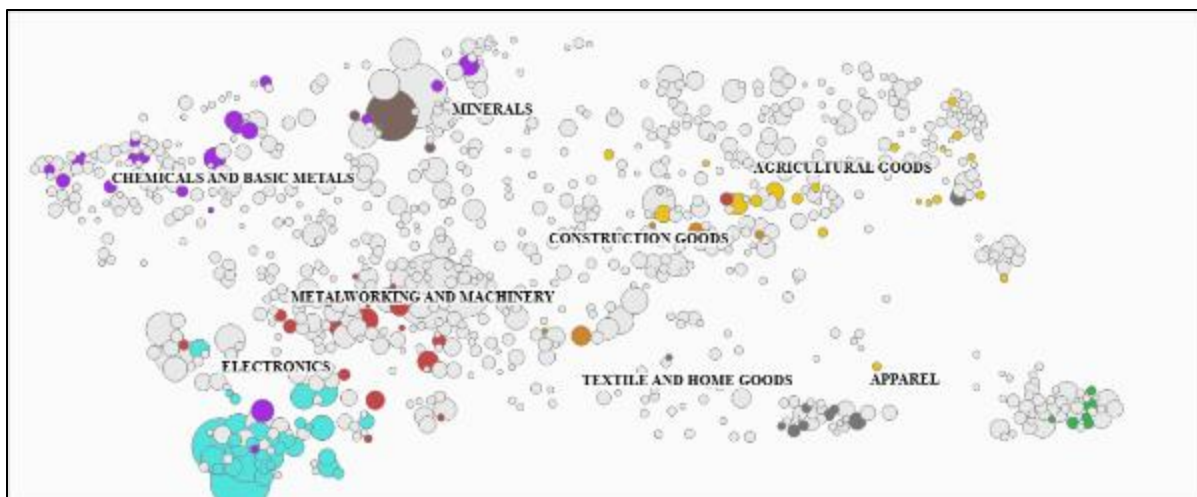
Mongolia

Mongolia has little concentration in the industries of interest for Vietnam and therefore is not a good comparator.



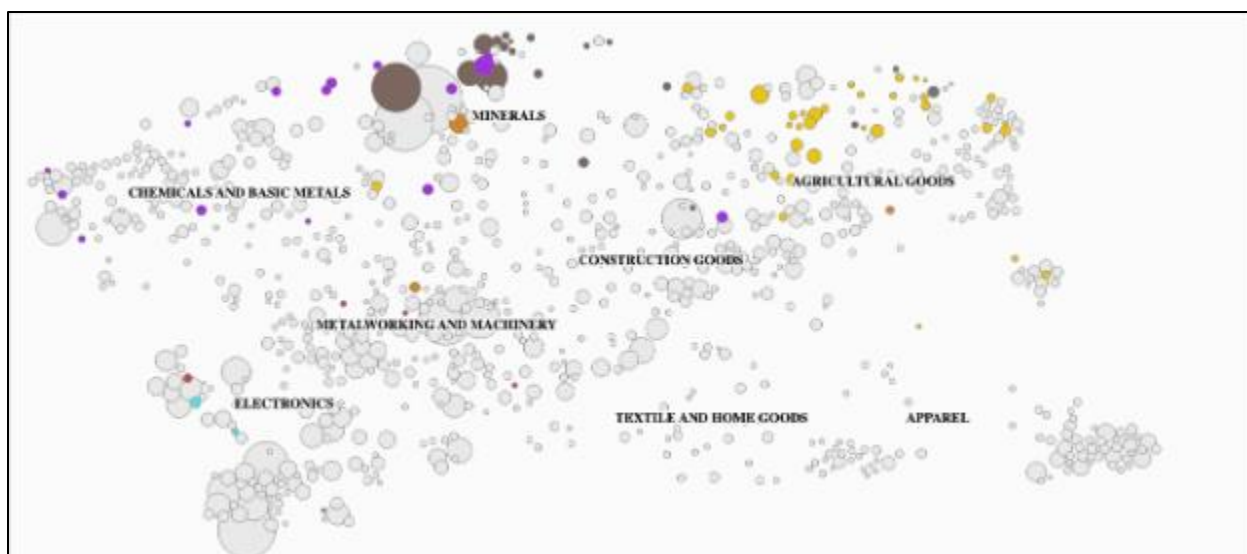
South Korea

South Korea has concentrations in electronics which make it have a similar product space to Vietnam. Its trade composition is made up of electronics, chemicals and vehicles, which put it at a larger stage of development than Vietnam, making it a poor comparator.



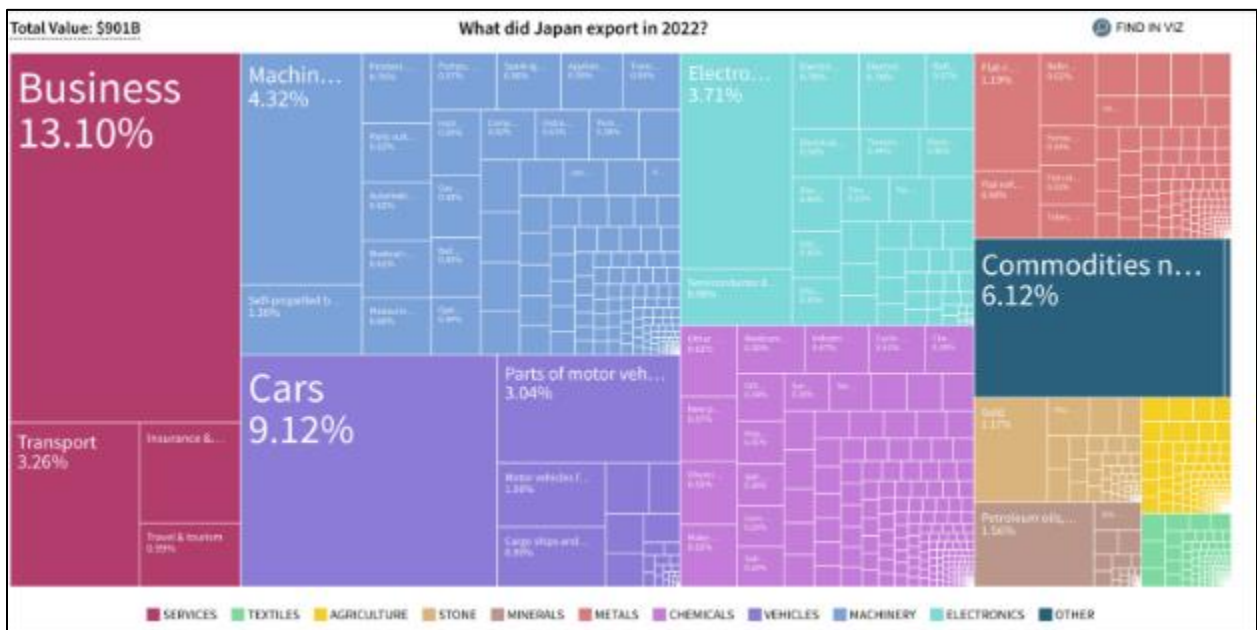
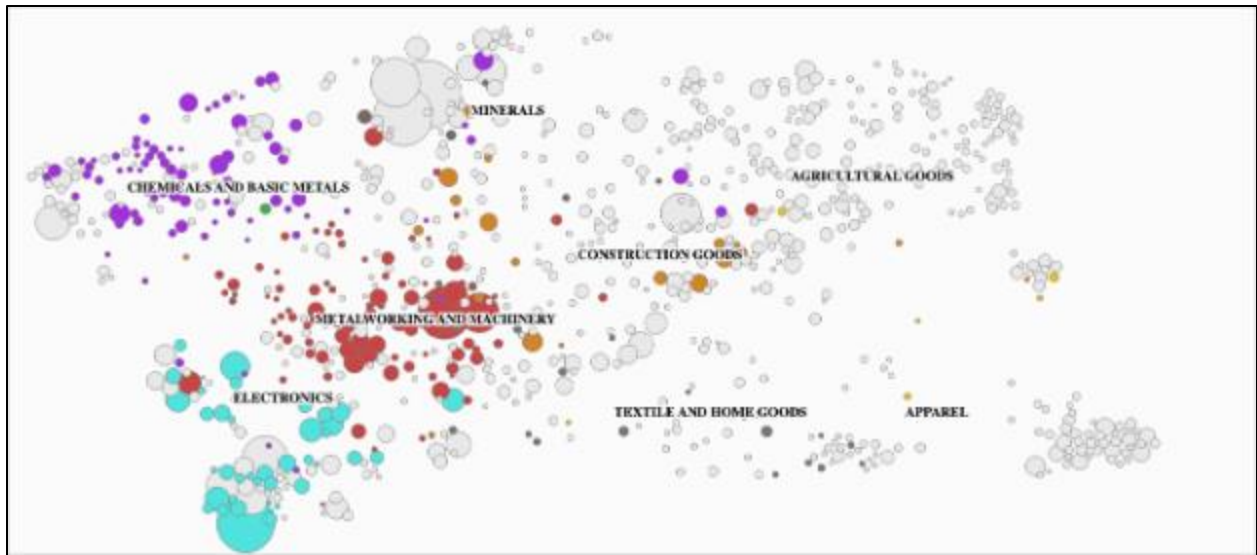
Australia

Australia has little concentration in the industries of interest for Vietnam and therefore is not a good comparator.



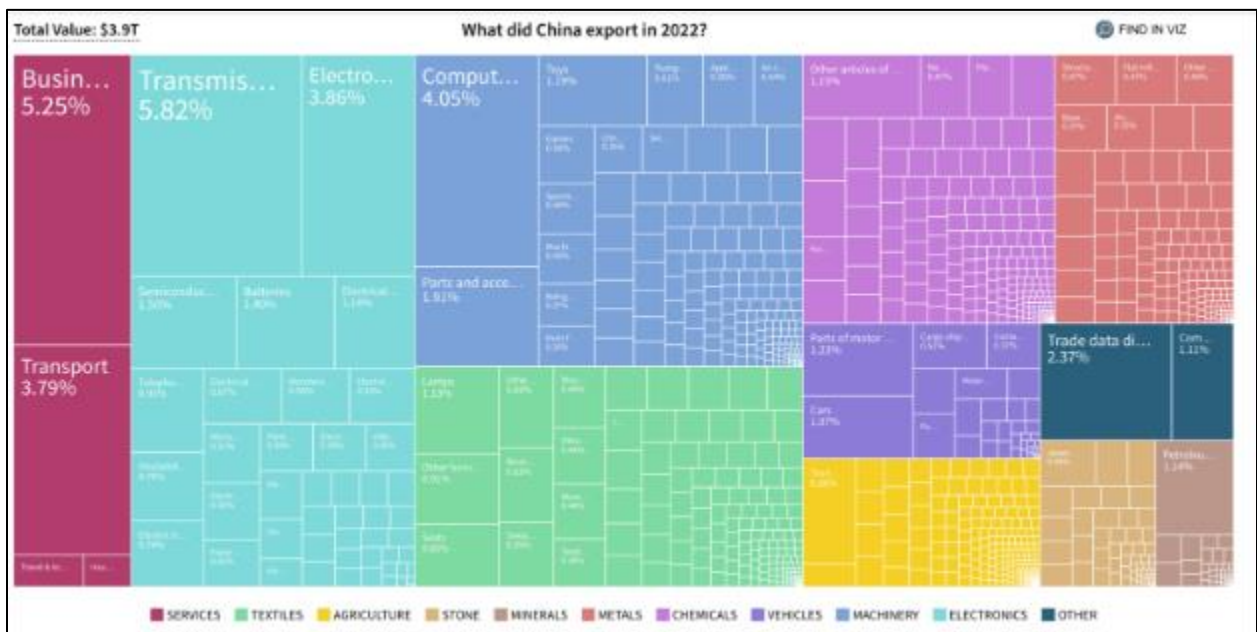
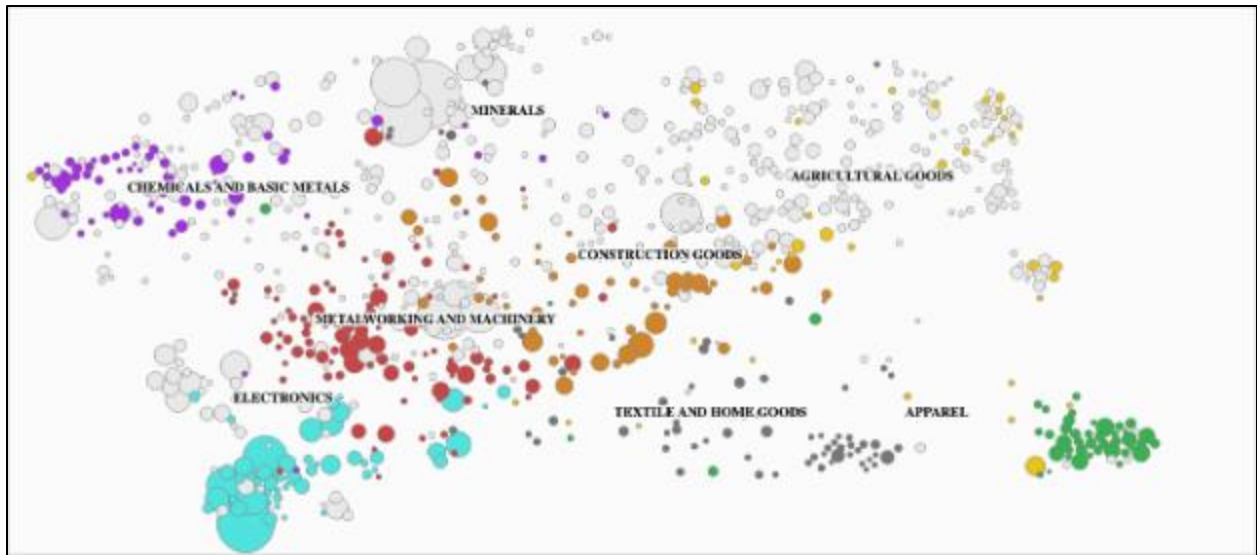
Japan

Japan is a large and highly diversified economy specializing in chemicals, metalworking and machinery as well as electronics. Its exports are driven by machinery, vehicles, and services putting it on a different growth path than Vietnam, making it a poor comparator.



China

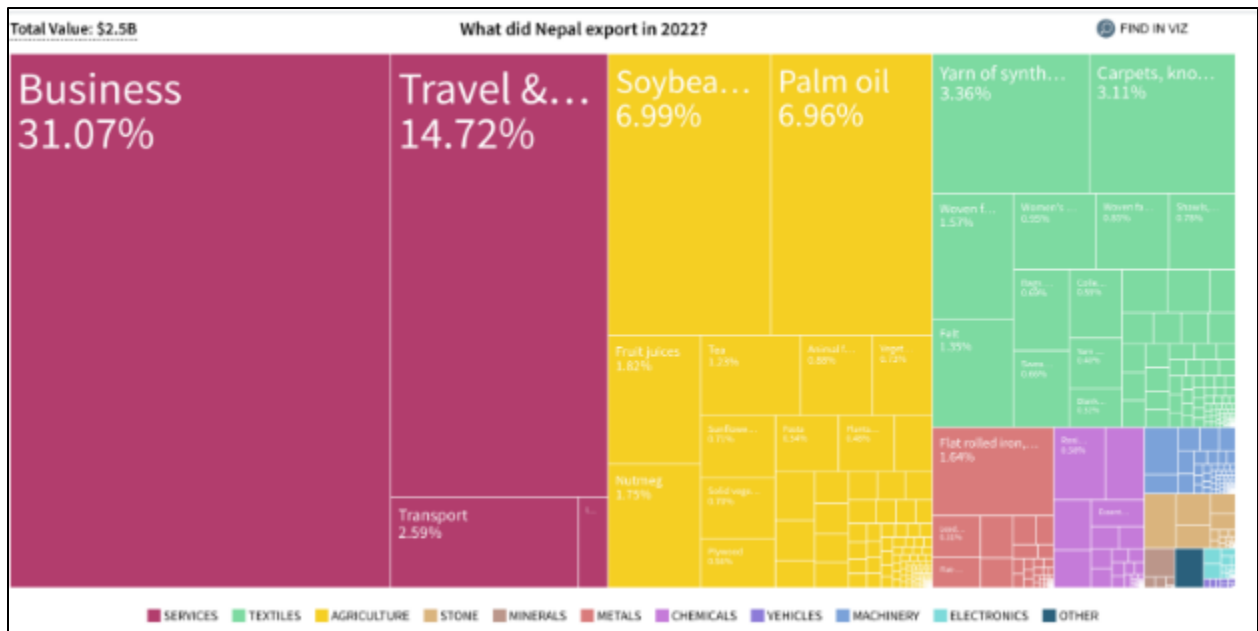
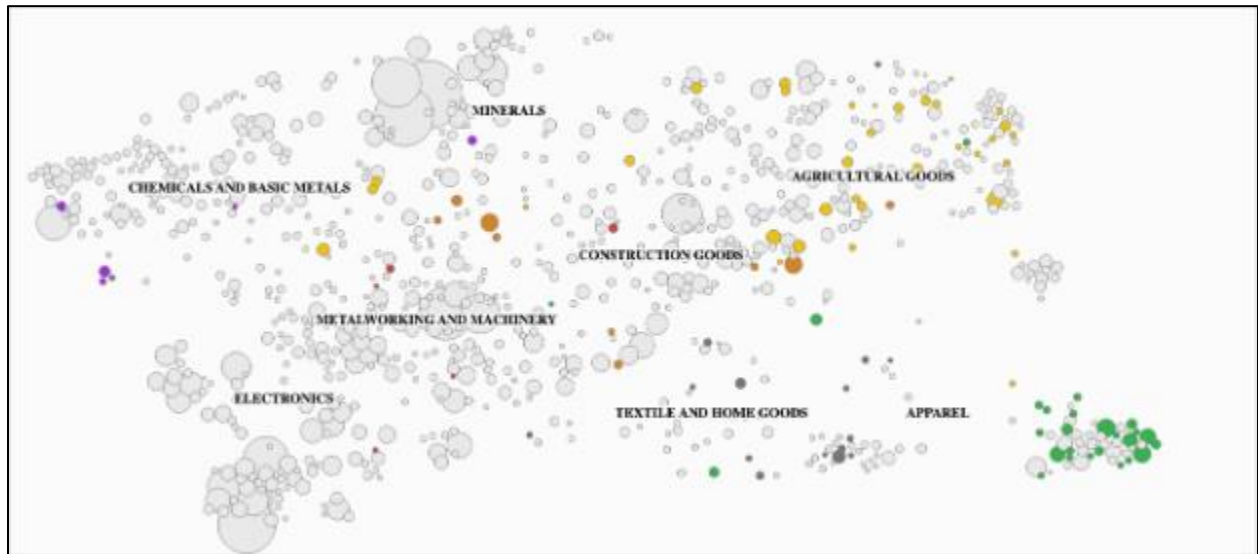
China is also a large and highly diversified economy, concentrating in two industries of interest: electronics and apparel. Its trade composition is driven by electronics, machinery, and textiles. We cannot reject China based on its product space and trade composition.



Nepal

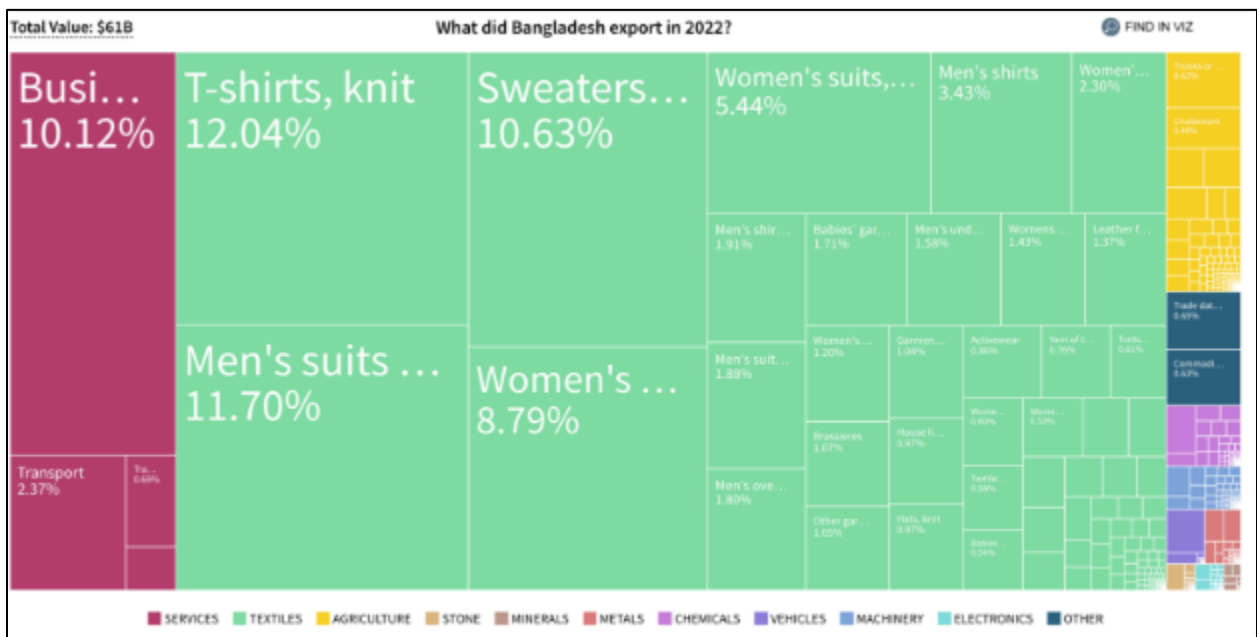
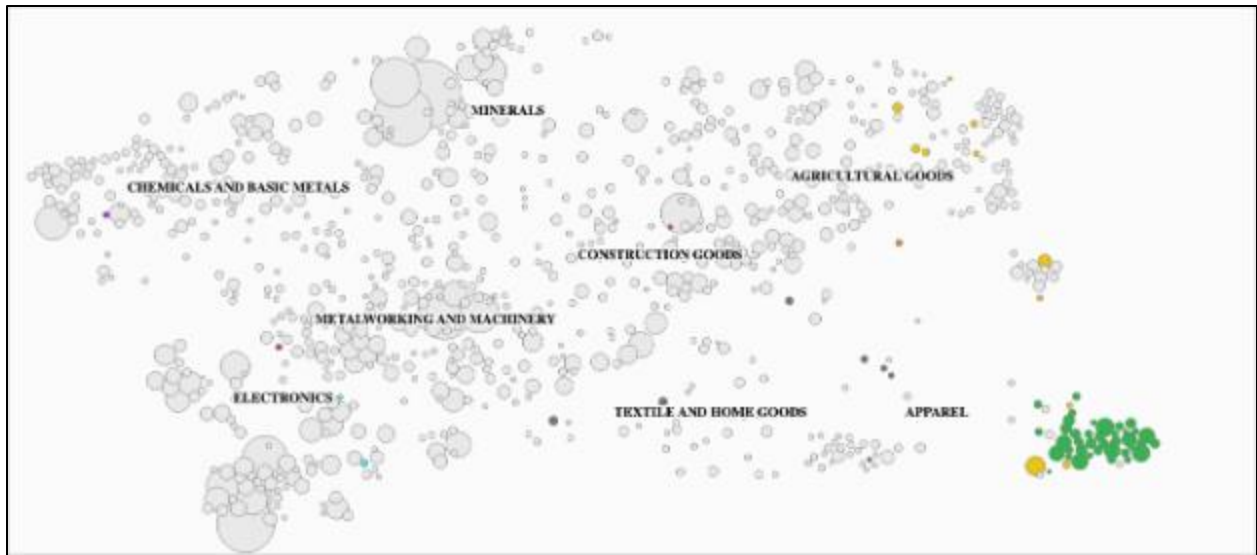
Nepal has budding specialization in apparel and agriculture. However, its exports are primarily comprised of services and agricultural products. This puts Nepal on a different growth trajectory

than Vietnam.



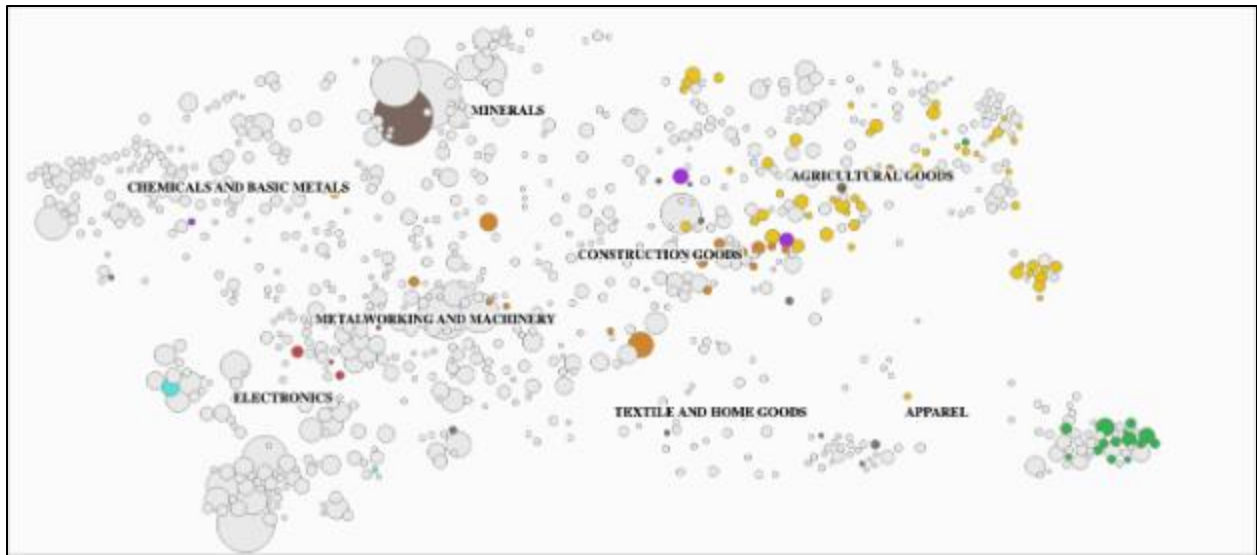
Bangladesh

Bangladesh is highly concentrated in apparel and has little specialization into any other industry. This is also reflected in its trade composition. This is informative of Vietnam's early 2000s growth, which may make Bangladesh a good lower-bound comparator. We will not reject Bangladesh based on its product space and trade composition.



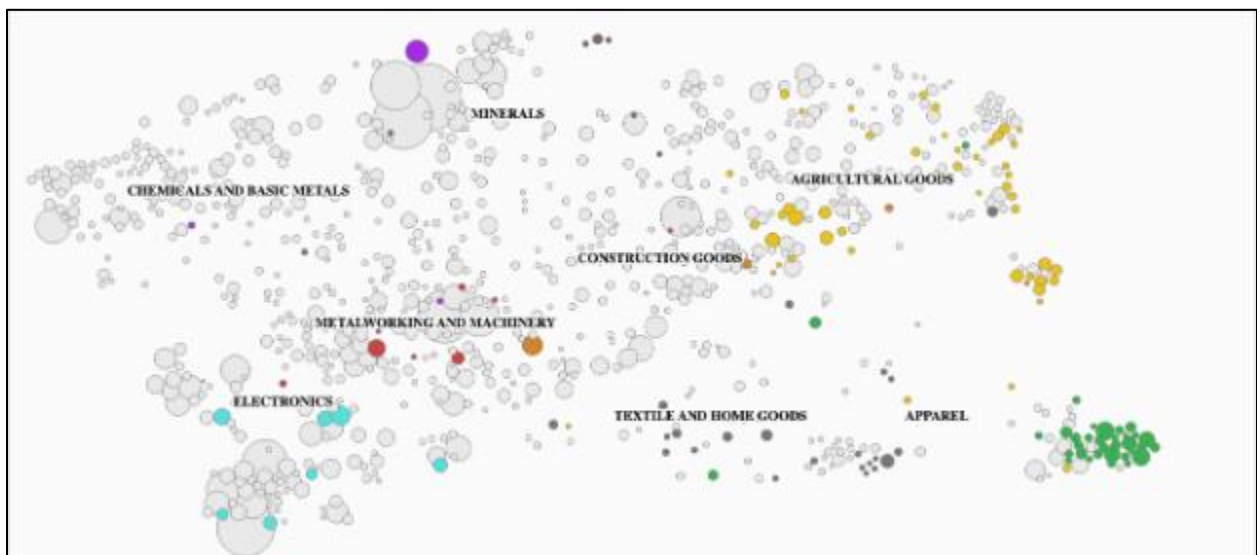
India

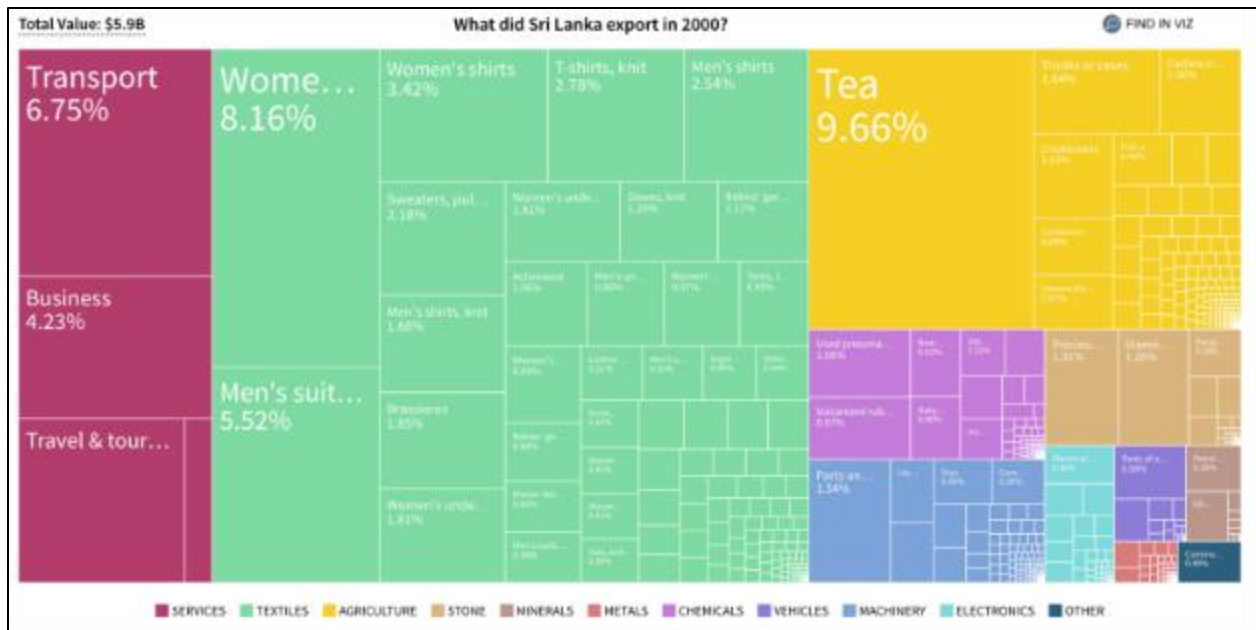
India is a large and highly diversified economy, with some concentration into electronics and apparel. Its trade composition is made of services, apparel and diamonds. This may make it a poor comparator of Vietnam because its services are driving its exports.



Sri Lanka

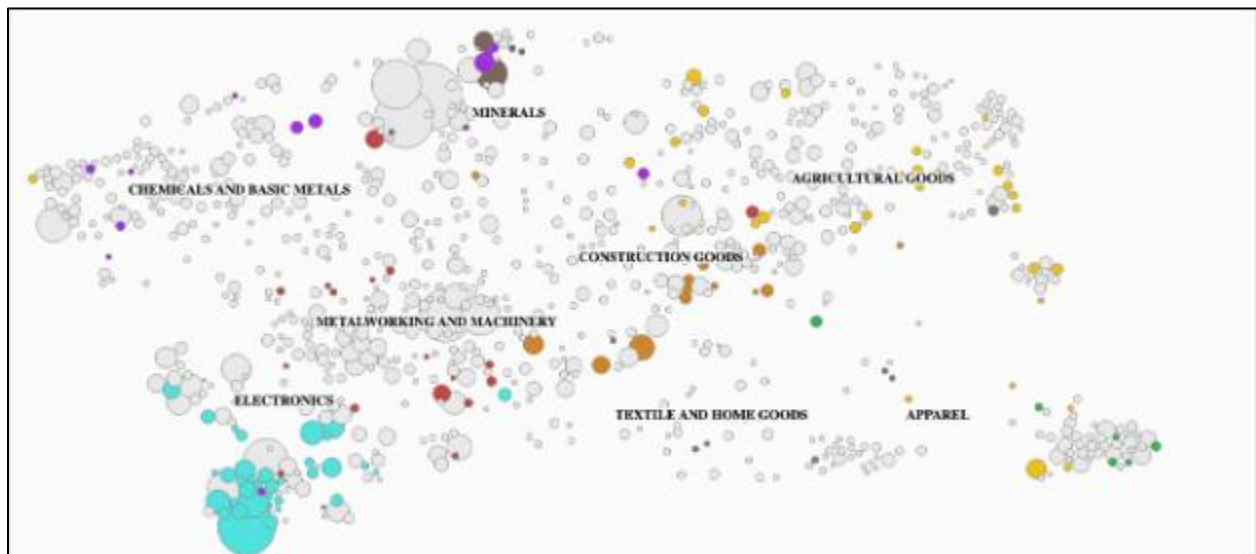
Sri Lanka has a large concentration of its product space in apparel and a budding concentration in electronics. Its trade composition is made up of services, textiles and agriculture. We do not reject Sri Lanka based on its trade composition and product space.

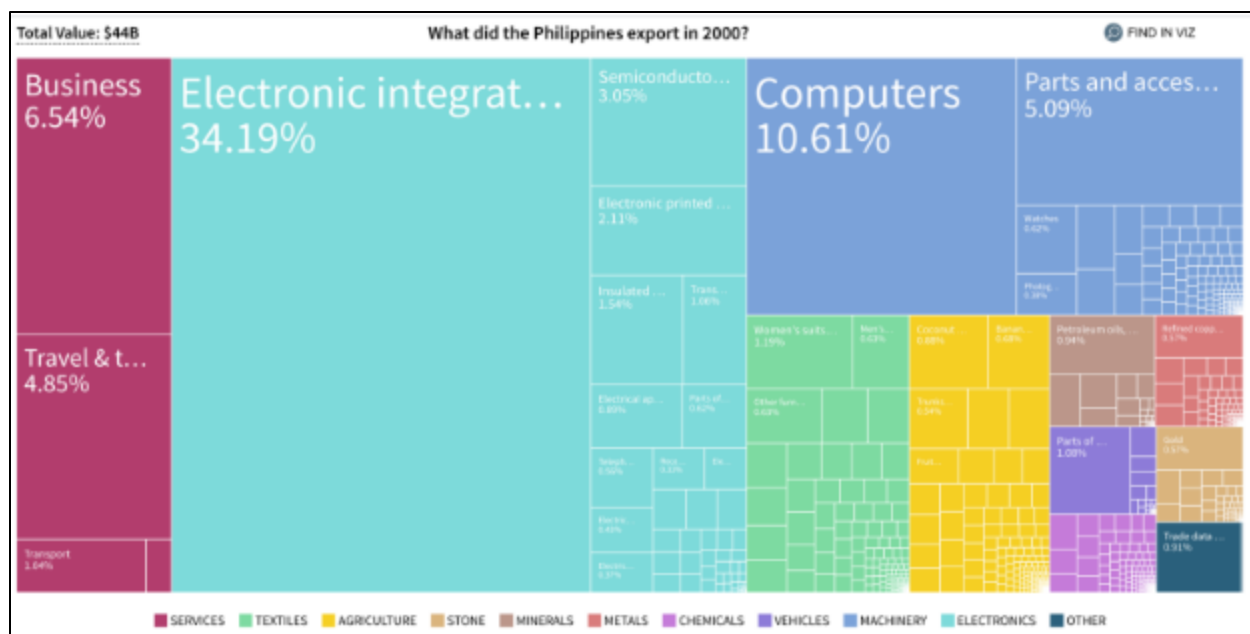




Philippines

Philippines product space reveals it specializes primarily in electronics. Its trade composition is made up of electronics, machinery and textiles. This shows that Philippines export growth has been driven by similar industries, therefore making Philippines a good comparator for Vietnam.





Appendix 5.3: Size of economy and GDP per capita

We include this in the analysis to supplement the CDA analysis. We consider overall GDP at PPPs and GDP per capita growth in figures 1 and 2 in Appendix 1.4 to eliminate too large/small economies that would not be comparable to Vietnam. Based on this analysis we reject: China, Sri Lanka.

GDP at 2023: China: \$31.23 trillion, Thailand: \$1.52 trillion, Bangladesh: \$1.41 trillion, Vietnam: \$1.35 trillion, Malaysia: \$1.15 trillion, Philippines: \$1.14 trillion, Singapore: \$754.75 billion, Sri Lanka \$287.13 billion, Cambodia: \$116.58 billion

Based on Figure A-3, we reject China because it is a much larger economy than Vietnam, despite having a similar product space.

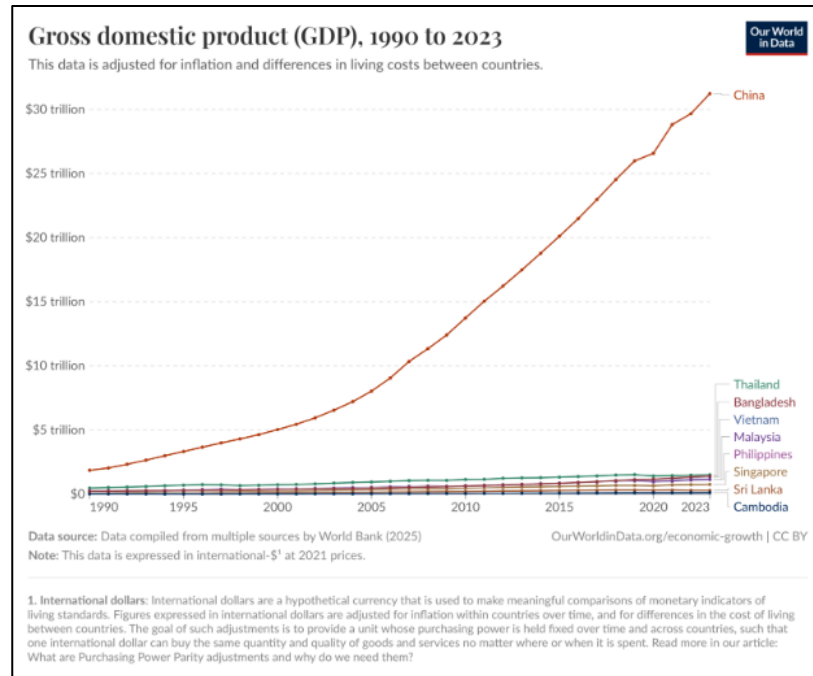


Figure 5.3a: GDP with Peer Countries

GDP per capita growth at 2023: Thailand: 1.96%, Bangladesh: 4.49%, Vietnam: 4.34% Malaysia: 2.29%, Philippines; 4.69%, Singapore: -3.72%, Sri Lanka: -1.66%, Cambodia: 3.62%

We can reject Sri Lanka based on falling GDP per capita growth which is not informative of Vietnam's growth trajectory.

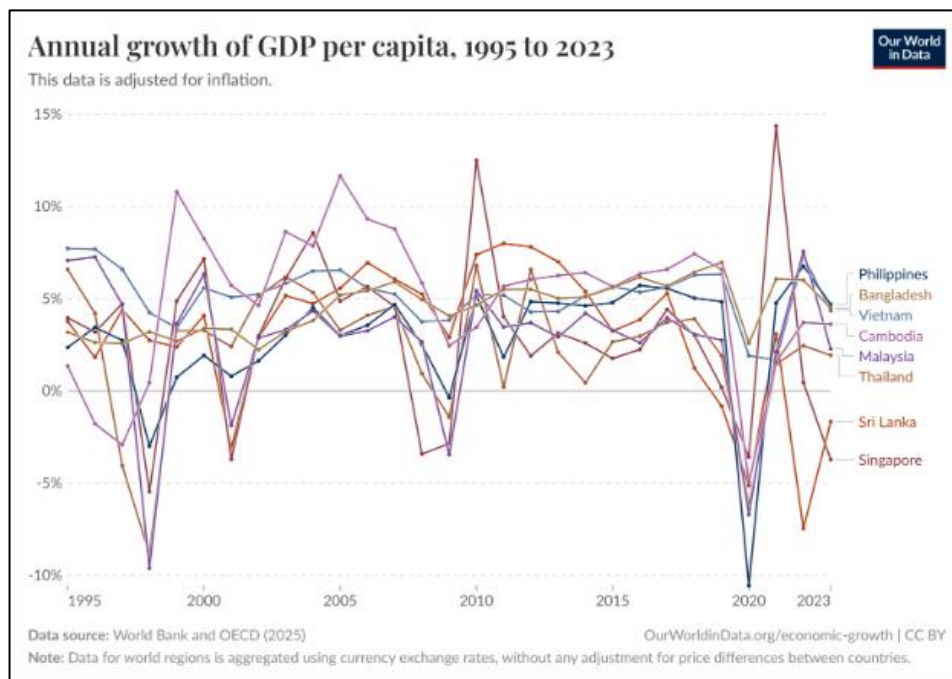


Figure 5.3b: Annual GDP per capita with Peers

Appendix 5.4: Aspirational Peer Analysis

Singapore is amongst the most economically advanced economies in the Indo-Pacific region. Both are similar growth trajectories in that both focused on foreign direct investments and exports to rapidly improve their living standards. Geographic proximity means both countries experienced similar regional shocks throughout their development.

It must be noted that there are political, economic, and demographic differences between the two countries. Vietnam faces unique challenges in a world that is deviating away from multilateral export and becoming more protectionist. Hence, the context of Singapore's growth solutions varies from that of Vietnam. Any extrapolation of Singapore's successful policies on the Vietnamese case must be conducted cautiously. Appendix 1.5 breaks down FDI flows and political similarities.

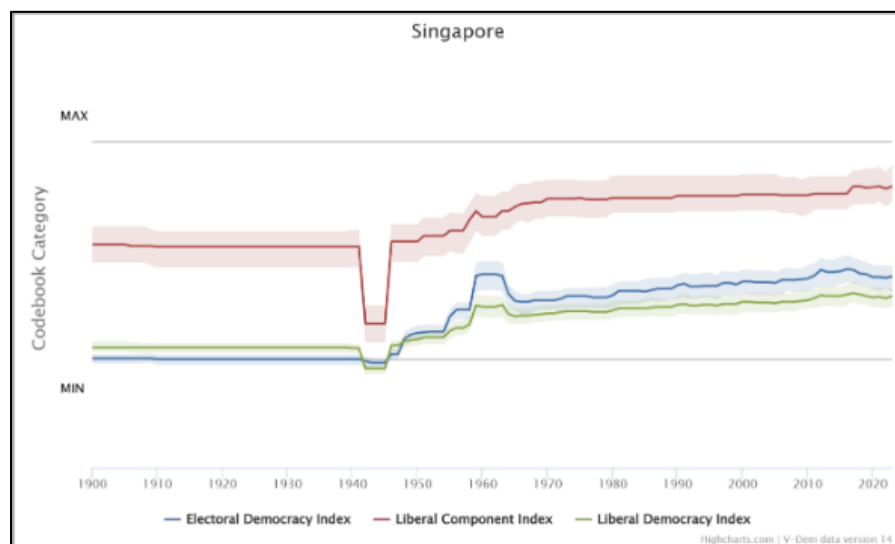


Figure 5.4a: Singapore Democratic Index

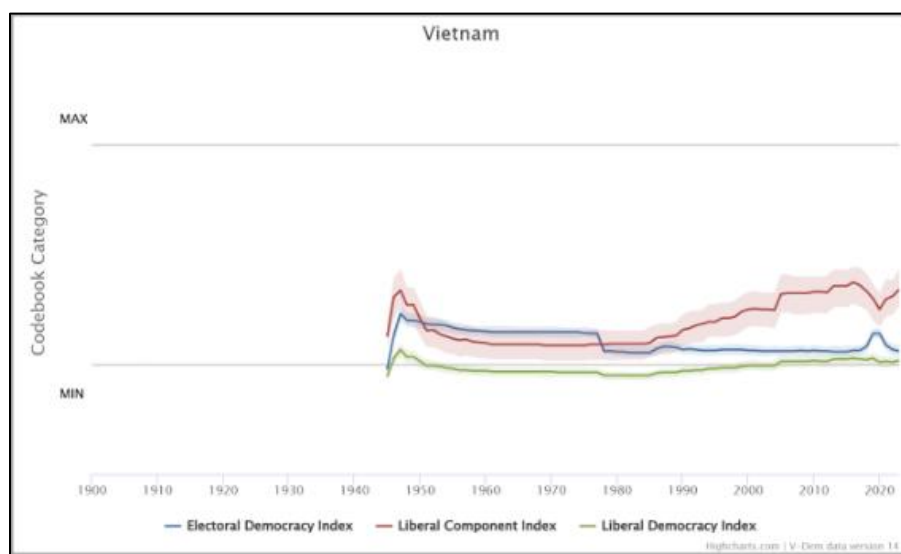


Figure 5.4b: Vietnam Democratic Index

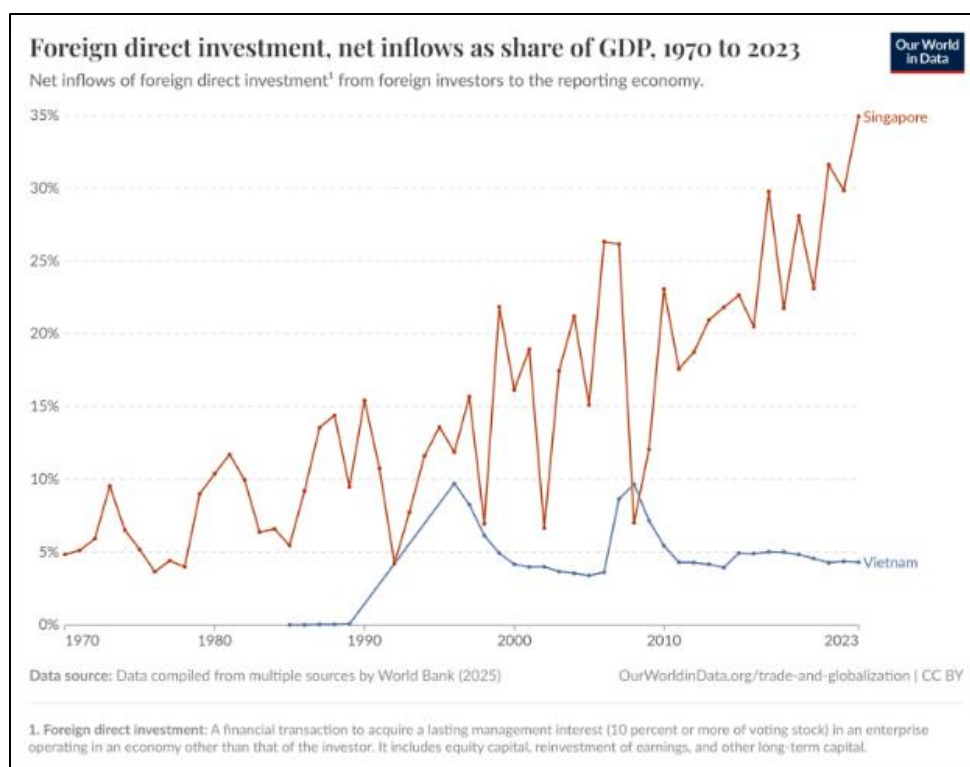


Figure A-7: Net Inflows of FDI as a share of GDP (Vietnam and Singapore)

Appendix 5.5: Final Selection

Country	Sample	Selection Criteria
Cambodia	World & Indo-Pacific	CA
Bangladesh	World & Indo-Pacific	CA and Commonality
Philippines	World & Indo-Pacific	CA and Commonality
Malaysia	Indo-Pacific	CA and Commonality
Thailand	Indo-Pacific	CA and Commonality
Singapore	Indo-Pacific	Aspirational Peer